GENERAL ORGANIC CHEMISTRY - II (GOC - II)

(I) Types of reaction intermediates

(II) Acidic Stength

(III) Basic Stength

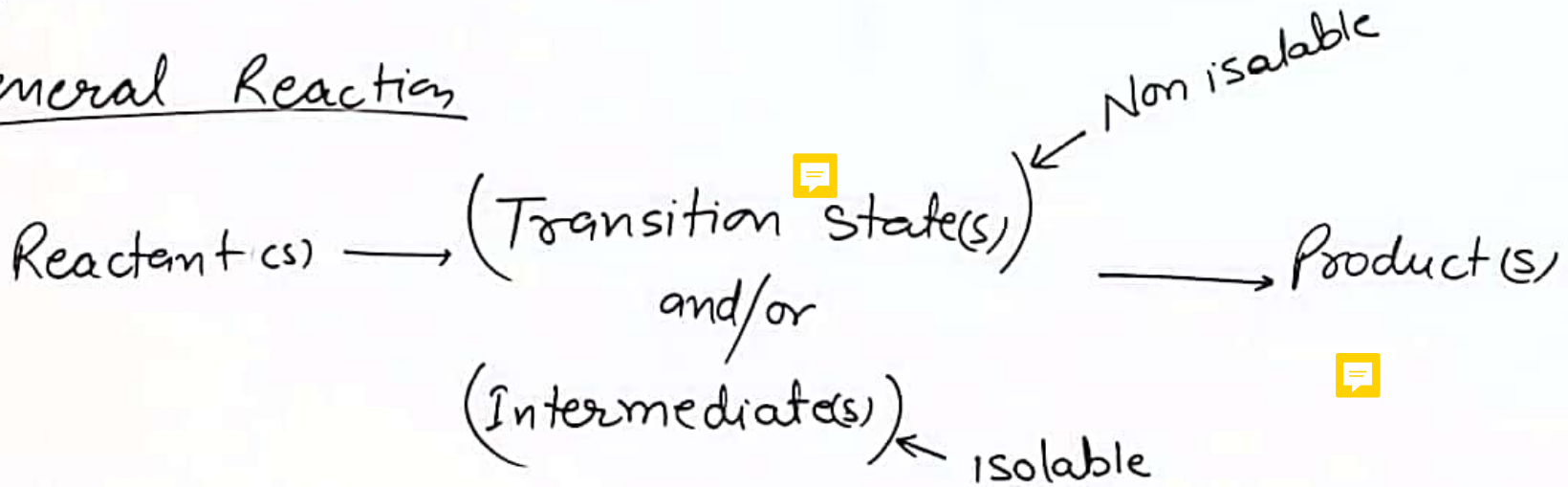
(IV) Acid – Base reactions

(V) Tautomerism.

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(I) Types of reaction intermediates

General Reaction



Types of intermediates:-

✓(i) Carbocation = Carbonium ion

✓(ii) Carbanion

✓(iii) Free radicals

(iv) Carbene

(v) Nitrene

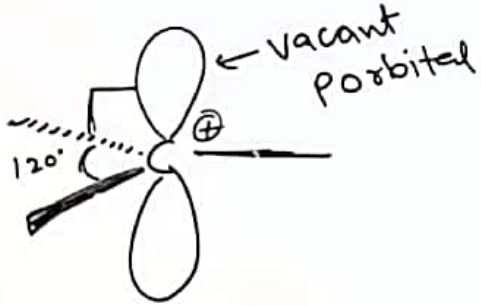
(vi) Benzyne

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(I) Types of reaction intermediates

(1) Carbonium ion = Carbocation

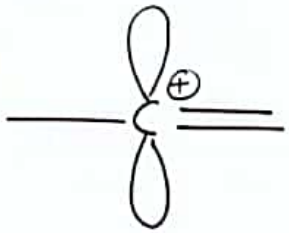
3σ



⇒ SP^2

⇒ Trigonal planar

$2\sigma + 1\pi$



⇒ SP

⇒ Linear

$1\sigma + 2\pi$



Unstable

⇒ incomplete octet

⇒ Electrophile

⇒ Lewis acid

GENERAL ORGANIC CHEMISTRY-II (GOC-II)

(I) Types of reaction intermediates

(2) Carbanions :-

⇒ complete octet

⇒ 6 e^-s in bonded + 2 e^- in non bonded (lone pair)

⇒ nucleophile

3σ bond



- sp^3
- Trigonal Pyramidal

2σ + 1π



sp^2

bent

1σ + 2π bond



sp

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(I) Types of reaction intermediates

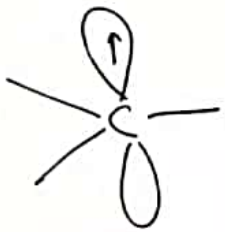
(3) Free radical :-

→ incomplete octet

→ 7 e^- containing (6 e^- in 3 bond + One non bond (unpaired))

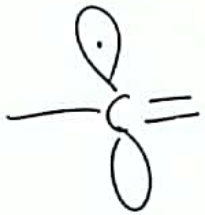
⇒ Paramag.

3σ $2\sigma + 1\pi$ $1\sigma + 2\pi$ bonds.



sp^2

Trigonal planar



sp

Linear



Unstable

GENERAL ORGANIC CHEMISTRY - II (GOC - II)

(I) Types of reaction intermediates

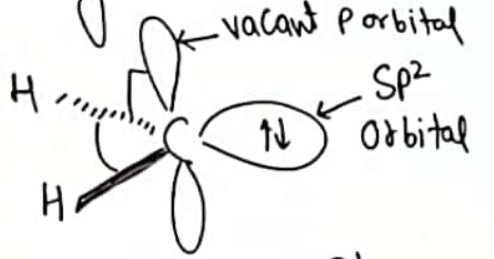
(4) Carbene :C<

⇒ incomplete Octed

⇒ 6 e^- (4 e^- in 2 bonds + 2 non bonded $e^\ominus$)

Types

Singlet Carbene $\uparrow\downarrow CH_2$



⇒ Sp^2, V Shape

⇒ Diamag.

Triplet Carbene $(\uparrow\uparrow CH_2)$



⇒ SP ⇒ Paramag.

= Linear

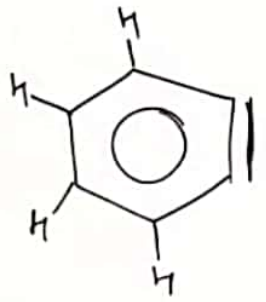
GENERAL ORGANIC CHEMISTRY - II (GOC - II)

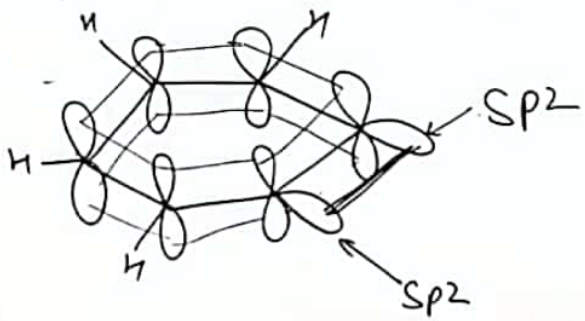
(I) Types of reaction intermediates

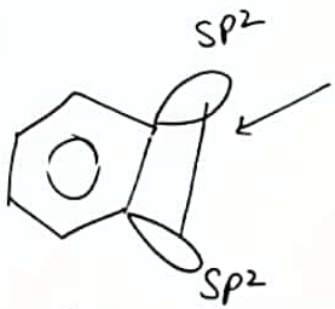
(5) Nitrene —N̈:

- incomple octet
- 6 e⁻(s) (2e⁻ in one bond + 4 non bonded e⁻(s))

(6) Benzyne:- (C_6H_4)







weak sp^2 - sp^2 (in same plan of ring)

6πe aromatic

GENERAL ORGANIC CHEMISTRY-II (GOC-II)

(I) Intermediates:-

Factors affecting stability of intermediates.

(1) When charge is present on different atoms of same period, then stability depends on electronegativity of atom.

(2.) When charge is present on different atoms of same group, then stability depends on size of atom.

(3) When charge is present on same type of atom in different species, then consider electronic effects.

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(I) Intermediates:-

⇒ Stability of ion $\propto \frac{1}{\text{Charge density}}$

$$\text{Charge density} = \frac{\text{charge}}{\text{Volume}}$$

⇒ " " " $\propto$ Delocalisation of charge
(Resonance)

⇒ For cations and free radicals :- Stability ↑es in presence of e^- donating groups (+M, +H, +I) and Stability ↓es in presence of e^- withdrawing groups (-M, -I)

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(I) Intermediates:-

⇒ For anions :- Stability ↑es in presence of e^- withdrowing groups ($-M, -I$)

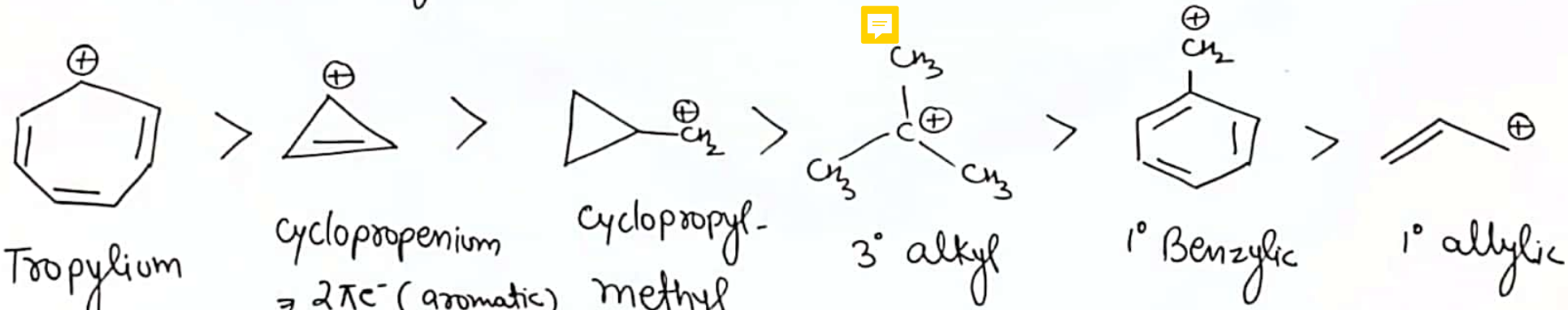
⇒ For cation and free radicals : Stability α $\frac{1}{\%.\ S\ character}$ (for same atoms)

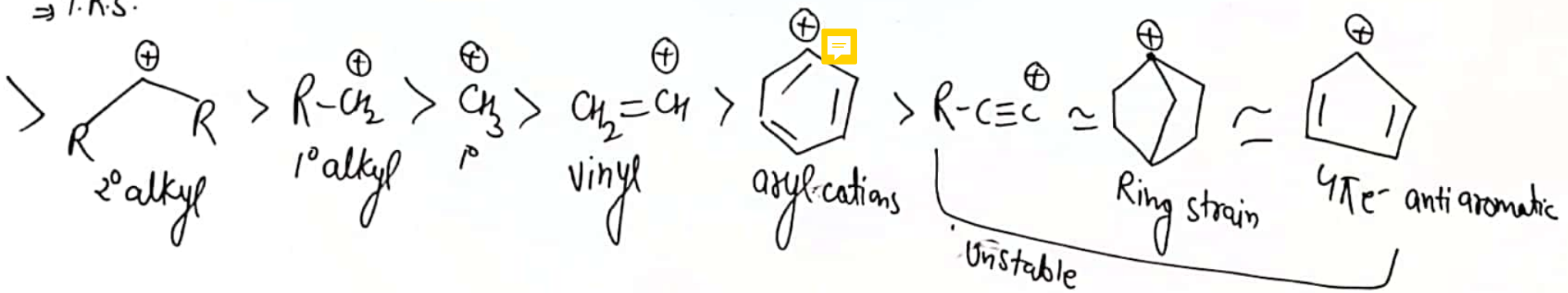
⇒ For anions : Stability α % S character (for same atoms)

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(I) Intermediates:-

General order of stability of carbocations.

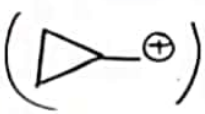




GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(I) Intermediates:-

General order of stability of carbocations.

Note 

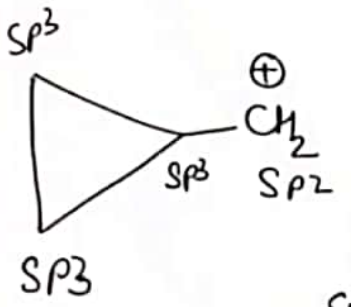
Cyclopropyl methyl cation is highly stable due to delocalisation of loosly bonded σ bond e$^-$(s) of ring with vacant p orbital of +ve charged carbon (Dancing resonance)

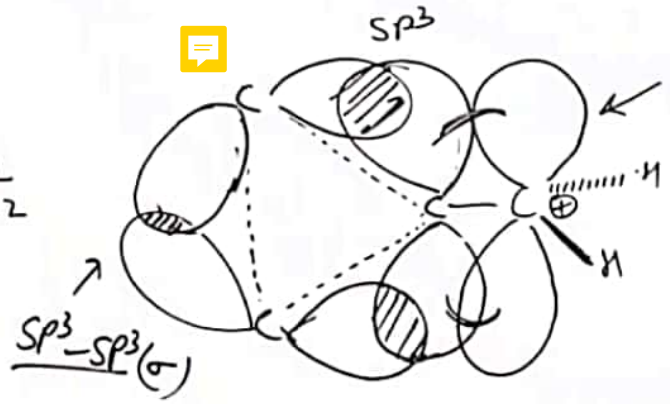
GENERAL ORGANIC CHEMISTRY - II (GOC-II)

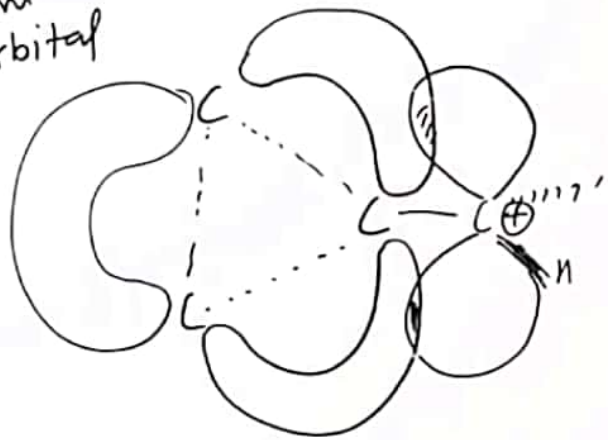
(I) Intermediates:-

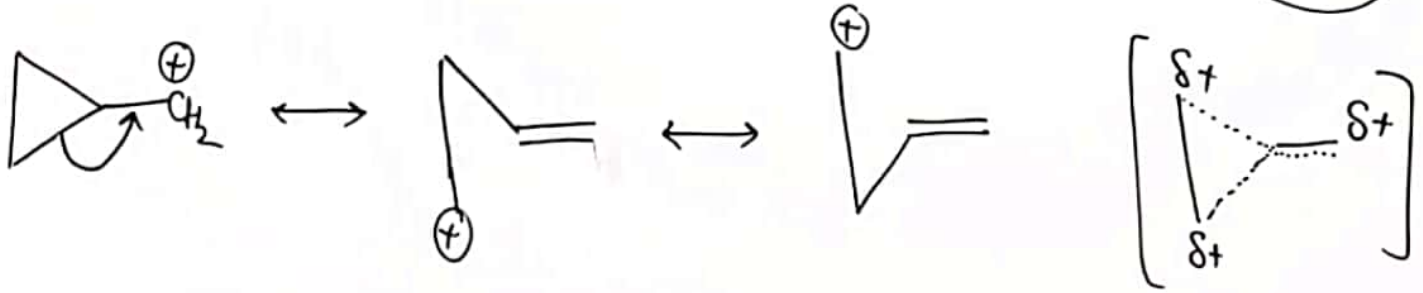
General order of stability of carbocations.

Note 





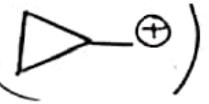


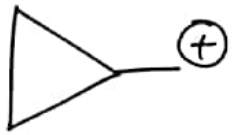
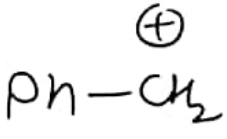


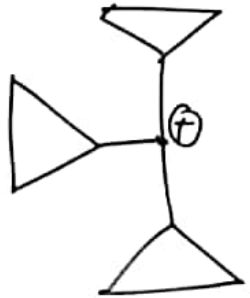
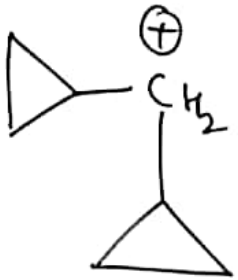
GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(I) Intermediates:-

General order of stability of carbocations.

Note 

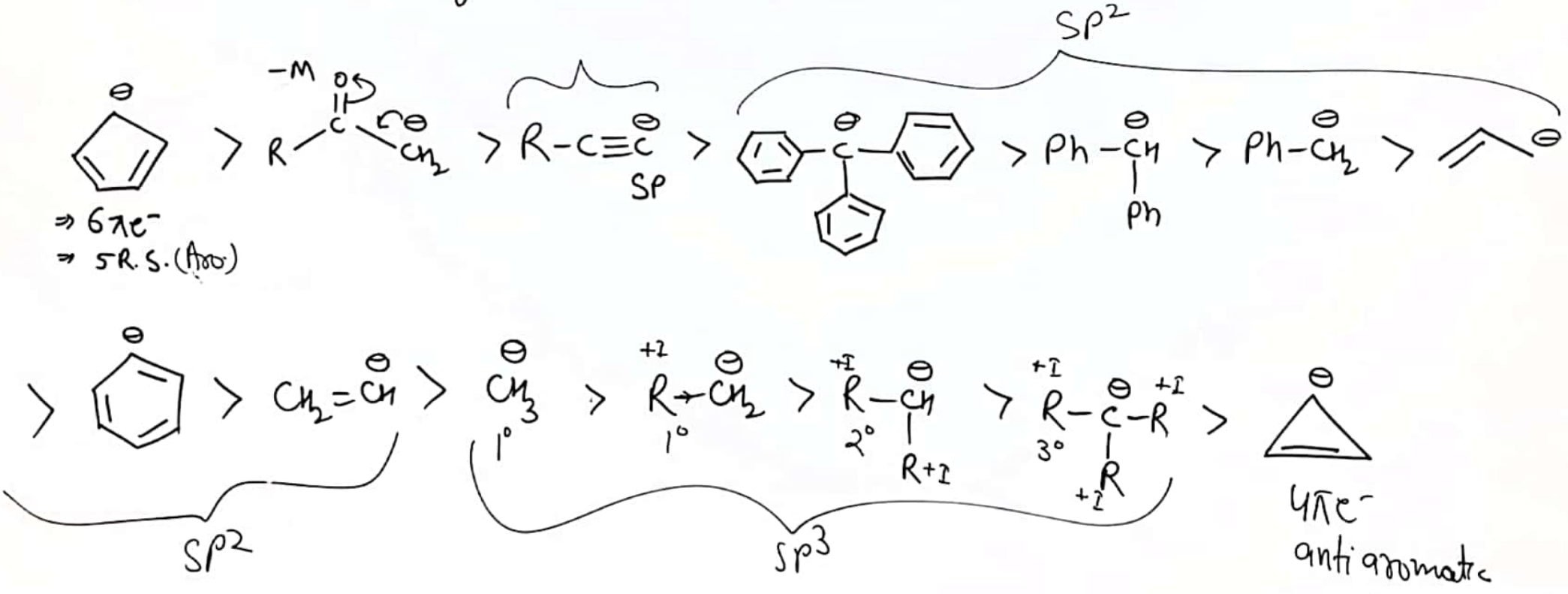
eg  >  Stability

eg.  >  > 

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(I) Intermediates:-

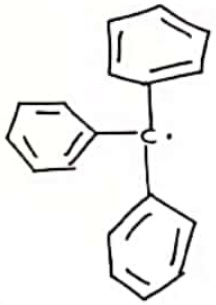
General order of stability of Carbanions

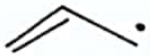


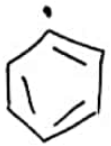
GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(I) Intermediates:-

General order of stability of free radicals.



$>$ Ph–$\dot{C}$H–Ph $>$ Ph–$\dot{C}H_2$ $>$  $>$ R–$\dot{C}$H–R (+H, +I) $>$ R–$\dot{C}$H–R $>$ R–$\dot{C}H_2$ $>$ $\dot{C}H_3$

$>CH_2=\dot{C}H >$  $>$ R–C≡$\dot{C}$

GENERAL ORGANIC CHEMISTRY - II (GOC- II)

(I) Stability Of intermediates:-

eg 1 $\overset{\ominus}{C}H_3 < \overset{\ominus}{N}H_2 < \overset{\ominus}{O}H < \overset{\ominus}{F}$ Stability

eg 2 $\overset{\ominus}{P}H_2 < \overset{\ominus}{S}H < \overset{\ominus}{Cl}$ "

eg 3 $\overset{\ominus}{F} < \overset{\ominus}{Cl} < \overset{\ominus}{Br} < \overset{\ominus}{I}$ "

eg 4 $\overset{\ominus}{O}H < \overset{\ominus}{S}H$

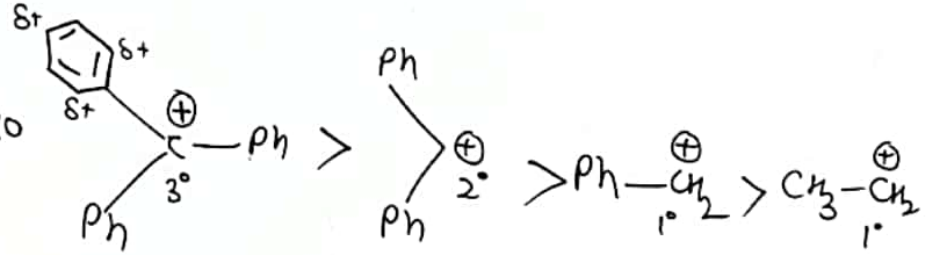
eg 5 $\overset{\ominus}{O}R < \overset{\ominus}{S}R$

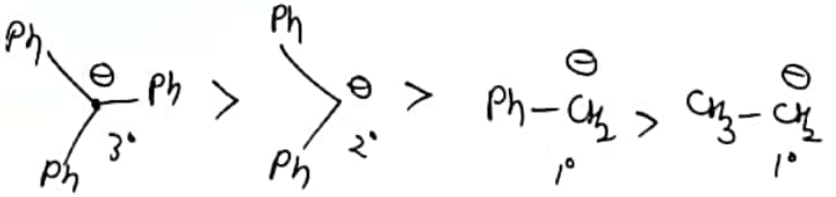
eg 6 $\overset{\ominus}{O}Ph < \overset{\ominus}{S}Ph$

eg 7 $R-\underset{SP^3}{\overset{\ominus}{C}H_2-CH_2} < R-\underset{SP^2}{\overset{\ominus}{C}H=CH} < R-\underset{SP}{C\equiv\overset{\ominus}{C}}$

eg 8 $R-CH_2-\overset{\oplus}{C}H_2 > R-CH=\overset{\oplus}{C}H > R-C\equiv\overset{\oplus}{C}$

eg 9 $R-CH_2-\dot{C}H_2 > R-CH=\dot{C}H > R-C\equiv\dot{C}$

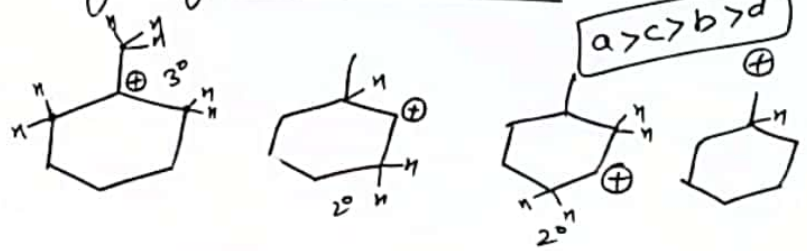
eg 10 

eg 11 

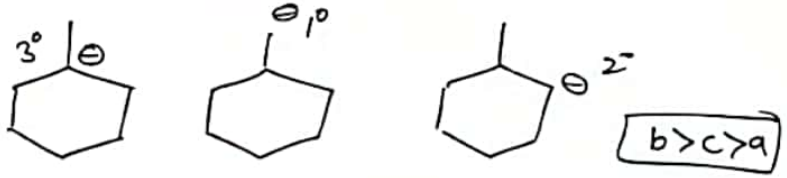
GENERAL ORGANIC CHEMISTRY - II (GOC- II)

(I) Stability Of intermediates:-

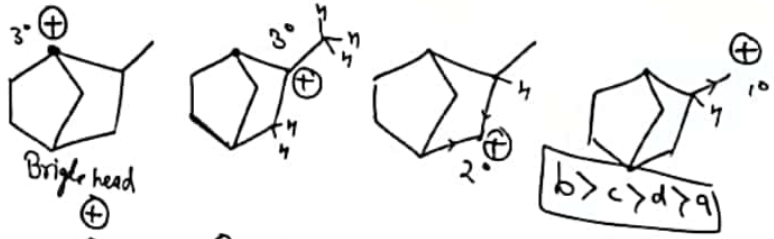
eg 12



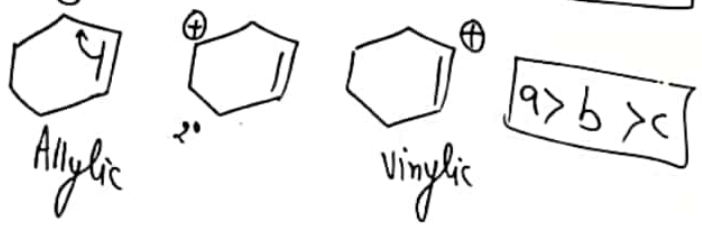
eg 13



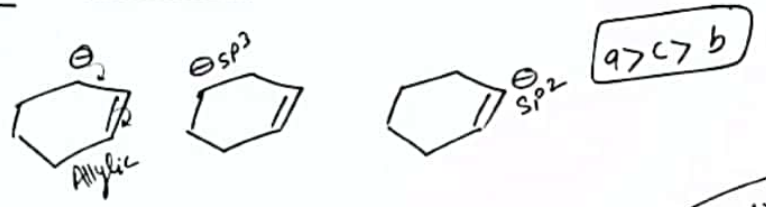
eg 14



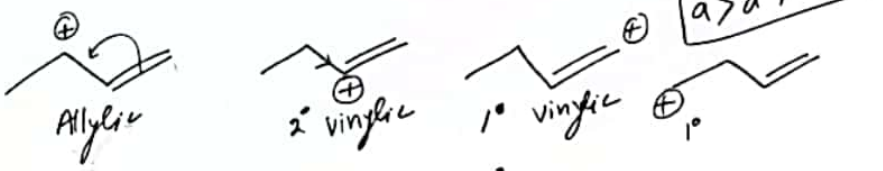
eg 15



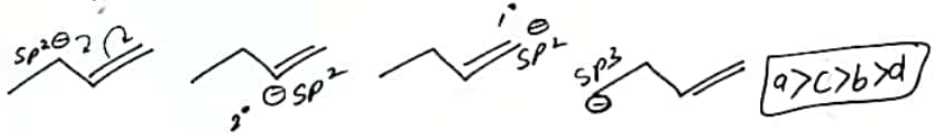
eg 16



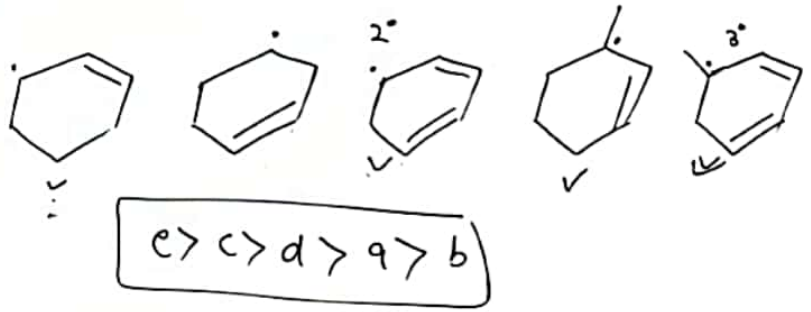
eg 17



eg 18



eg 19



GENERAL ORGANIC CHEMISTRY - II (GOC - II)

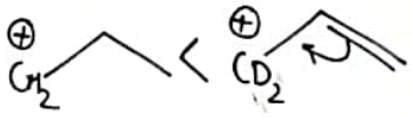
(I) Stability Of intermediates:-

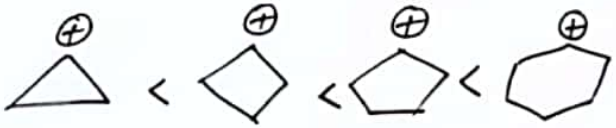
eg 20 $CH_3^{\oplus} < CD_3^{\oplus}$ **+I D>H**

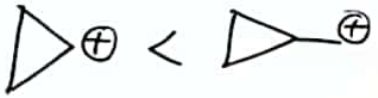
eg 21 $CH_2(H)-CH_2^{\oplus} > CD_2(D)-CD_2^{\oplus}$ C-H > C-D H.C

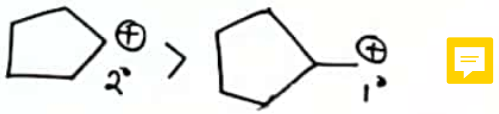
(C-D > C-H Bond energy)

eg 22 $CH_2^{\oplus}-C(CH_3)(CH_3)(CH_3) < CD_2^{\oplus}-C(CD_3)(CD_3)(CD_3)$

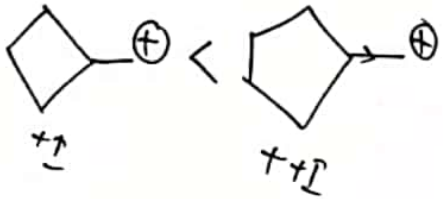
eg 23 

eg 24 

eg 25 

eg 26 

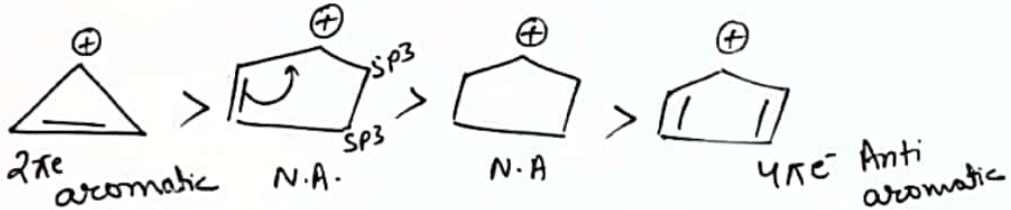
eg 27 

eg 28 

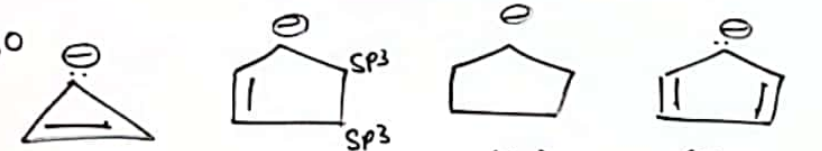
GENERAL ORGANIC CHEMISTRY - II (GOC - II)

(I) Stability Of intermediates:-

eg 29

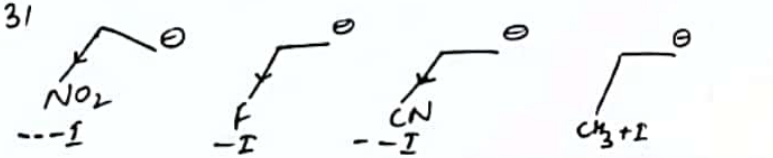


eg 30



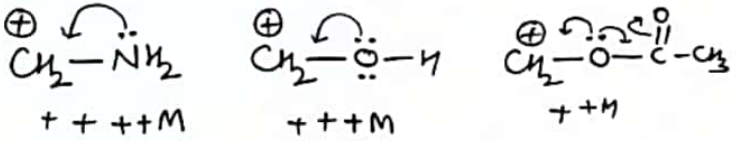
d > b > c > a

eg 31



a > c > b > d

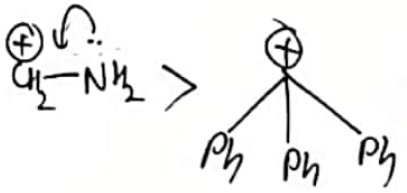
eg 32





a > b > c > d

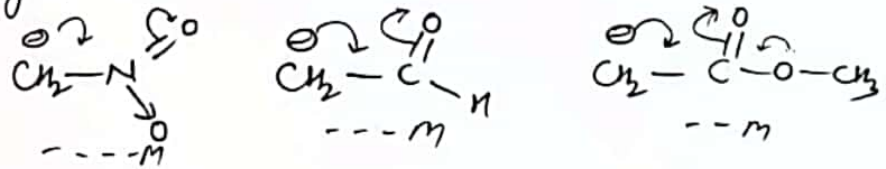
eg 33



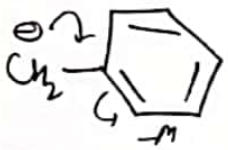
GENERAL ORGANIC CHEMISTRY - II (GOC - II)

(I) Stability Of intermediates:-

eg 34



a > b > c > d

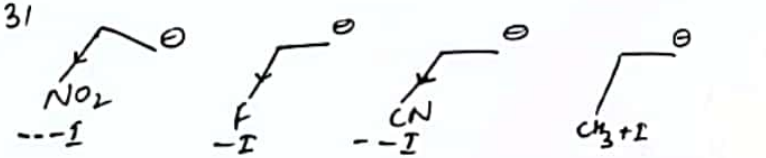


-M

$-\overset{\overset{O}{\|}}{N}\rightarrow O > -C\equiv N > -\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-OH > -\overset{\overset{O}{\|}}{C}-H > -\overset{\overset{O}{\|}}{C}-R$

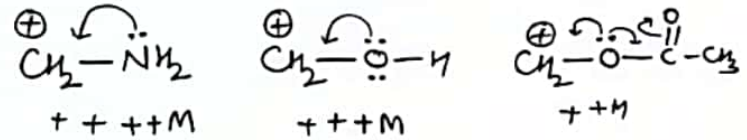
$> -\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R > -\overset{\overset{O}{\|}}{C}-OR > -\overset{\overset{O}{\|}}{C}-NH_2 > -\overset{\overset{O}{\|}}{C}-O^{\ominus} > Ph$

eg 31



a > c > b > d

eg 32

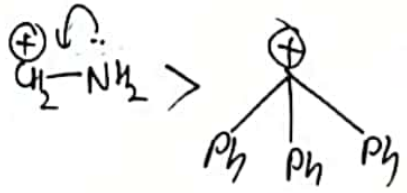




+M

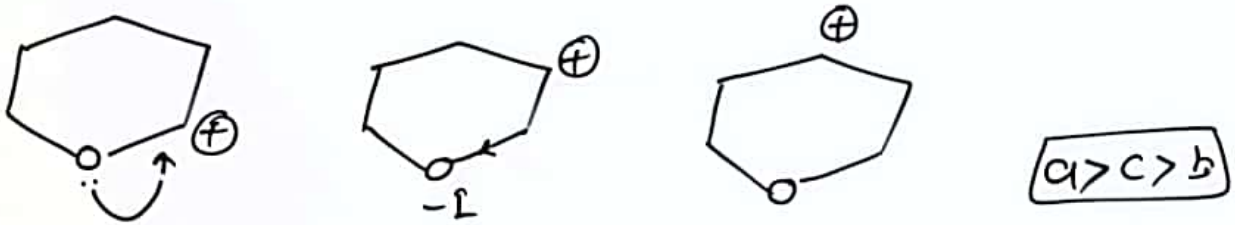
a > b > c > d

eg 33



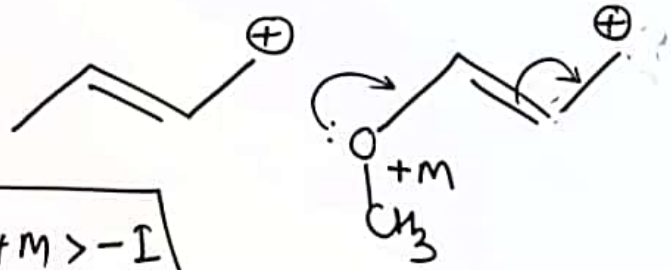
(I) Stability Of intermediates:-

eg 35



a > c > b

eg 36

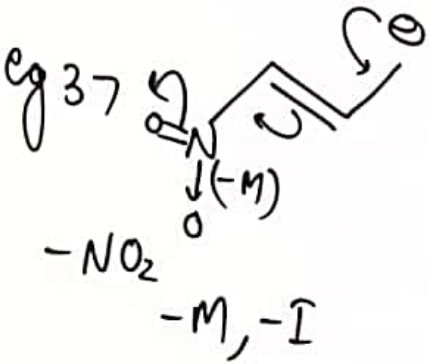


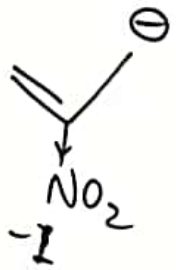
$-\ddot{O}CH_3$ +m > -I

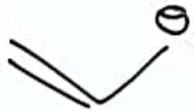


b > a > c

eg 37



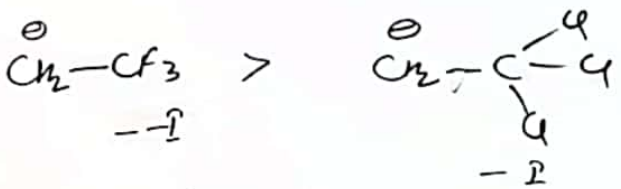


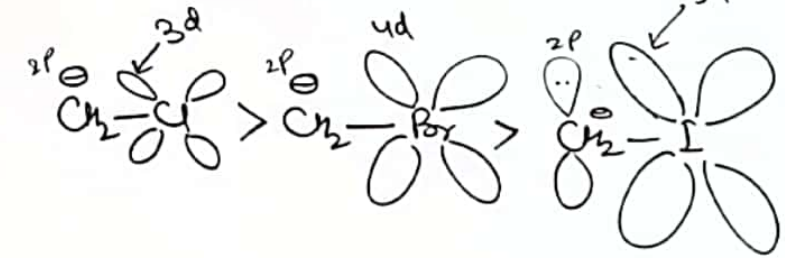


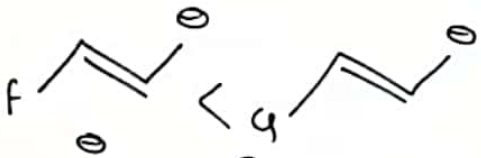
a > b > c

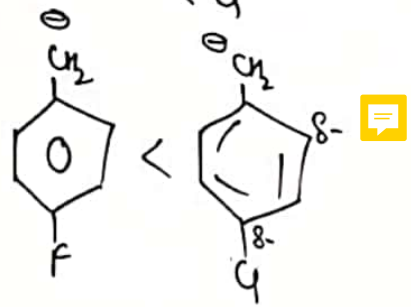
GENERAL ORGANIC CHEMISTRY - II (GOC - II)

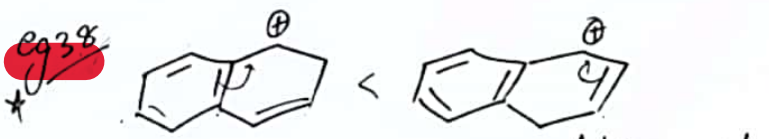
(I) Stability Of intermediates:-

eg 42 

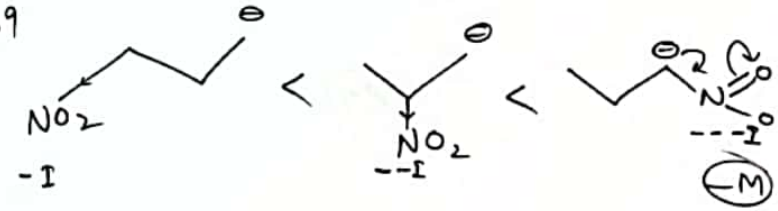
eg 43 

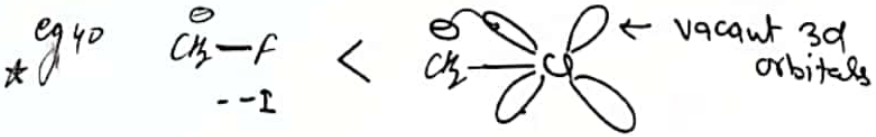
eg 44 

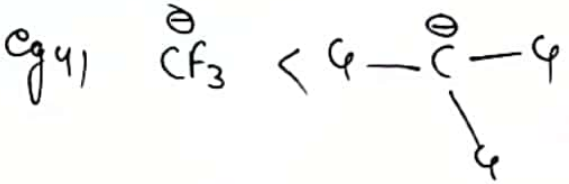
eg 45 

eg 38 

more no. of benzenoid Reso. structures.

eg 39 

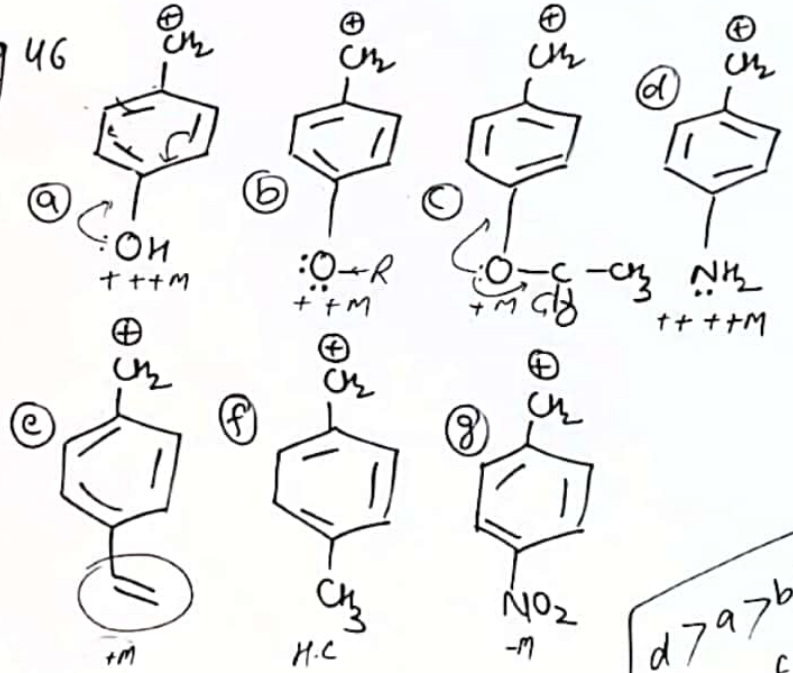
eg 40 

eg 41 

GENERAL ORGANIC CHEMISTRY - II (GOC - II)

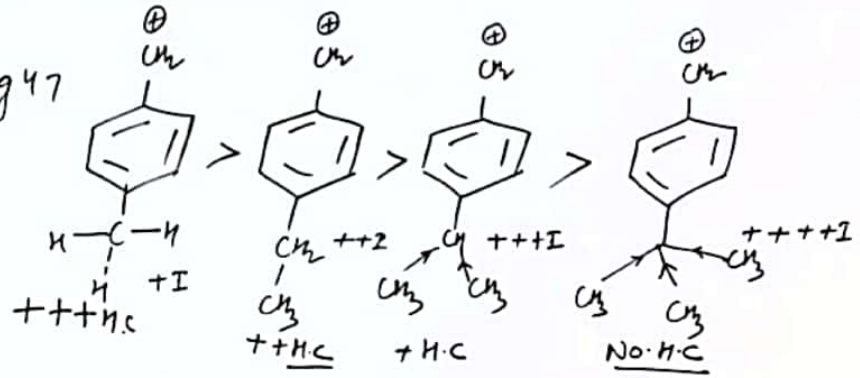
(I) Stability Of intermediates:-

eg 46

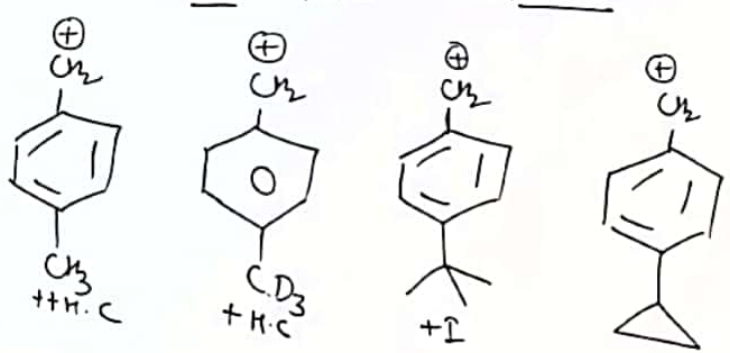


d > a > b > c > e > f > g

eg 47



eg 48

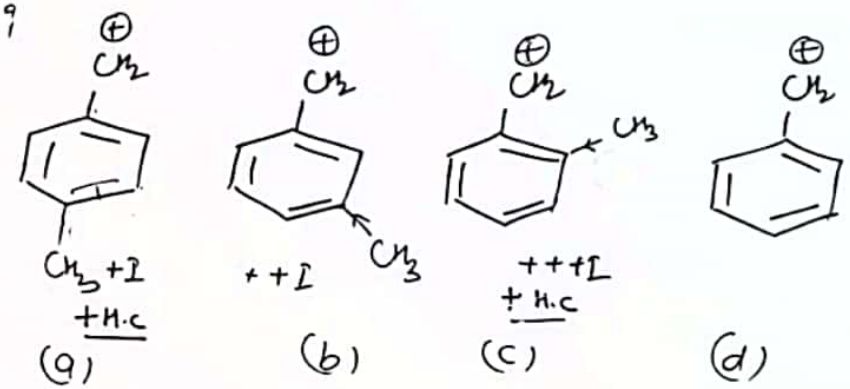


d > a > b > c

GENERAL ORGANIC CHEMISTRY - II (GOC- II)

(I) Stability Of intermediates:-

eg 49



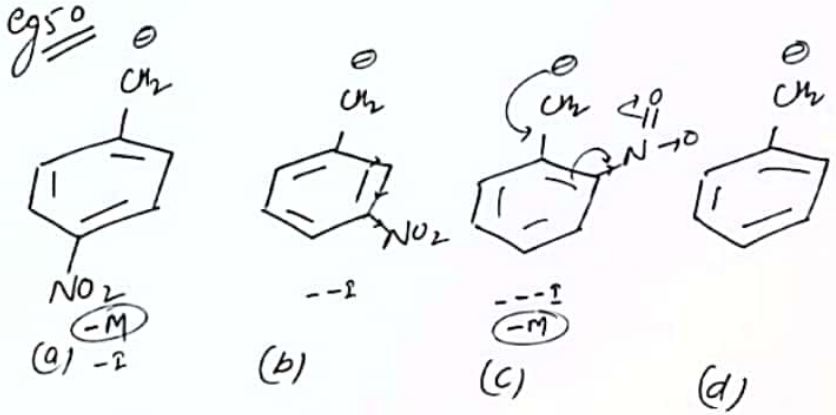
Hint $-CH_3$ $+H.C > +I$

($O=P$) ($O>m>P$)

Stability

C > a > b > d

eg 50



Hint $-NO_2$ $-M > -I$

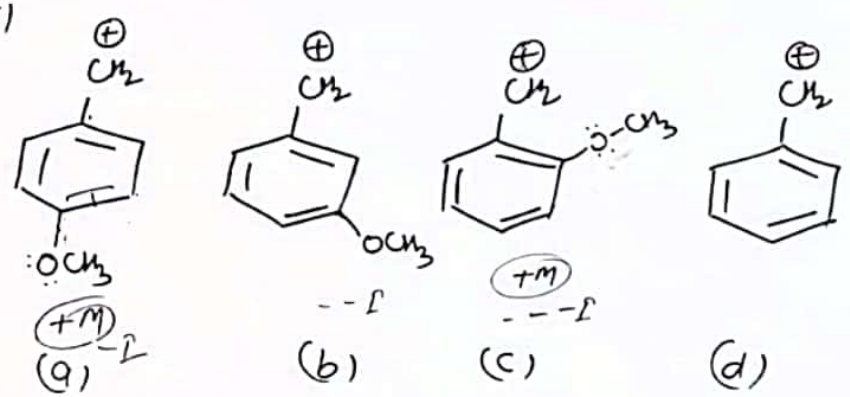
($O=P$) $O>m>P$

C > a > b > d

GENERAL ORGANIC CHEMISTRY - II (GOC - II)

(I) Stability Of intermediates:-

eg 51

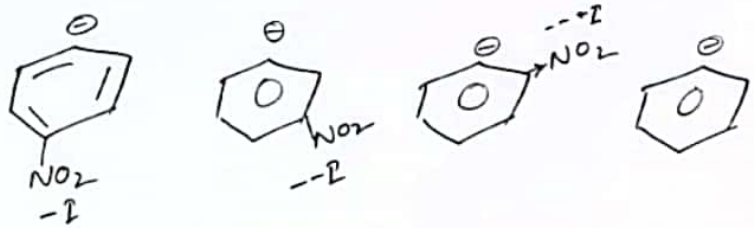


Hint $-\ddot{O}-CH_3$ +M > -I

(O = P) (O > m > P)

a > c > d > b

eg 52



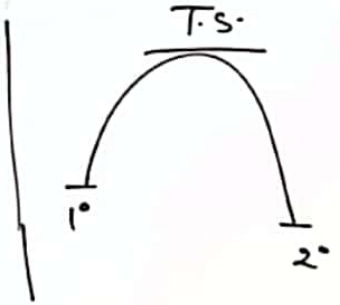
c > b > a > d

GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

Migration of a group or an atom with bond electrons on adjacent electron deficient atom.





Rearrangement common in ⇒ carbocations, carbenes, Nitrenes..

" " " rare in ⇒ carbanions, free radicals...

Daughter cation should be more stable than parent cation.

Only 1,2 shift possible in rearrangement

GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

Types of shift

(i) Hydride shift

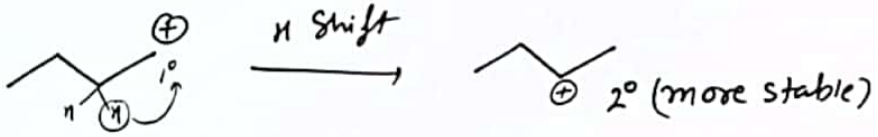
(ii) alkyl "

(iii) aryl "

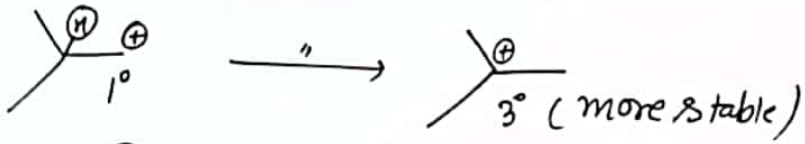
(iv) ring expansion

(v) ring contraction etc.

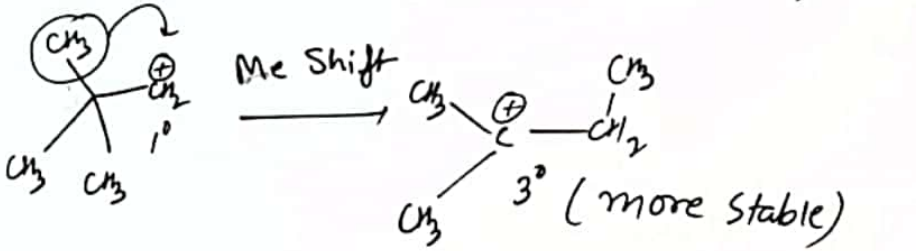
eg 1



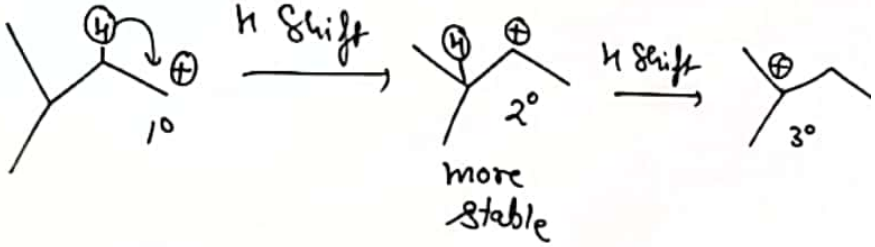
eg 2



eg 3



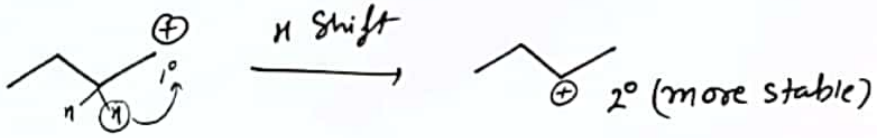
eg 4



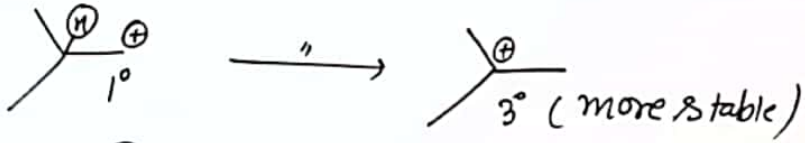
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

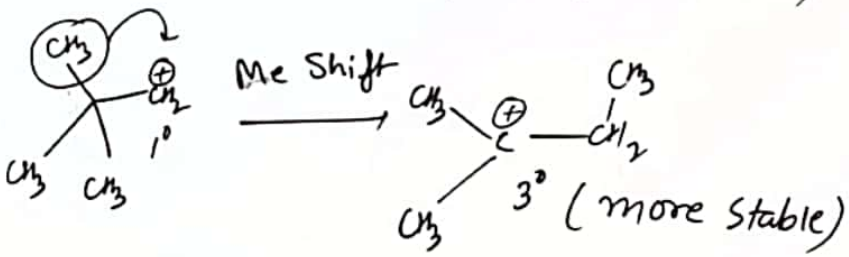
eg 1



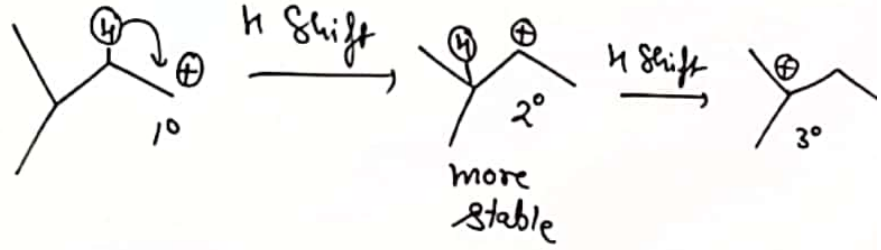
eg 2



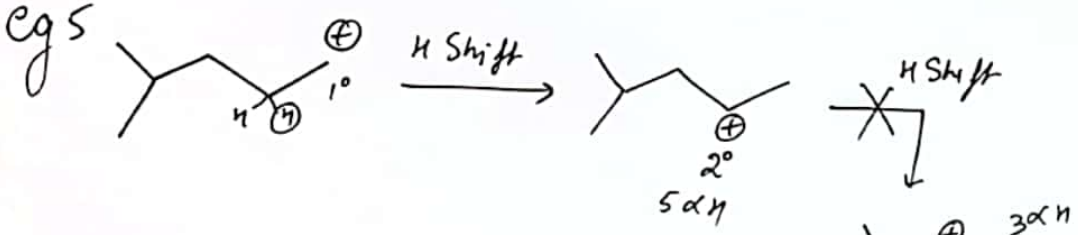
eg 3



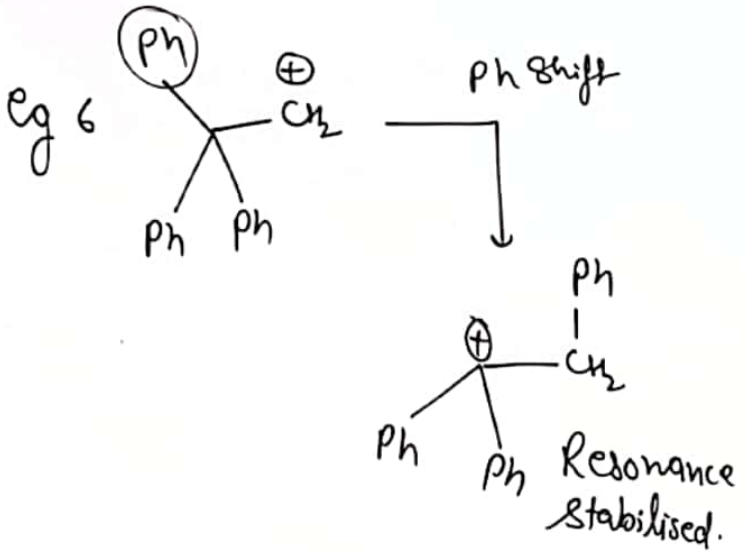
eg 4



eg 5



eg 6



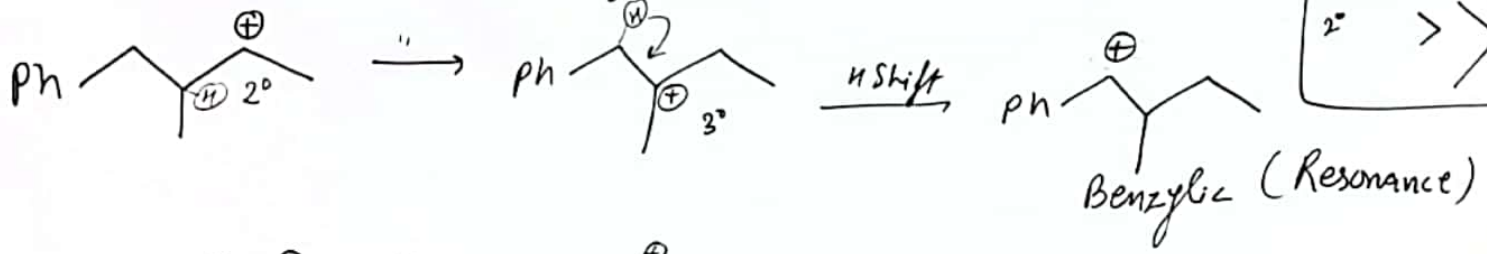
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

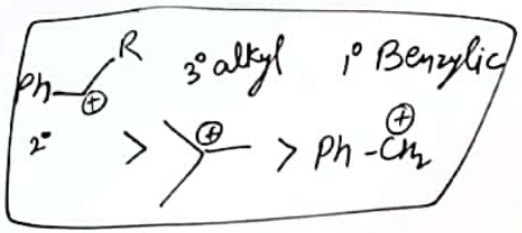
Rearrangement in carbonium ions

eg 7

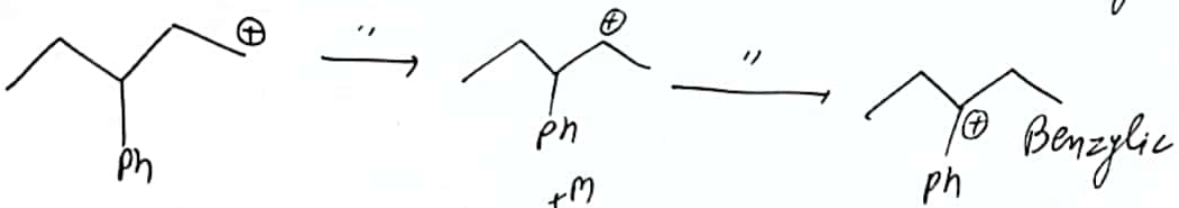


eg 8

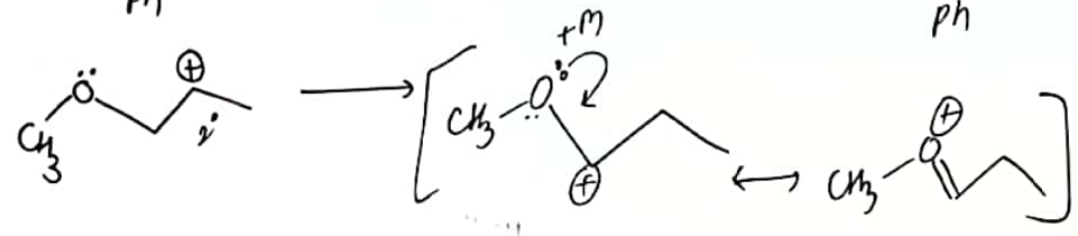




eg 9



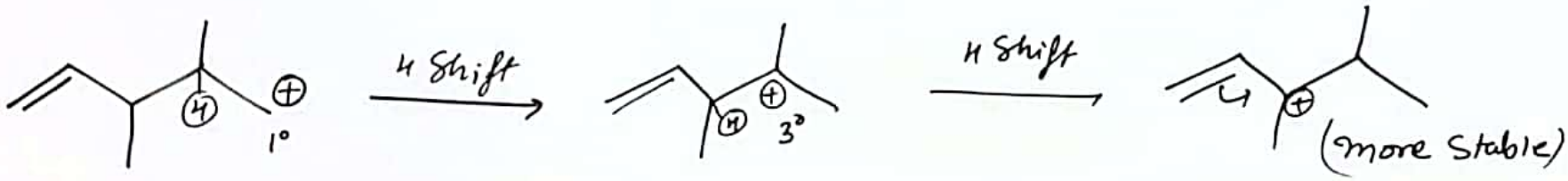
eg 10



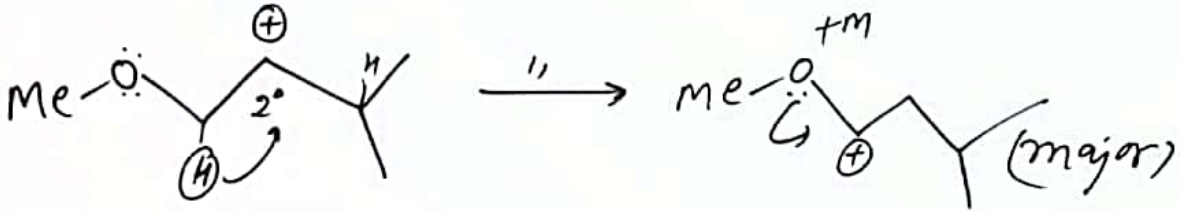
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

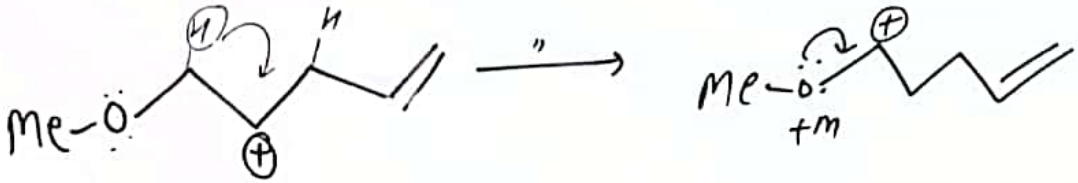
Eg 11



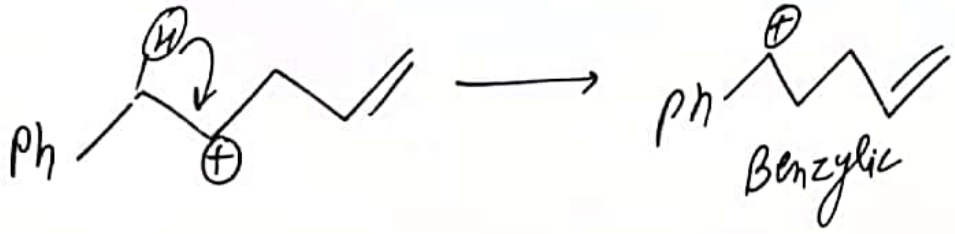
Eg 12



Eg 13



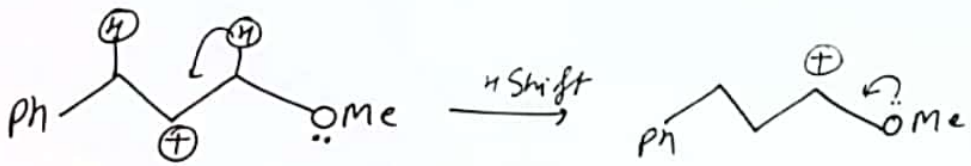
Eg 14



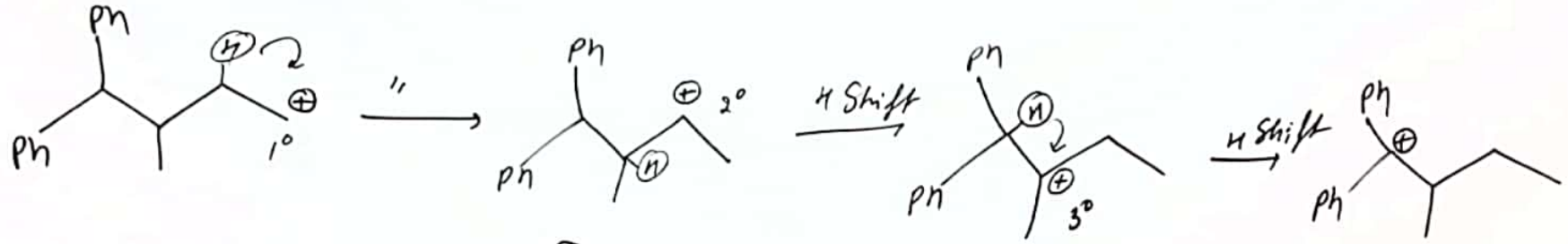
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

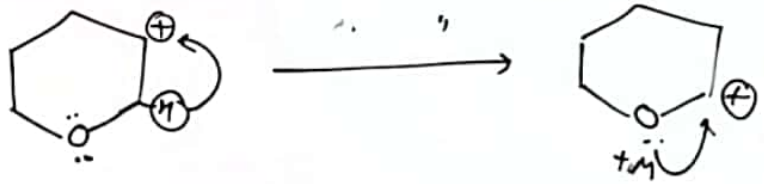
eg 15



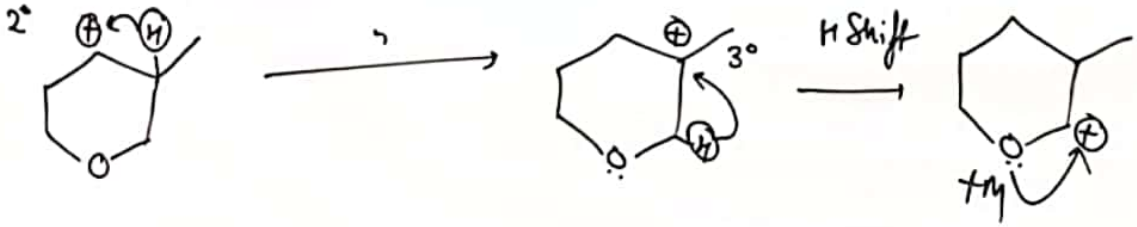
eg 16



eg 17



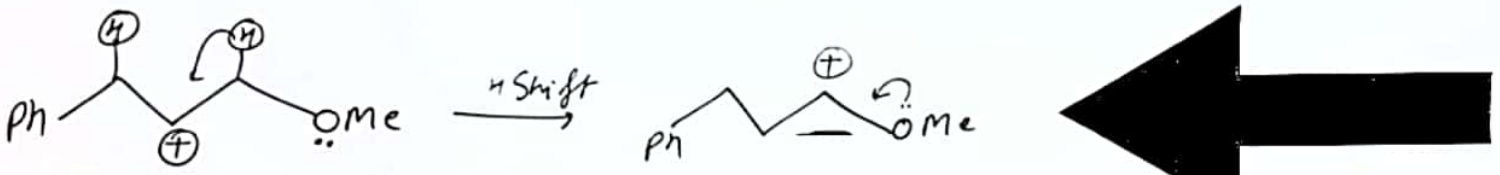
eg 18



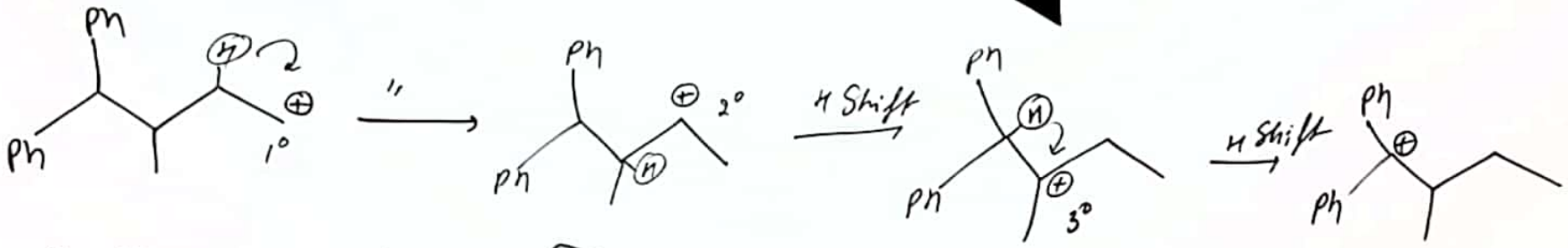
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

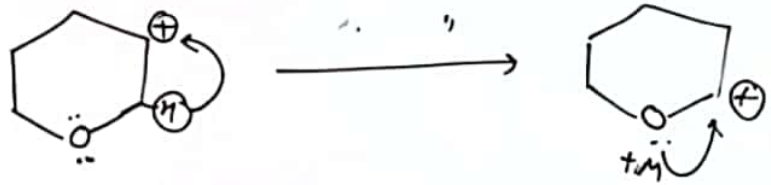
eg 15



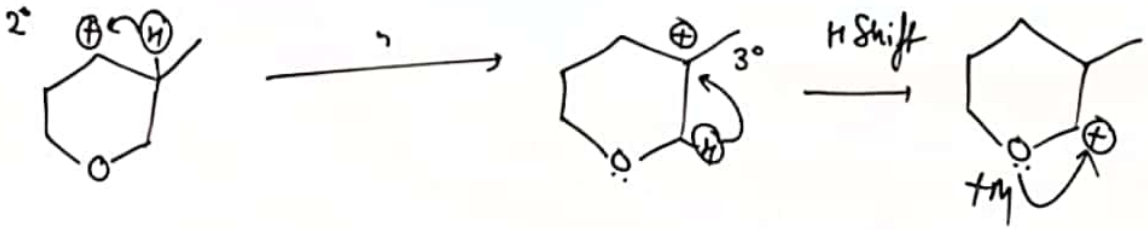
eg 16



eg 17



eg 18



GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

eg 19

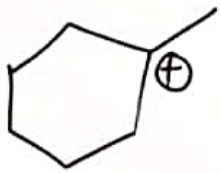


eg 20



⟶ No Rearrang.-

eg 21



⟶ No Rearr---

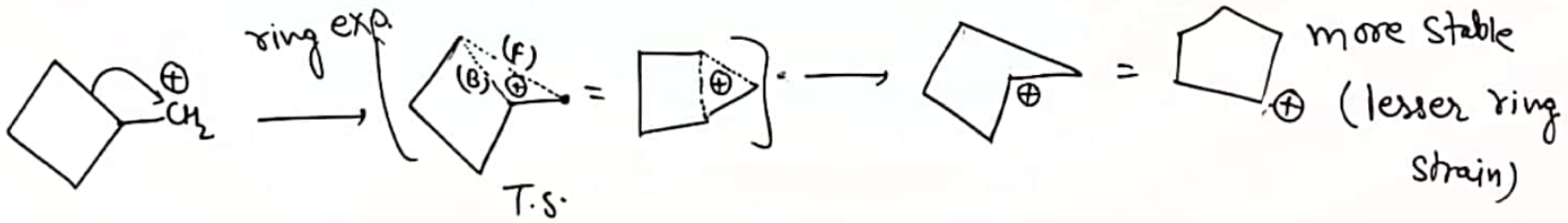
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

⇒ When +ve charge (in carbonium ion) is present on adjacent position of four or five membered ring then ring expansion possible.

⇒ When +ve charge is present on atom of four or seven membered ring then ring contraction possible.

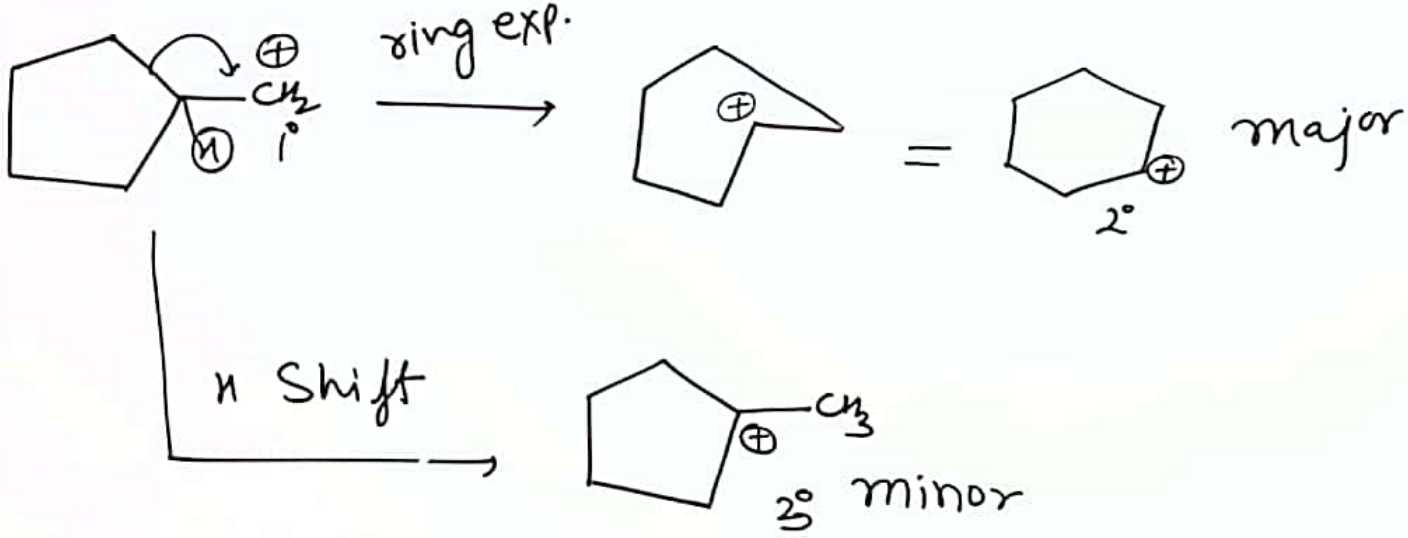
Eg 22



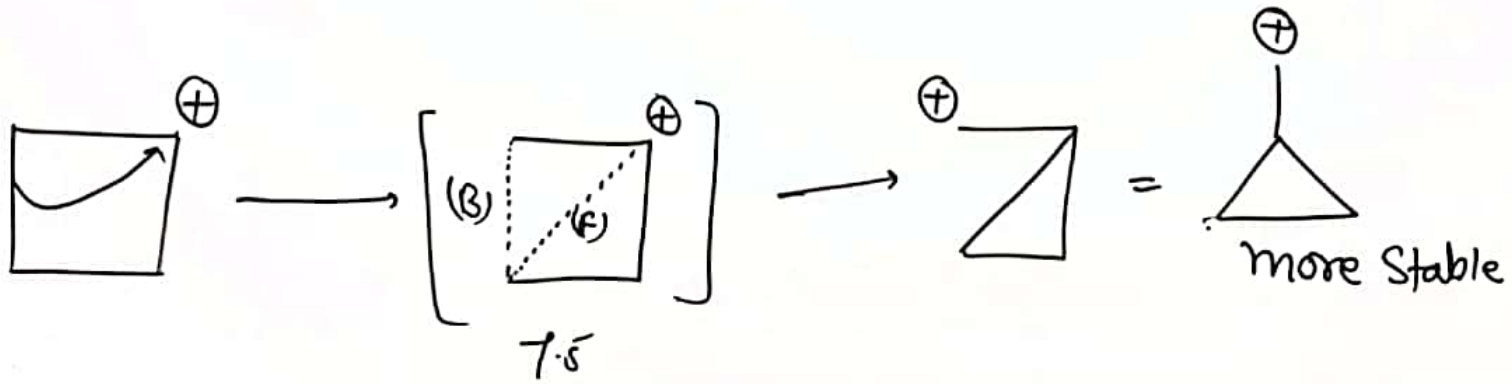
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

eg 23



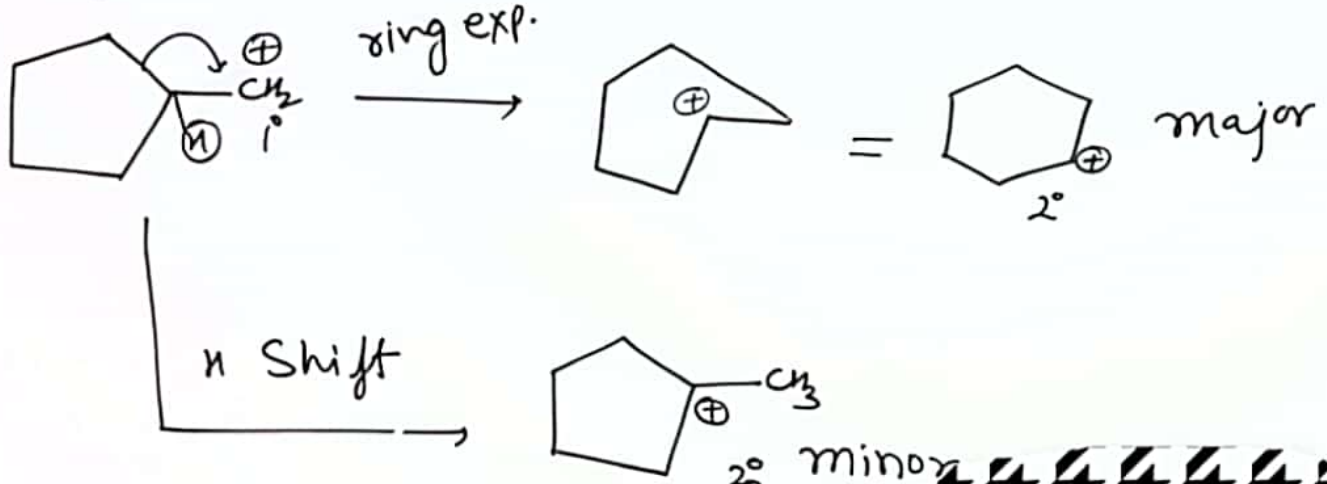
eg 24



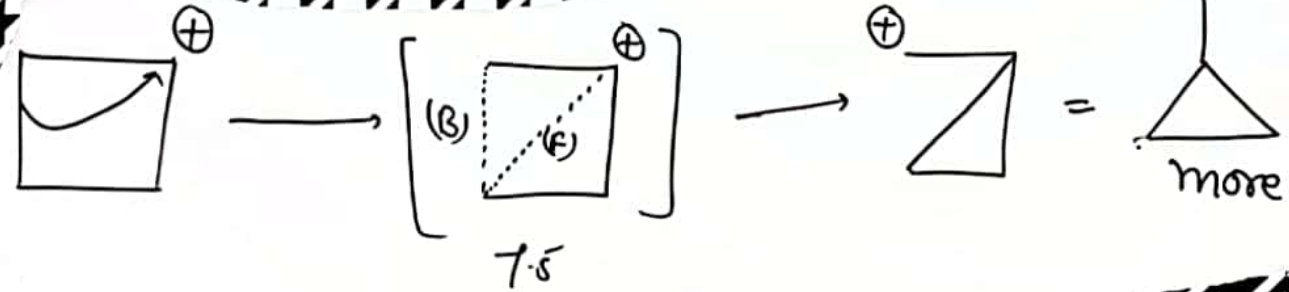
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

eg 23



eg 24

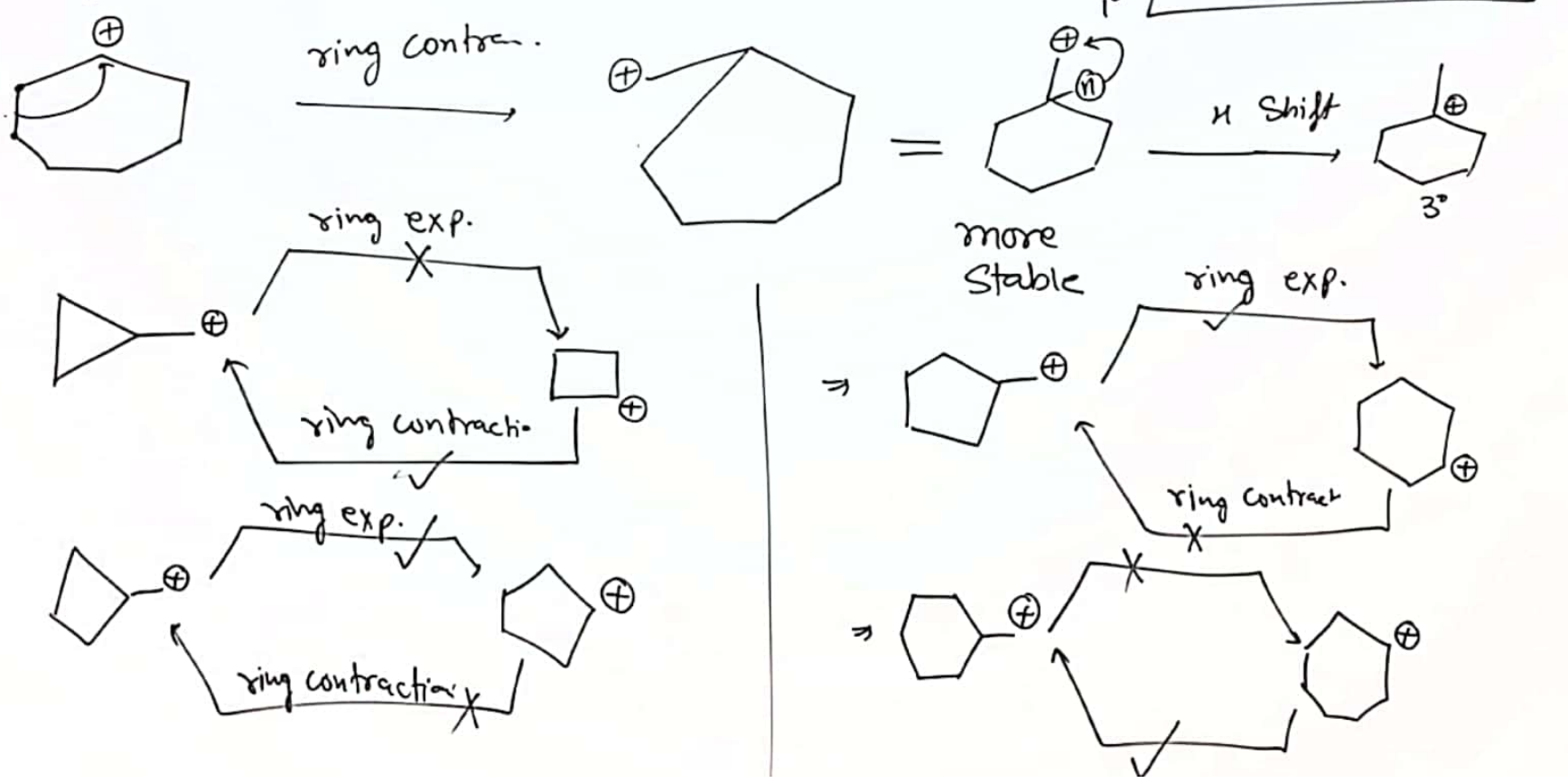


GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

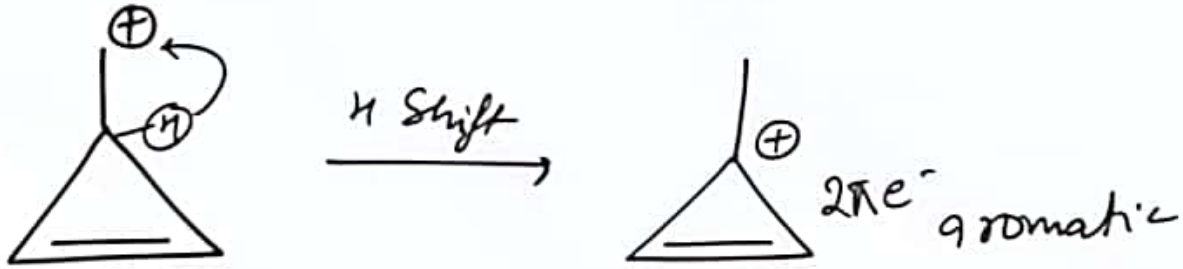
△ < □ < ⬠ < ⬡

eg 25 ≡

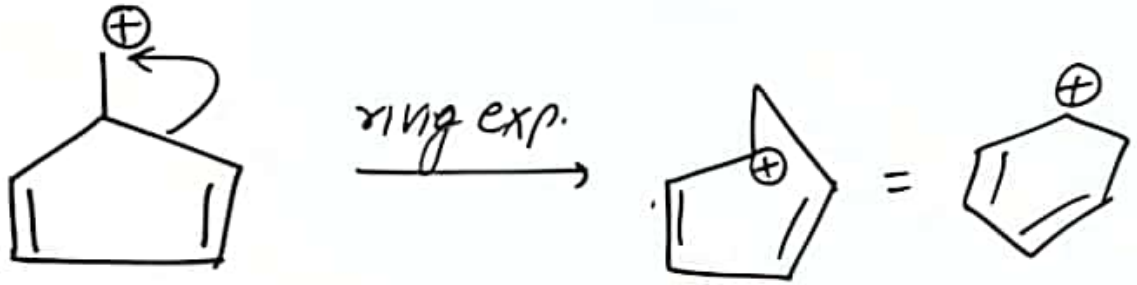


Rearrangement in carbonium ions

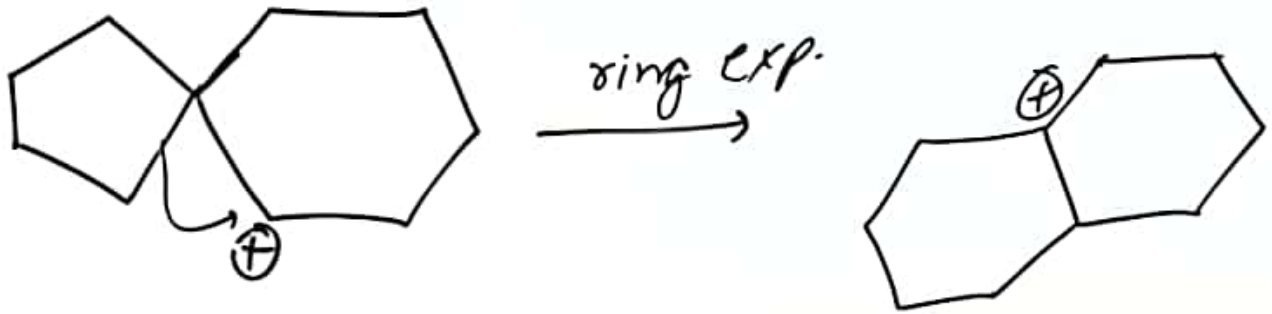
eg



eg

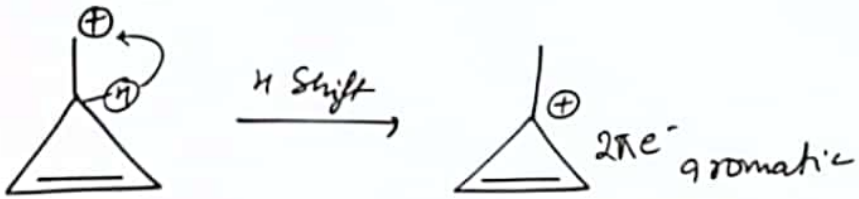


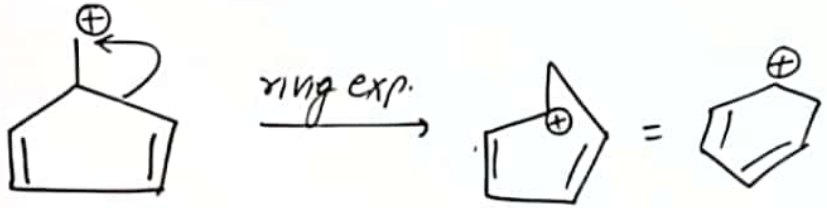
eg

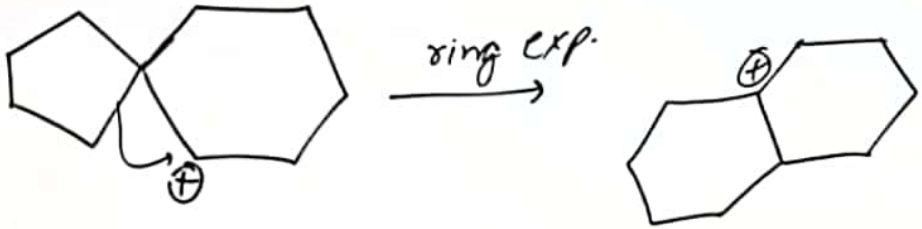


GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

eg 

eg 

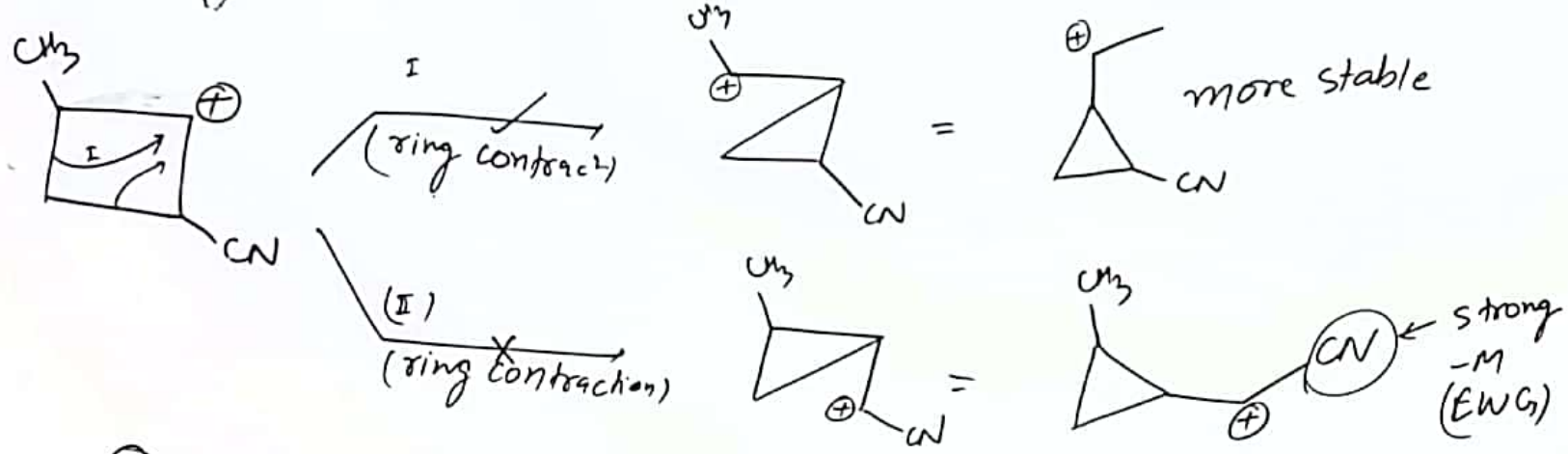
eg 

eg. 

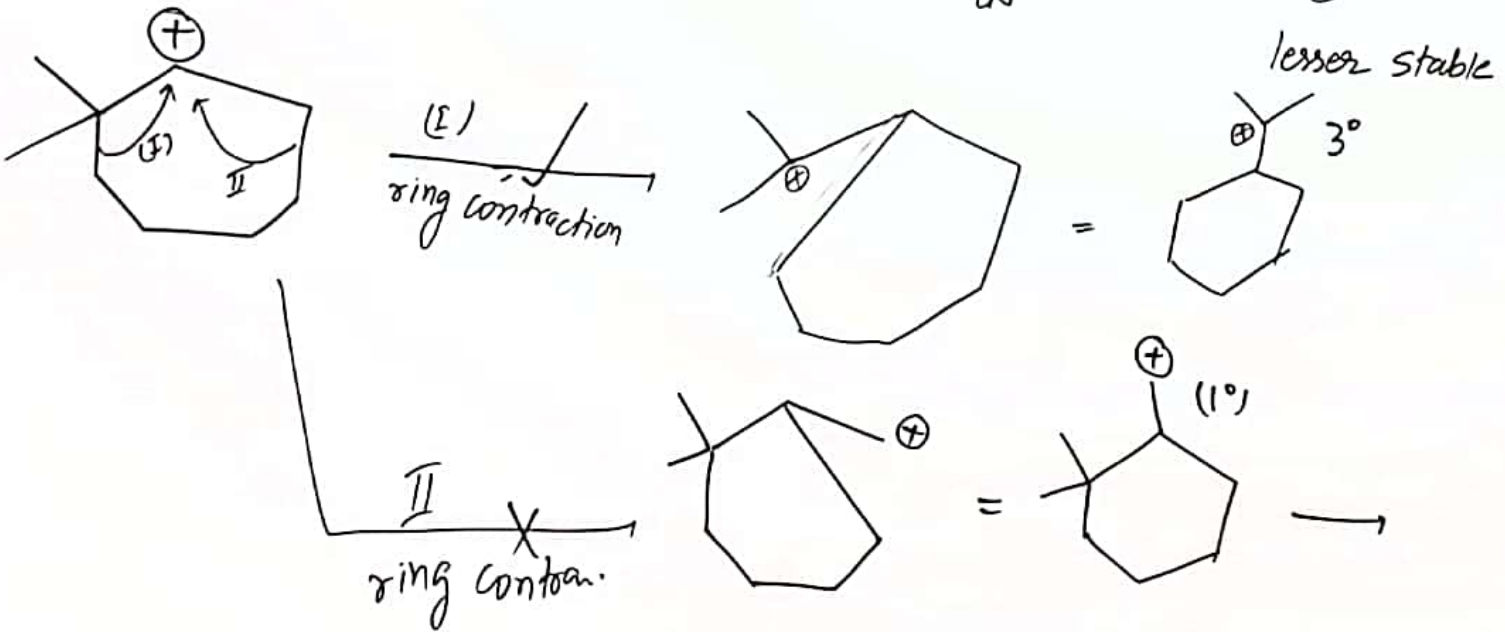
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

Rearrangement in carbonium ions

eg



eg



GENERAL ORGANIC CHEMISTRY – II (GOC-II)

(II) Acidic Strength :–

Arrhenius acid :– H^+ donor in aq. medium

eg HCl, H_2SO_4, HNO_3, R-COOH etc.

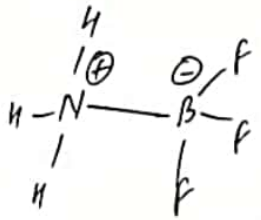
Brosted - lowry acid :– Proton donating species in (aq. or non aq. medium)

$$\underset{\text{Acid}}{HCl} + \underset{\text{Base}}{H_2\ddot{O}:} \rightleftharpoons \underset{\text{Base}}{\ddot{Cl}^{\ominus}} + \underset{\text{Acid}}{H_3O^{\oplus}}$$

$$\underset{\text{Acid}}{H_2O} + \underset{\text{Base}}{\ddot{N}H_2^{\ominus}} \rightleftharpoons \underset{\text{Base}}{OH^{\ominus}} + \underset{\text{Acid}}{NH_3}$$

Lewis acid :– lone pair accepting species

$$\underset{\text{Base}}{\ddot{N}H_3} + \underset{\text{Acid}}{BF_3} \longrightarrow$$



GENERAL ORGANIC CHEMISTRY -II GOC-II

(II) ACIDIC STRENGTH:-

Factors affecting acidic strength

$$A-H \underset{}{\overset{K_a}{\rightleftharpoons}} \underset{\text{conjugate base}}{A^{\ominus}} + H^{\oplus}$$

⇒ Acidic strength ↑es ------ K_a ↑es ----- pK_a ↓es

⇒ " " " " ↑es ---- Stability of conjugate anion A^- ↑es

⇒ " " " ↓es -- — Bond stength of A–H bond ↑es

⇒ " " " " ↑es ------ Oxi. no. of central atom of A ↑es

⇒ " " " ↑a in presence of EWG (–M, –I)

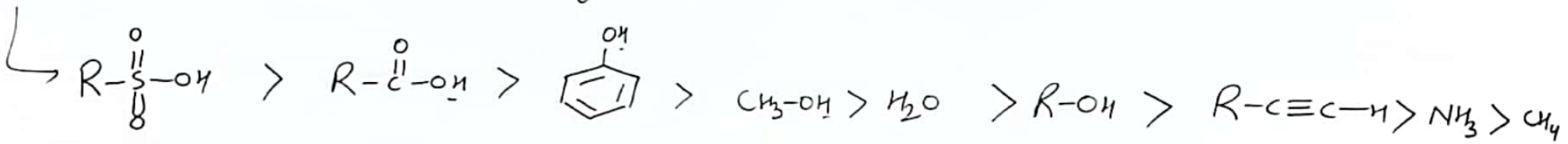
⇒ " " " ↓es " " " " EDG (+M, +HC, +I)

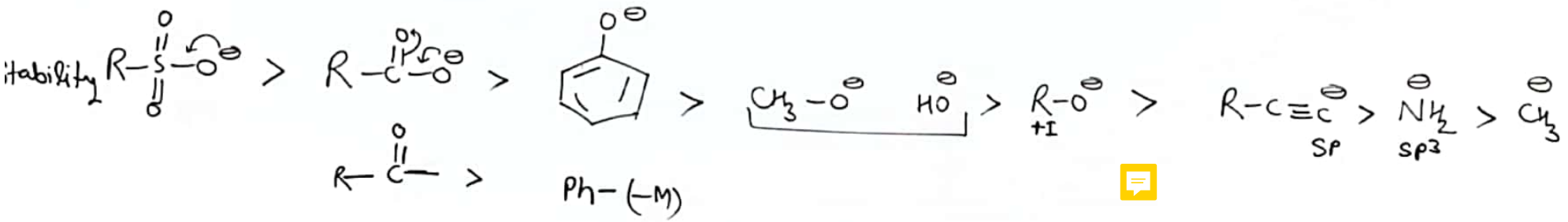
⇒ Acidic strength also depends upon nature of solvent

GENERAL ORGANIC CHEMISTRY -II GOC-II

(II) ACIDIC STRENGTH:-

General order of acidic strength of organic acids.







(II) ACIDIC STRENGTH:-

eg 1 $CH_3-H < NH_3 < H_2O < HF$

CH_3^- F^-

eg 2 $PH_3 < H_2S < HCl$

eg 3 $H-F < HCl < HBr < H-I$

Weak

$F^{\ominus}$ I^-

eg 4 $H_2O < H_2S$

eg 5 $Me-OH < Me-SH$

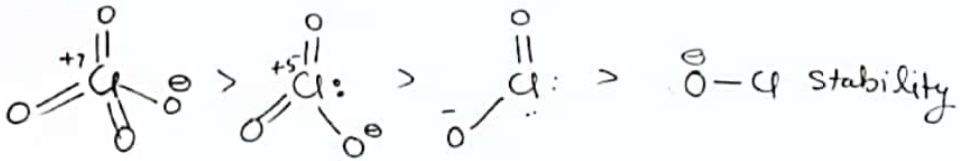
eg 6 $Ph-O^-_{2p} < Ph-S^-_{3p}$

GENERAL ORGANIC CHEMISTRY -II GOC-II

(II) ACIDIC STRENGTH:-

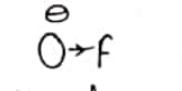
eg 10 $\overset{+7}{HClO_4} > \overset{+5}{HClO_3} > \overset{+3}{HClO_2} > \overset{+1}{HOCl}$

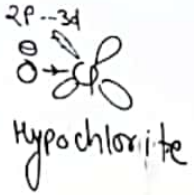
Perchloric acid | Chloric acid | Chlous acid | Hypochlorous acid



stability

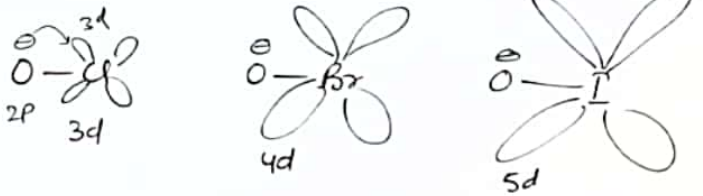
eg 11 H–O–F < H–O–Cl acid strength.


Hypofluorite



Hypochlorite

eg 12 H–O–Cl > H–O–Br > H–O–I

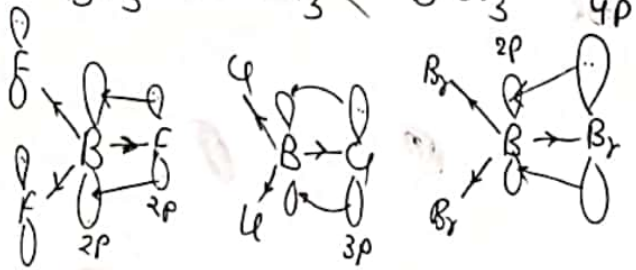


eg 13 $HClO_3 > HBrO_3 > HIO_3$

eg 14 $HClO_4 > HBrO_4 > HIO_4$

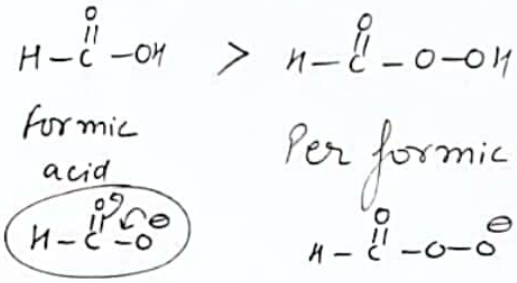


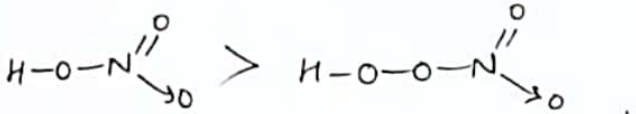
Imp eg 15 $BF_3 < BCl_3 < BBr_3$

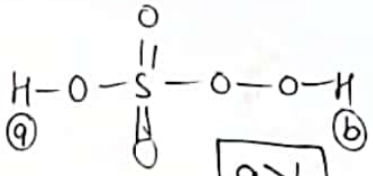


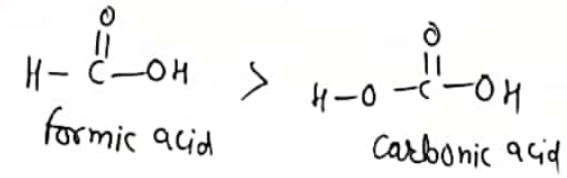
GENERAL ORGANIC CHEMISTRY -II GOC-II

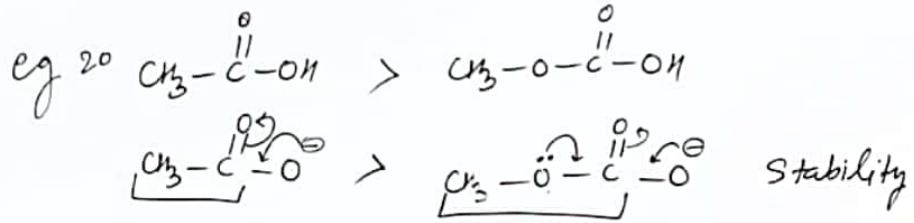
(II) ACIDIC STRENGTH:-

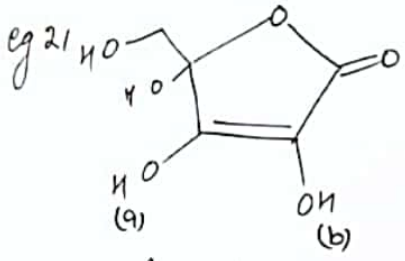
eg 16 

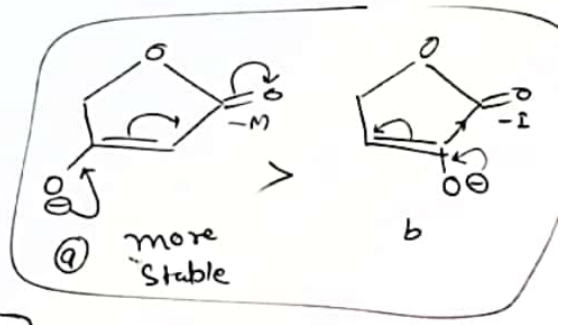
eg 17 

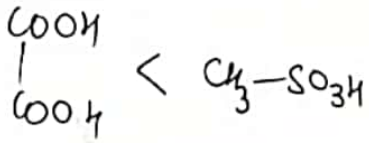
eg 18 

eg 19 

eg 20 

eg 21 

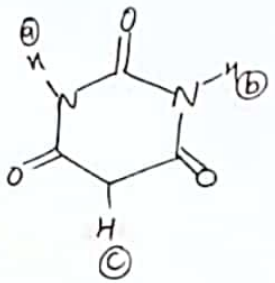


eg 2 

GENERAL ORGANIC CHEMISTRY -II GOC-II

(II) ACIDIC STRENGTH:-

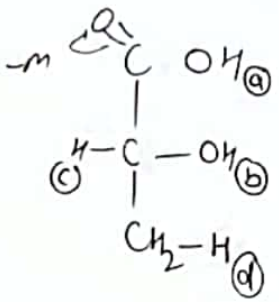
eg 23



Barbuturic acid

a=b>c

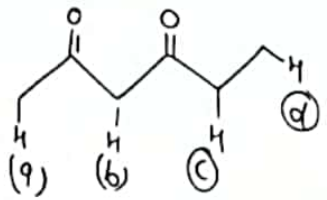
eg 24



Lactic acid

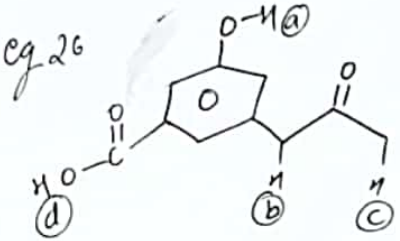
a>b>c>d

eg 25



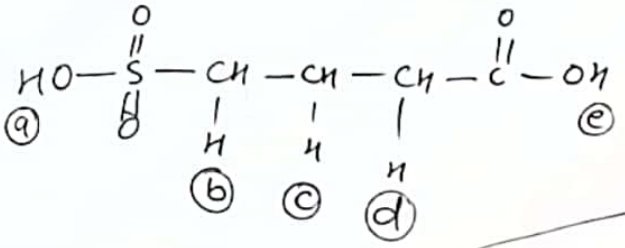
b>a>c>d

eg 26



d>a>b>c

eg 27

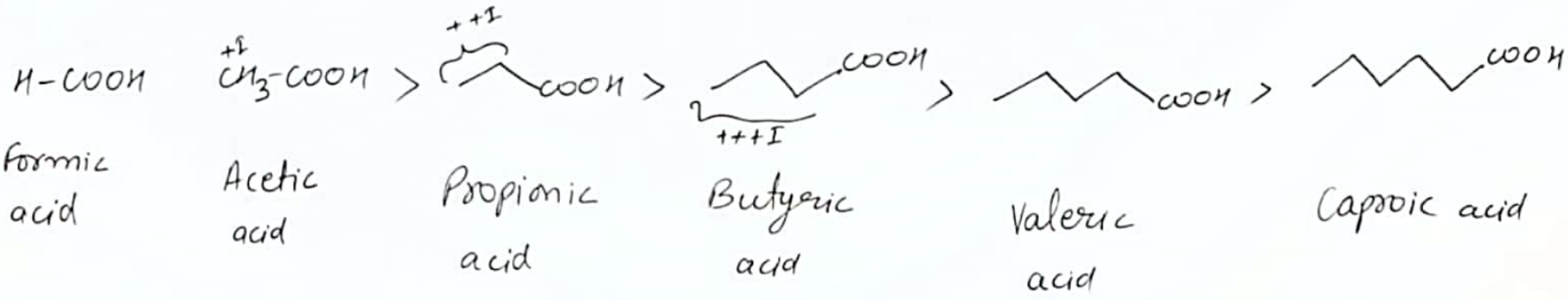


a>e>b>d>c

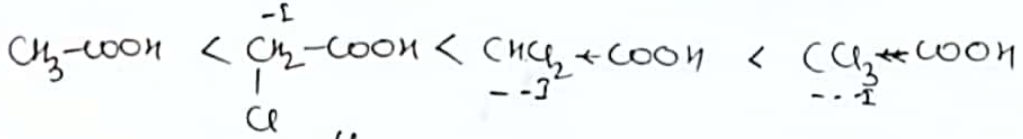
$-SO_3H > -\overset{O}{\overset{\|}{C}}-OH > -M/-I$

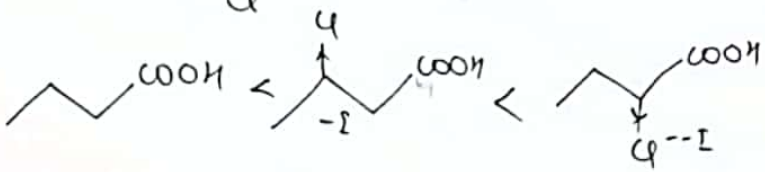
GENERAL ORGANIC CHEMISTRY -II GOC-II

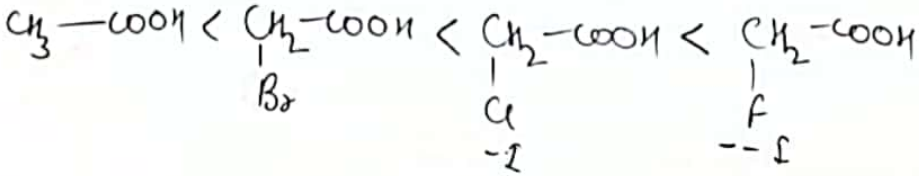
(II) ACIDIC STRENGTH:-

eg 28 

Formic acid — Acetic acid — Propionic acid — Butyric acid — Valeric acid — Caproic acid

eg 29 

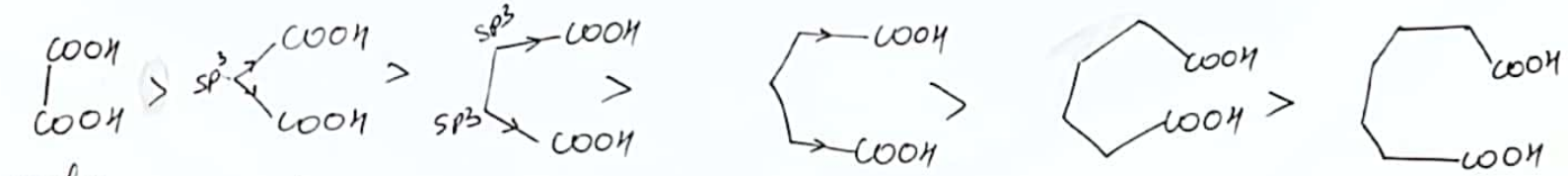
eg 30 

eg 31 

GENERAL ORGANIC CHEMISTRY -II GOC-II

(II) ACIDIC STRENGTH:-

eg 32



oxalic acid O

Malonic acid M

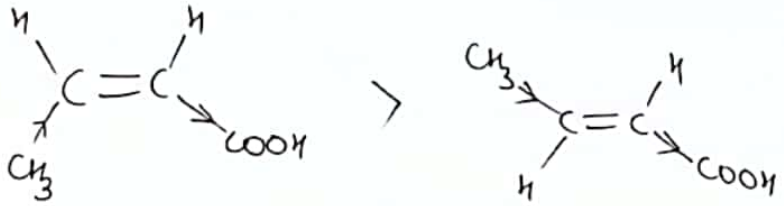
Succinic acia S

Glutaric acid G

Adipic acia A

Pimalic acid P

eg 33



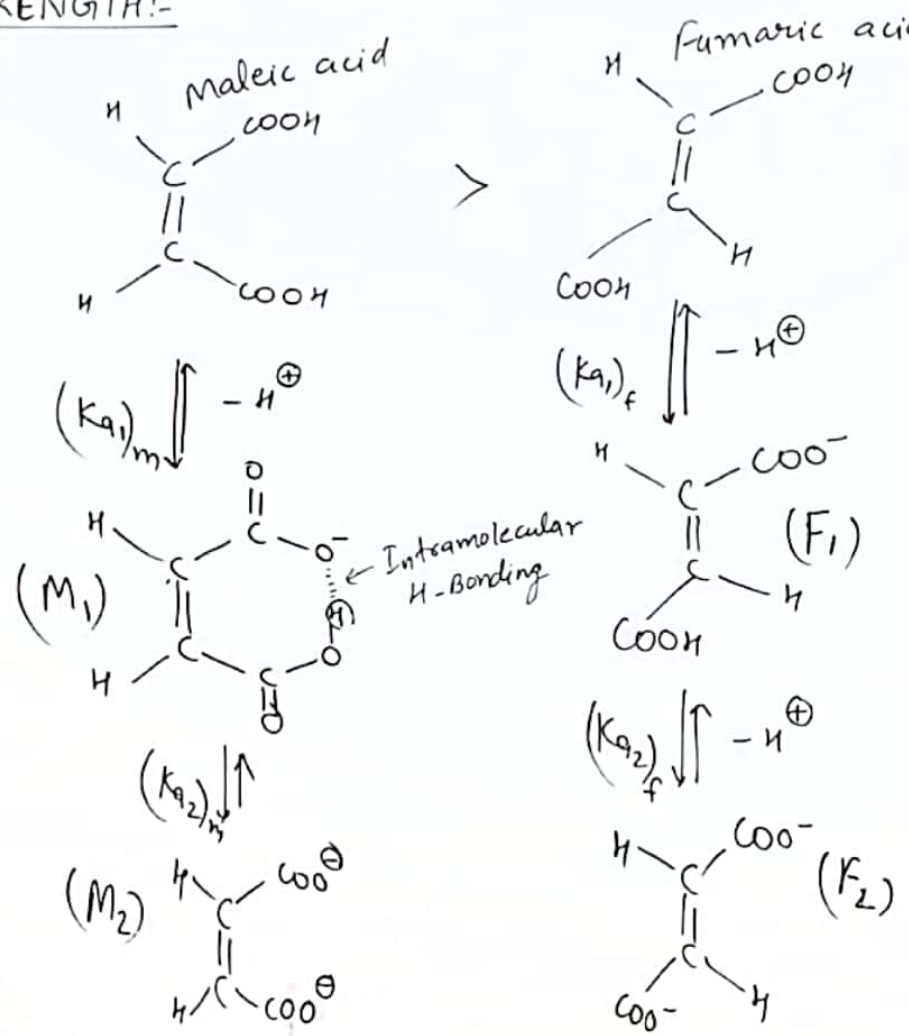
Cis

Trans

GENERAL ORGANIC CHEMISTRY -II GOC-II

(II) ACIDIC STRENGTH:-

Eg 34



$\Rightarrow (Ka_1)_m > (Ka_2)_m$

$\Rightarrow (Ka_1)_f > (Ka_2)_f$

$\Rightarrow (Ka_1)_m > (Ka_1)_f >> (Ka_2)_f > (Ka_2)_m$

$\Rightarrow$ Stability $= M_1 > F_1$

Stability $= M_2 < F_2$

GENERAL ORGANIC CHEMISTRY – II (GOC-II)

(II) Acidic Strength:-

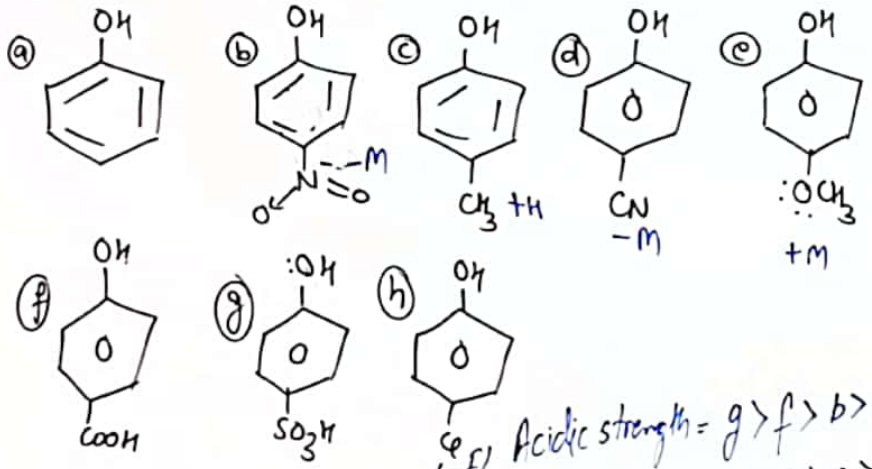
⇒ Acidic strength of phenol and derivatives of phenols

Phenols > Alcohols

$Ar-O^{\ominus}$ > $R-O^{\ominus}$

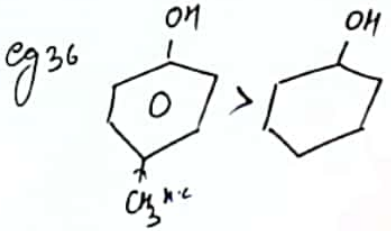
more Stable

Eg 35

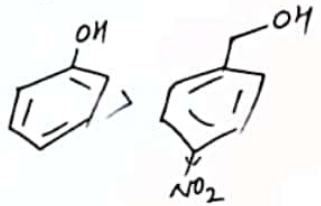


Acidic strength = g > f > b > d > h > a > c > e

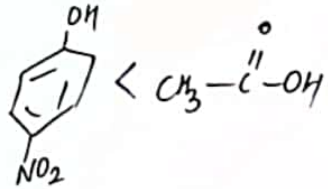
Eg 36



Eg 37



Eg 38

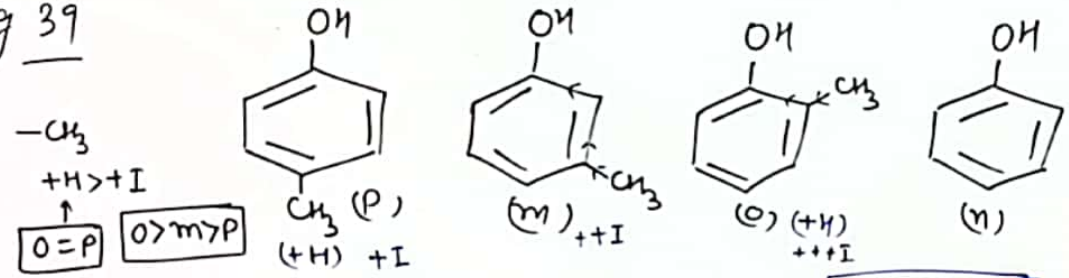


GENERAL ORGANIC CHEMISTRY – II (GOC-II)

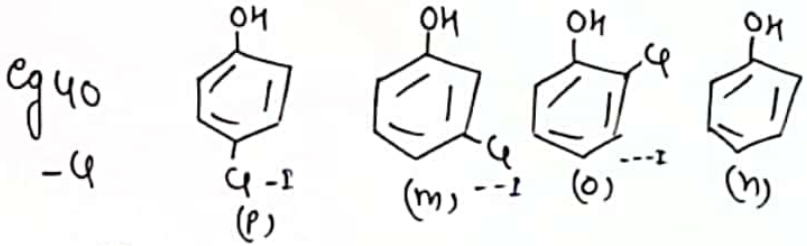
(II) Acidic Strength:-

⇒ Acidic strength of phenol and derivatives of phenols

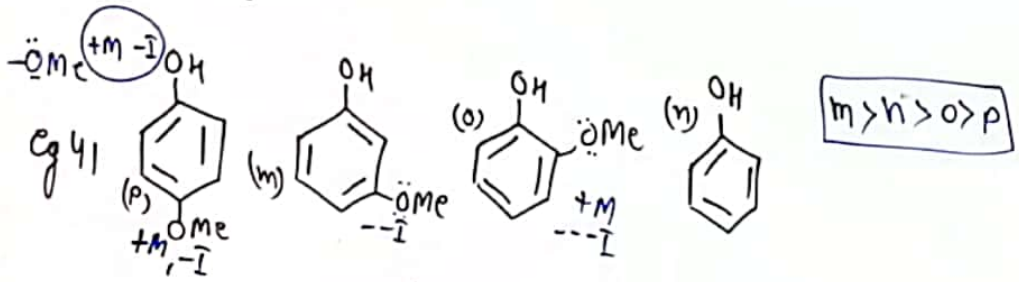
eg 39



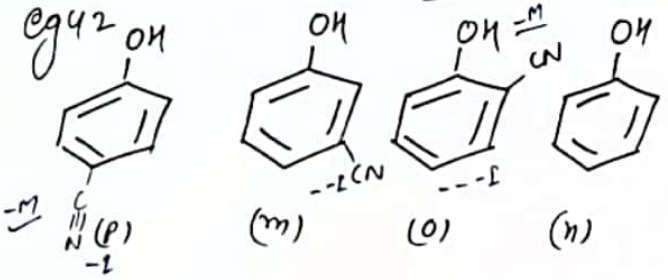
eg 40



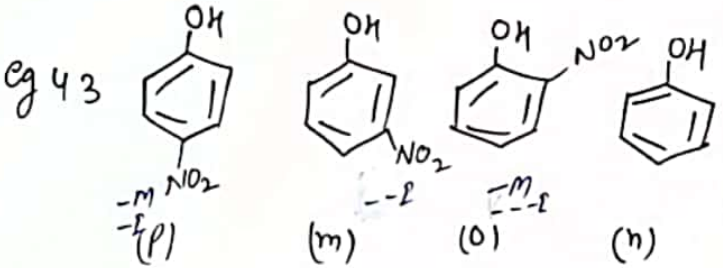
eg 41



eg 42



eg 43



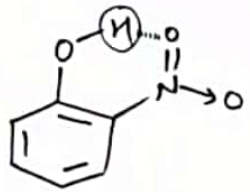
O > P > m > n ✗

P > O > m > n ✓

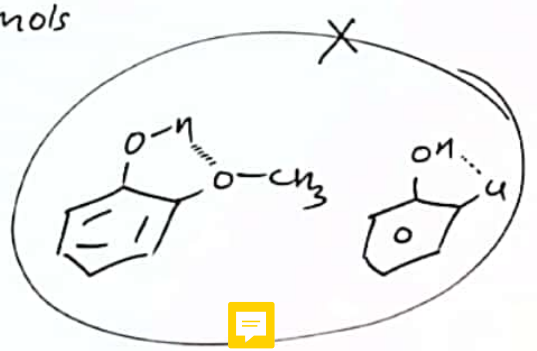
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

(II) Acidic Strength:-

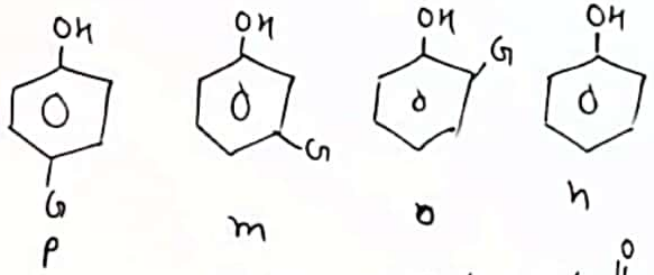
⇒ Acidic strength of phenol and derivatives of phenols



due to intramolecular H. bonding acidic strength of o. nitro phenole is lesser than p-nitro phenol.



Imp.

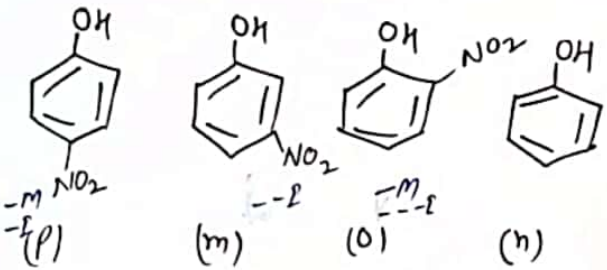


if G = $-NO_2$ / $-CHO$ / $-\overset{O}{\overset{\|}{C}}-R$ / $-\overset{O}{\overset{\|}{C}}-OR$ ⇒ p > o > m > n

intramolecular H bonding

$-NO_2$: $-M > -I$

Eg 43



(p) (m) (o) (n)

o > p > m > n ✗

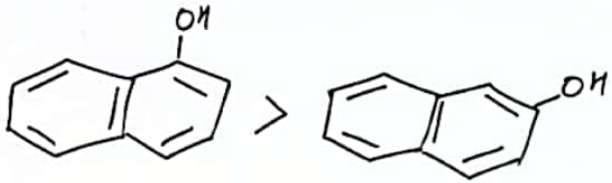
p > o > m > n ✓

GENERAL ORGANIC CHEMISTRY – II (GOC-II)

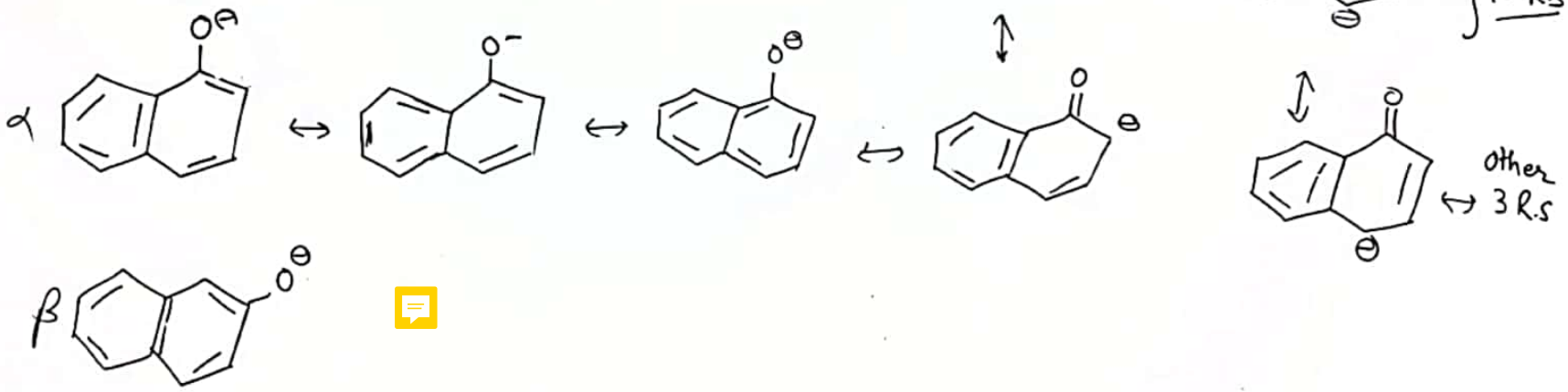
(II) Acidic Strength:-

⇒ Acidic strength of phenol and derivatives of phenols

eg 44



α-Naphthol > β-Naphthol



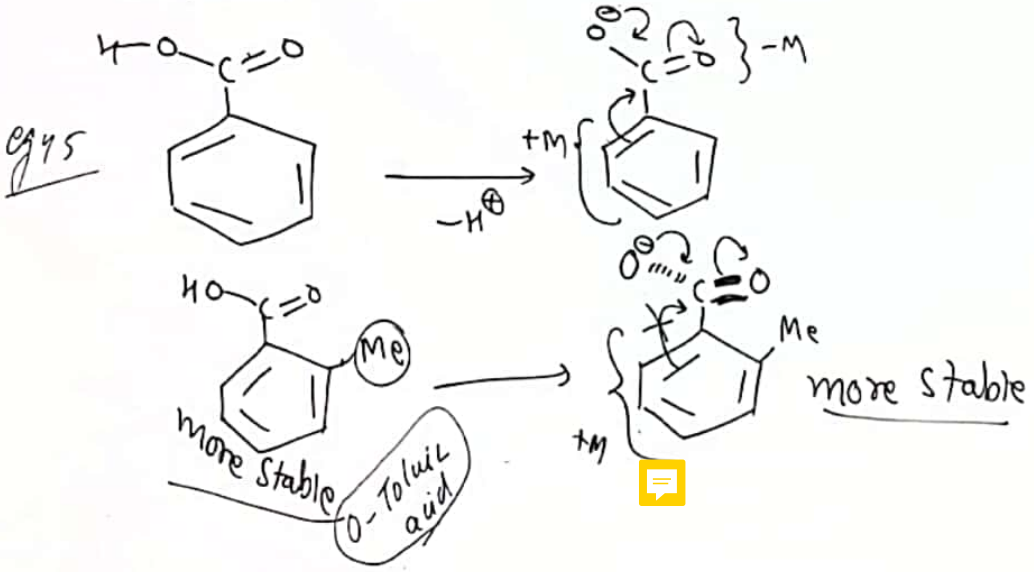
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

(II) Acidic Strength:-

Acidic strength of benzoic acid and derivative ...

Note = Ortho effect / Ortho Rule :-

Ortho substituted benzoic acid is most acidic than benzoic acid and meta, Para isomers. (It is due to SIR effect)



GENERAL ORGANIC CHEMISTRY – II (GOC-II)

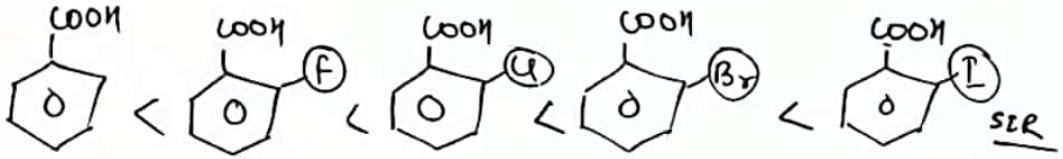
(II) Acidic Strength:-

Acidic strength of benzoic acid and derivative ...

eg 46

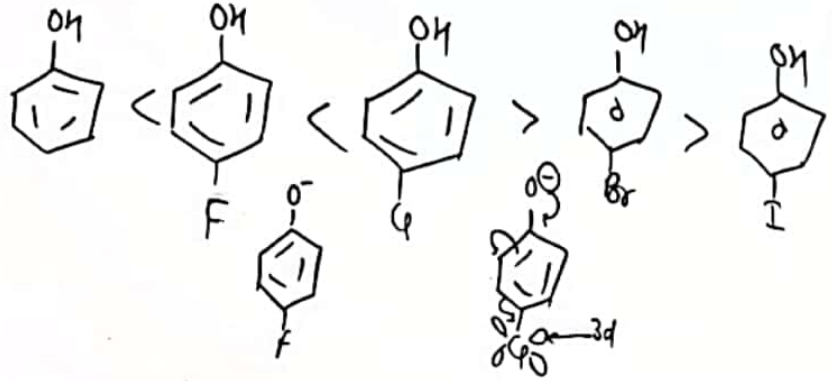


eg 47



SLR

eg 48



GENERAL ORGANIC CHEMISTRY – II (GOC-II)

(II) Acidic Strength :–

$$HCl \xrightarrow{aq.} H^{\oplus} + Cl^{\ominus}$$

$$\overset{\delta-}{A} - \overset{\delta+}{H} \rightleftharpoons A^{\ominus} + H^{\oplus}$$

weak acid — conjugate base

$$K_a = \frac{[A^-]\,[H^+]}{[A-H]}$$

$$pH = -\log [H^+]$$

$$pOH = -\log [OH^-]$$

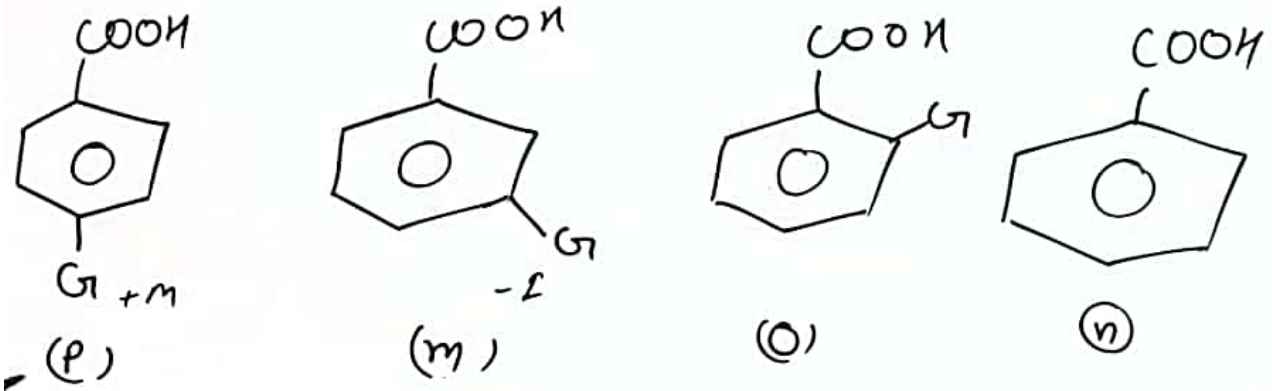
$$pK_a = -\log K_a$$

$$pK_b = -\log K_b$$

$$pK_a + pK_b = pK_w = 14 \text{ at } 25^\circ C$$

(for conjugate acid-base)

strength of benzoic acid and derivative ...



if $G = -CH_3$ $(+H > +I)$ $\Rightarrow$ o > n > m > p

$G = -Cl$ $(-I)$ $\Rightarrow$ o > m > p > n

$G = -CN$ $(-M, -I)$ $\Rightarrow$ o > p > m > n

$G = -NO_2$ $(-M\ -I)$ $\Rightarrow$ o > p > m > n

$G = -\ddot{O}CH_3$ $(+M\ -I)$ $\Rightarrow$ o > m > n > p

* $G = -\overset{+M,-I}{\ddot{O}}H$ $\Rightarrow$ o > m > n > p

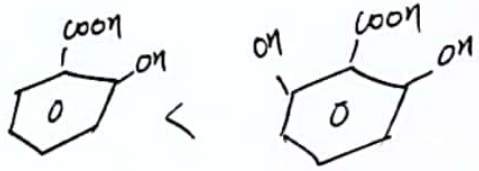
NO ortho effect

$\Rightarrow$ Salicylic acid is most acid than benzoic acid and other isomer due to intramolecular H bonding.

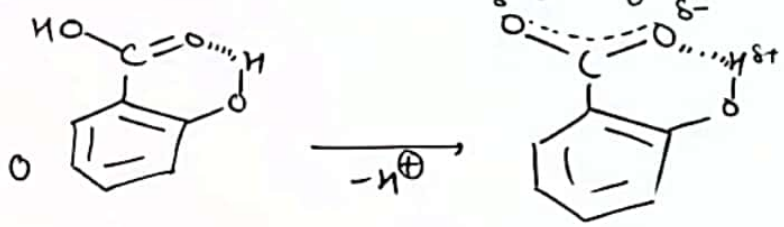
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

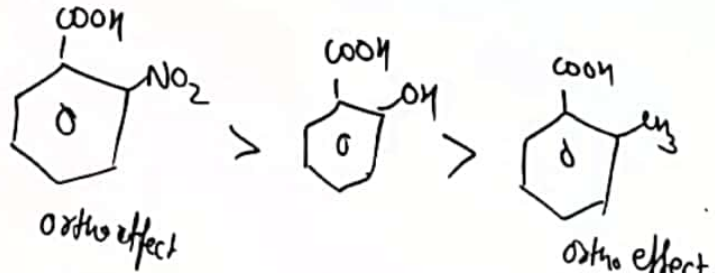
(II) Acidic Strength:-

Acidic strength of benzoic acid and derivative ...

47 

* Salicylate ion is most stable than other due to intramolecular hydrogen bonding



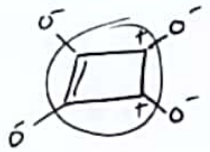
Eg 46 

ortho effect

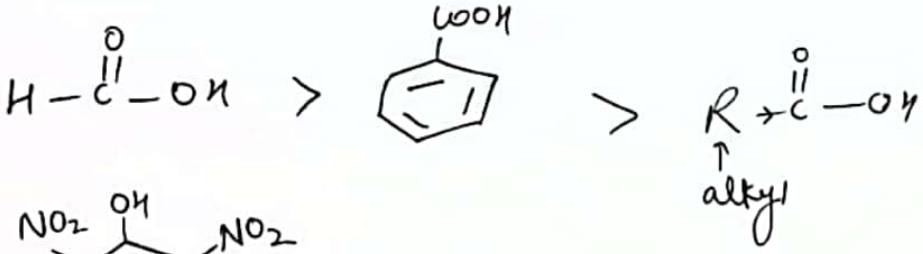
ortho effect

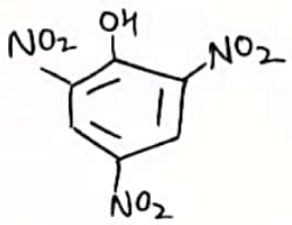
GENERAL ORGANIC CHEMISTRY – II (GOC-II)

(II) Acidic Strength:-



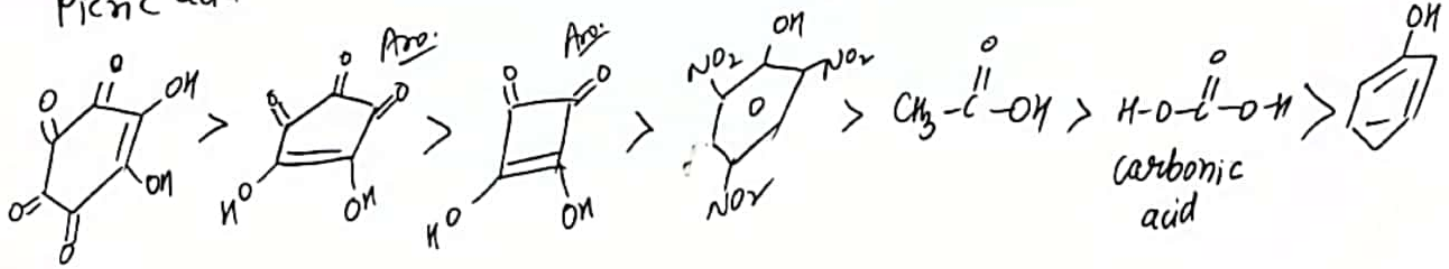
Some important orders of acidic strength

① 

② 

Picric acid

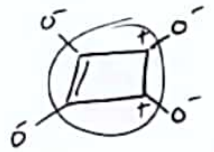


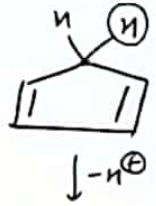
③ 

GENERAL ORGANIC CHEMISTRY – II (GOC-II)

(II) Acidic Strength:-

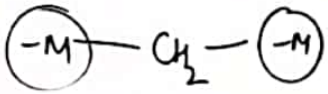
Some important orders of acidic strength



(4)  > H_2O

$6\pi e$

aro.

(5)  > H_2O

Active methylene group.

eg 6 

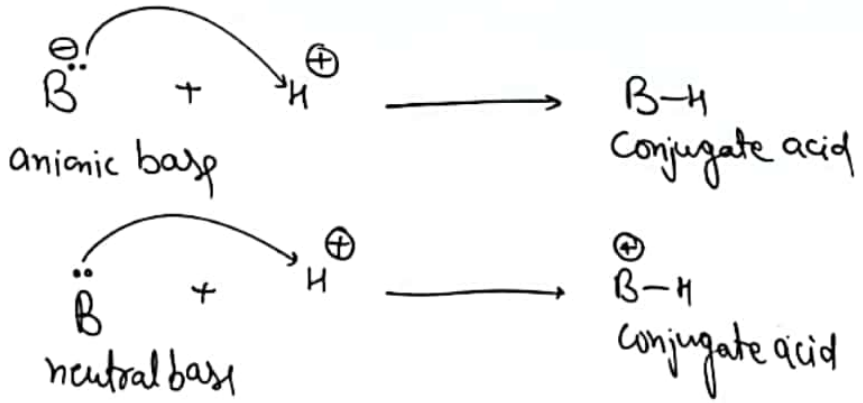
GENERAL ORGANIC CHEMISTRY -II GOC-II

(III) Basic Strength

(i) Arrhenius base:- $\overset{\ominus}{O}H$ donor in aq. medium.

(ii) Bronsted-lowary base:- Proton accepting species.

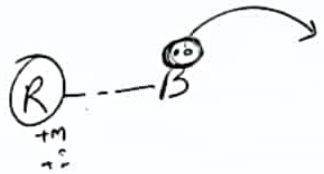
(iii) Lewis base:- lone pair donating species.



GENERAL ORGANIC CHEMISTRY -II GOC-II

(III) Basic Strength

Factors affecting basic strength.



(i) anion base > neutral base

(ii) For anionic bases

$$\text{basic strength} \propto \frac{1}{\text{Stability of anionic base}}$$

(iii) Basic Strength $\propto$ Stability conjugate acid

(iv) Basic Strength $\propto$ lone pair donating nature to proton ($H^{\oplus}$)
or
e^- availability on e^- donor atom.

(v) Basic strength ↑es in presence of EDG (+M, +H, +I) on lone pair donating atom.

(vi) Basic strength ↓es in presence of EWG (-M, -I) on lone pair donar atom

(vii) Other factors ⇒ H. Bonding, SIR, Steric hindrance, Solvation etc.

GENERAL ORGANIC CHEMISTRY -II GOC-II

(III) Basic Strength

eg 1 $\overset{\ominus}{N}H_2 > \ddot{N}H_3$

amide ion

eg 2 $:\overset{\ominus}{O}H > H_2\ddot{O}$

eg 3 $\overset{\ominus}{C}H_3 > \overset{\ominus}{N}H_2 > \overset{\ominus}{O}H > \overset{\ominus}{F}$ Basic Strength

$CH_4 < NH_3 < H_2O < HF$ acidic strength

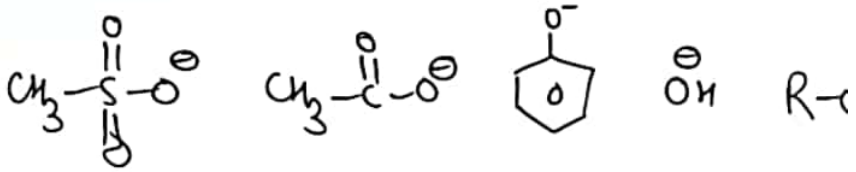
$Na\overset{\oplus}{C}H_3\overset{\ominus}{} > Na\overset{+}{N}H_2\overset{-}{} > Na\overset{+}{O}H\overset{-}{}$ Basic strength

eg 4

$R-C\equiv\overset{\ominus}{C}$ $R-CH=\overset{\ominus}{C}H$ $R-CH_2-\overset{\ominus}{C}H_2$

eg 5 $NaCH_3$ $LiCH_3$

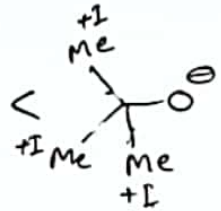
eg 6 NH_3 H_2O

eg 7 

eg 8 

GENERAL ORGANIC CHEMISTRY -II GOC-II

(III) Basic Strength

eg 9 $CH_3-O^{\ominus}$ <  <  Basic

eg 10 $\underset{+I}{Me}-\overset{\ominus}{N}H > \overset{\ominus}{N}H_2 > Ph-\overset{\ominus}{N}H > R-\underset{-M}{\underset{\|}{C}}(=O)-\overset{\ominus}{N}H$ Basic

eg 11 $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$

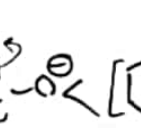
eg 12 $HO^{\ominus} > HS^{\ominus}$

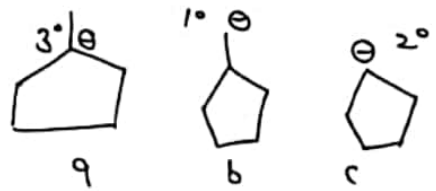
eg 13 $CH_3-O^{\ominus} > CH_3-S^{\ominus}$

eg 4 $R-\underset{SP}{C}\equiv\overset{\ominus}{C}$ (50%) < $R-\underset{SP^2}{CH}=\overset{\ominus}{C}H$ (33.3% ⊖) < $R-CH_2-\underset{SP^3}{CH_2}^{\ominus}$ (25% ⊖) Basic strength

eg 5 $\overset{\delta+}{Na}\,\overset{\ominus}{C}H_3$ (ionic) > $\overset{\delta+}{Li}\,\overset{\delta-}{C}H_3$

eg 6 $\ddot{N}H_3 > H_2\ddot{O}:$

eg 7 $CH_3-\underset{\|}{\overset{O}{\overset{\|}{S}}}(=O)-O^{\ominus}$ < $CH_3-\underset{-M}{\underset{\|}{C}}(=O)-O^{\ominus}$ <  < $OH^{\ominus}$ < $R-O^{\ominus}$ Basic

eg 8 

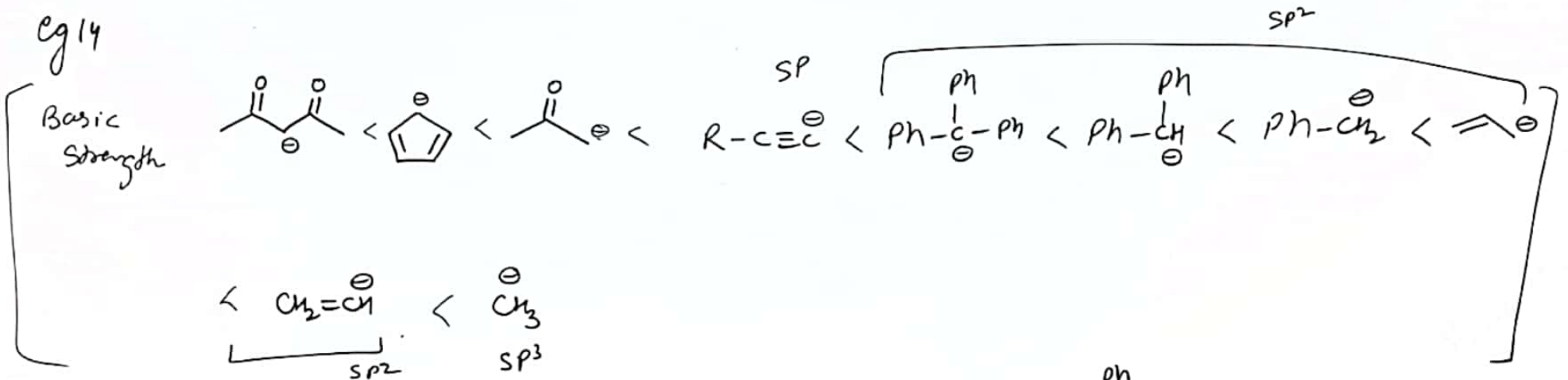
Basic → a > c > b

GENERAL ORGANIC CHEMISTRY -II GOC-II

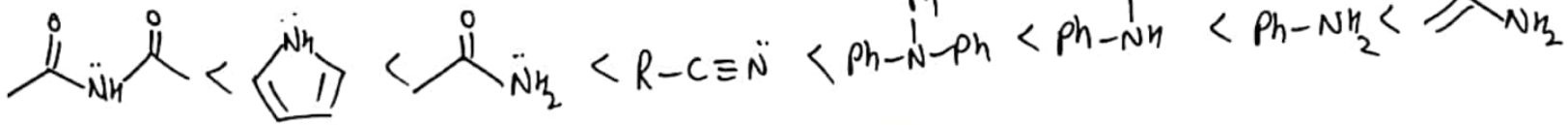
(III) Basic Strength

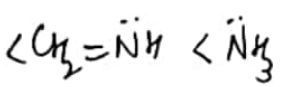
Eg 14

Basic Strength



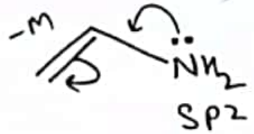
Eg 15

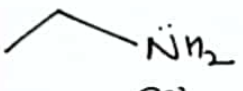


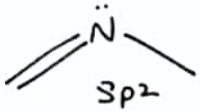


(III) Basic Strength

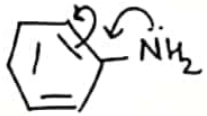
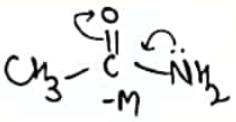
eg 16 $CH_3-CH_2-\ddot{N}H_2$ (sp^3) > $CH_3-CH=\ddot{N}H$ (sp^2, Imine) > $CH_3-C\equiv\ddot{N}$ (sp)

eg 17 





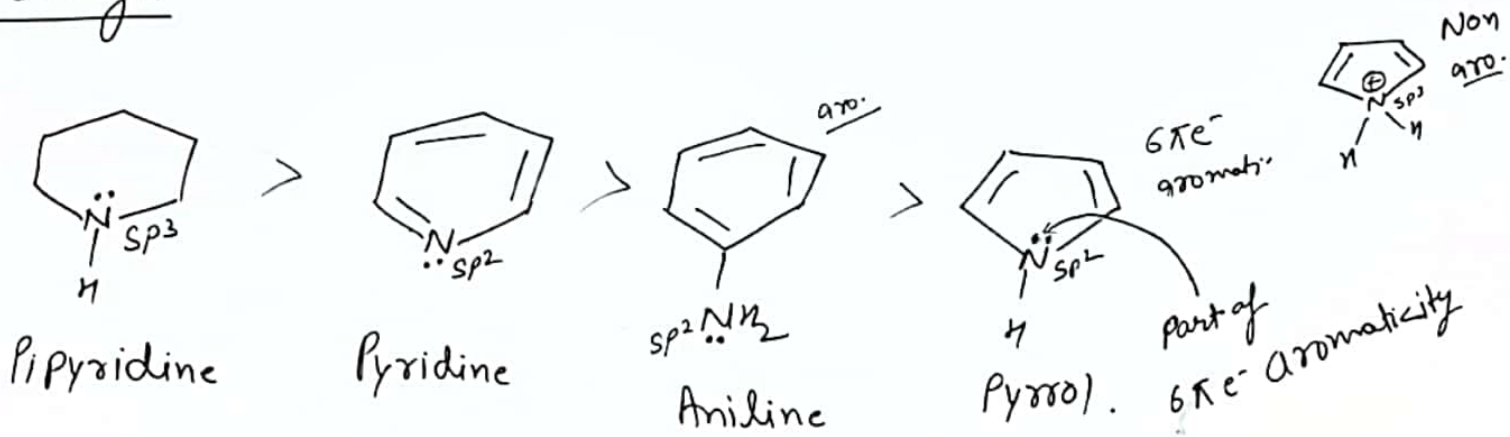
b > c > a

eg 18 $CH_3-\ddot{N}H_2$ >  > 

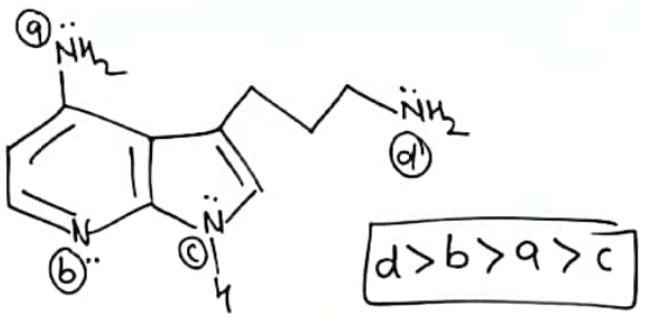
Ali amine | aro. amine | amide

(III) Basic Strength

eg 19



eg 20

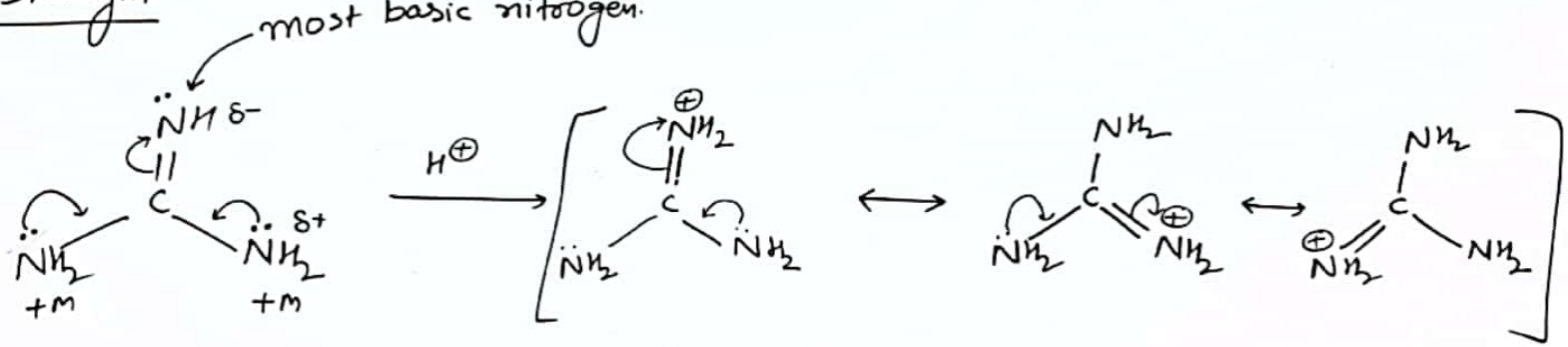


d > b > a > c

(III) Basic Strength

eg 21

Imp



Guanidine

(highly basic)

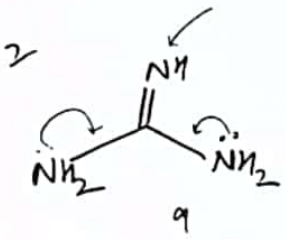
Conjugate acid is highly stable due to 3 equal resonating structures.

⇒ due +M effect of two $-NH_2$ group.

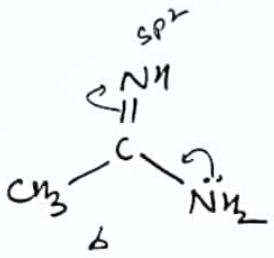
⇒ Conjugate acid is highly stable.

(III) Basic Strength

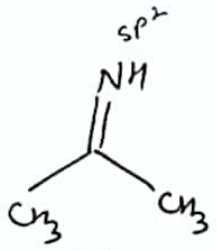
Eg 22



a



b



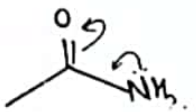
c



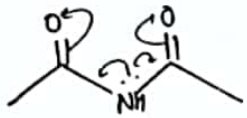
d

a > b > d > c

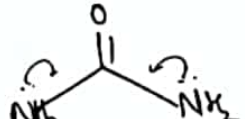
Eg 23



a



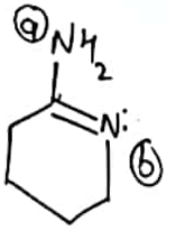
b

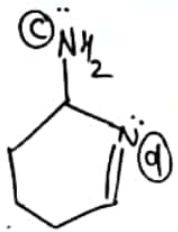


c Ureq

c > a > b

Eg 24





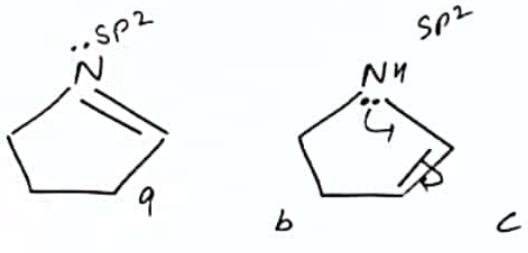
GENERAL ORGANIC CHEMISTRY -II GOC-II

(III) Basic Strength



Imidine

eg 25



c > a > b

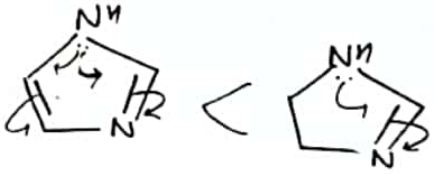
eg 25 Hw

Cl – NH_2 — Chloramine

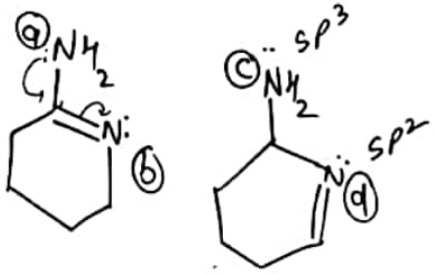
HO – NH_2 — Hydroxyl amine

NH_2 – NH_2 — Hydrazine

eg 26



eg 24



b > c > d > a

GENERAL ORGANIC CHEMISTRY -II GOC-II

(III) Basic Strength

Aliphatic amines

eg 2s

	1°	2°	3°
NH_3	$R-NH_2$	$R-NH-R$	R_3N

* [Basic strength. in polar solvent]

$R = Me \Rightarrow 2° > 1° > 3° > NH_3$

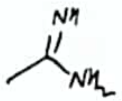
$R = Et \Rightarrow 2° > 3° > 1° > NH_3$
(in maximum cases)

* [Basic strength in gas phase & in non polar solvent]

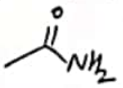
$3° > 2° > 1° > NH_3$

General order of basic strength for N. ... base.

imidine > Aliphatic amine > aromatic amine > amide



$Ph-NH_2$



GENERAL ORGANIC CHEMISTRY -II GOC-II

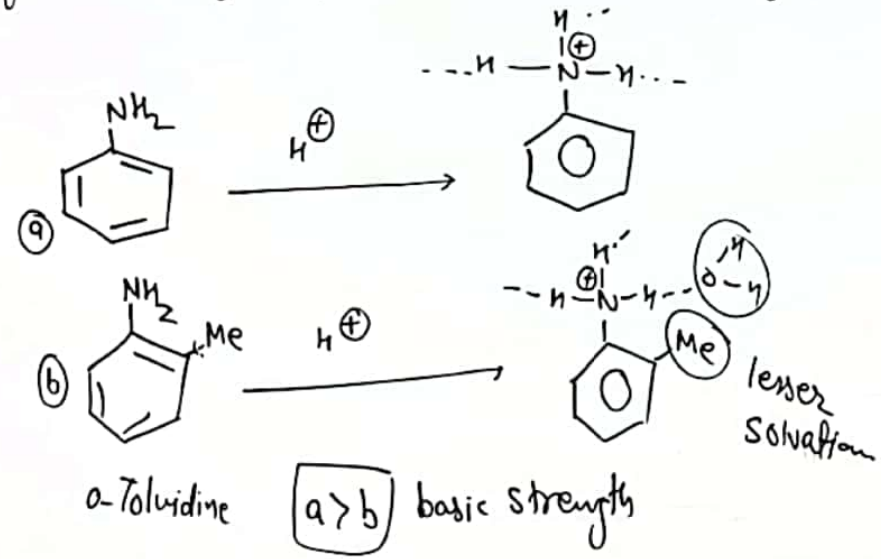
(III) Basic Strength

Basic strength of aromatic amine.

$R-NH_2 > Ar-\ddot{N}H_2$

⇒ Ortho effect:- When a substituent is present on ortho position of $-NH_2$ group, then basic strength ↓ due to steric hindrance.

Eg 29 —



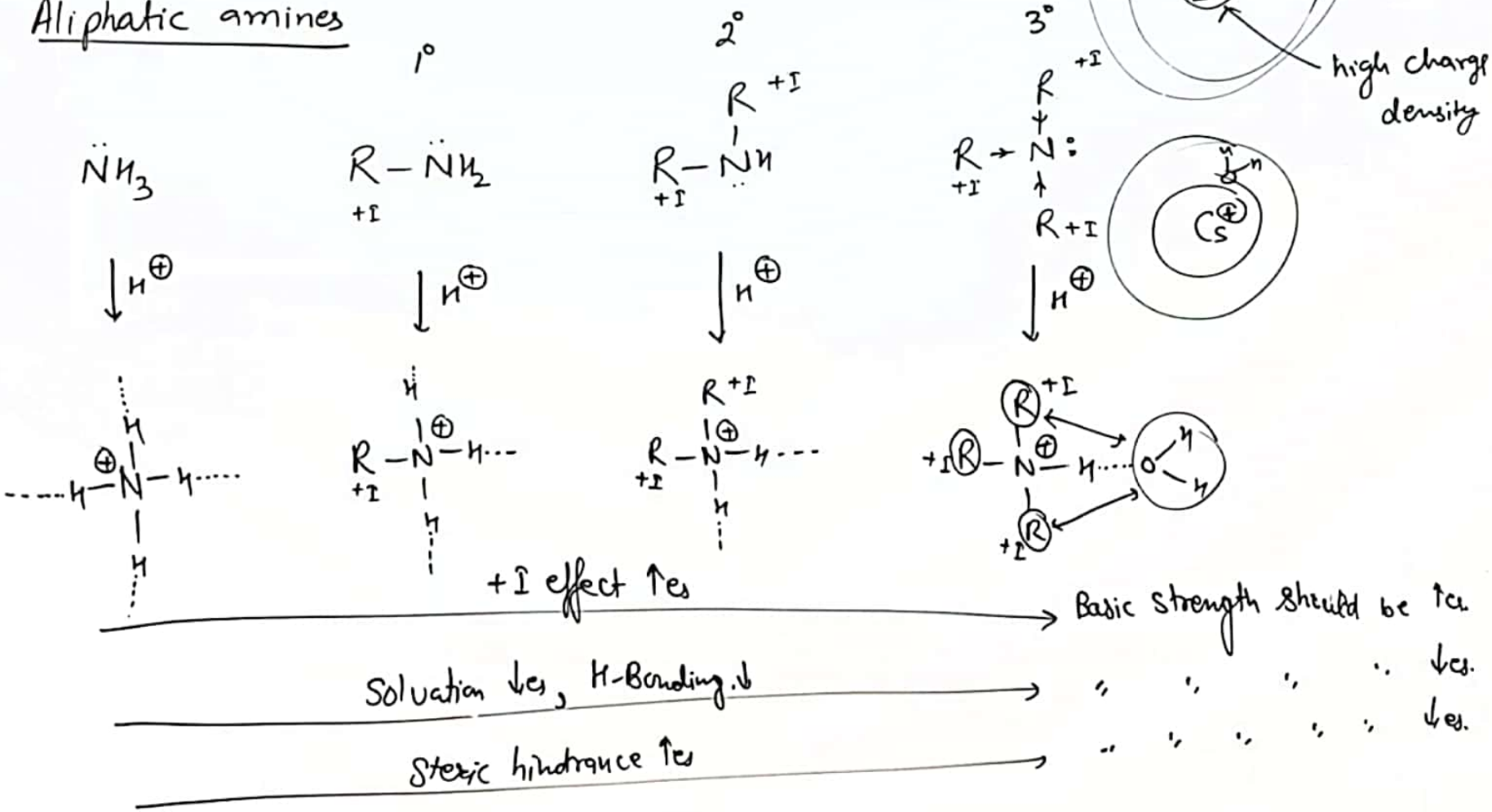
a) b)

o-Toluidine a > b basic strength

(III) Basic Strength

Aliphatic amines

Eg 2s



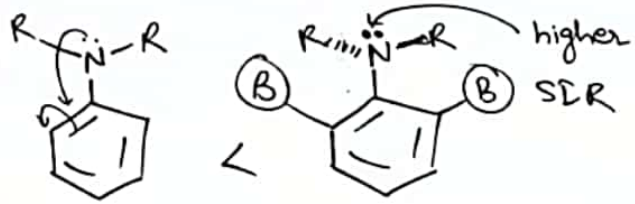
GENERAL ORGANIC CHEMISTRY -II GOC-II

(III) Basic Strength

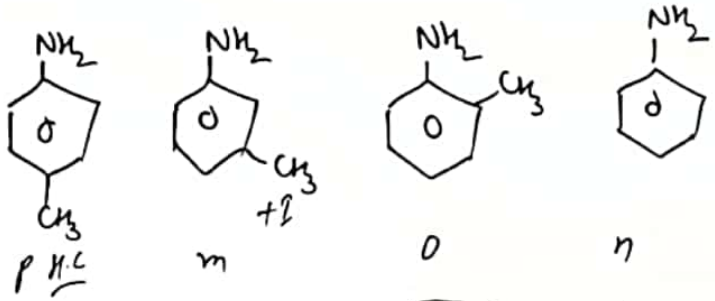
Basic strength of aromatic amine.

→ **SIR effect:-** When a substituent is present on ortho position of $-NR_2$ then basic strength is due to SIR effect.

eg 30

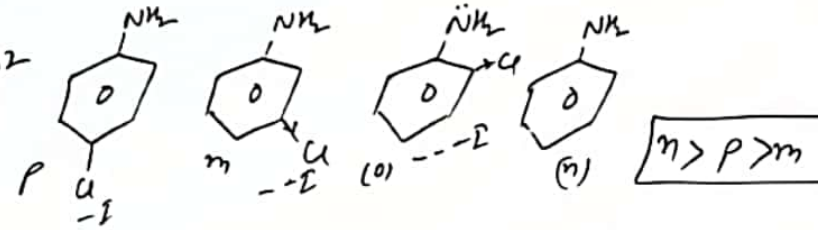


eg 31



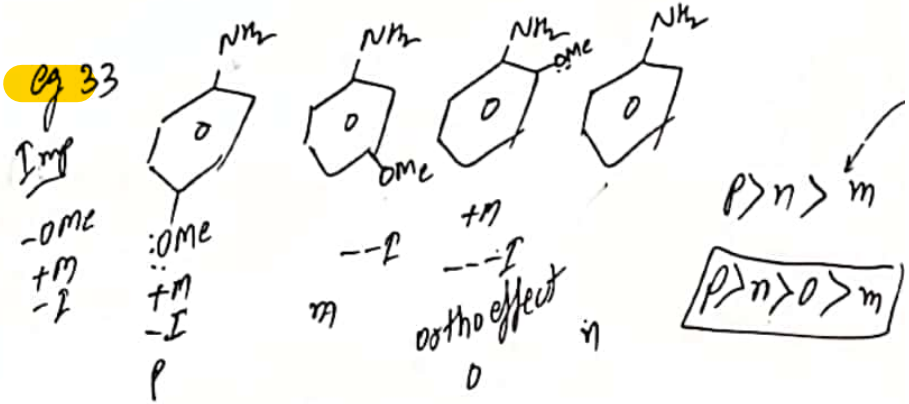
basic strength $p > m > n > o$

eg 32



$n > p > m$

eg 33



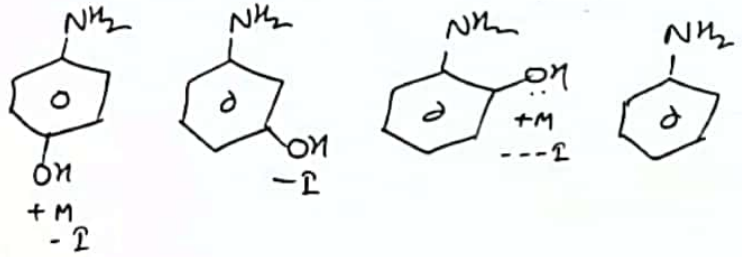
$p > n > m$

$p > n > o > m$

(III) Basic Strength

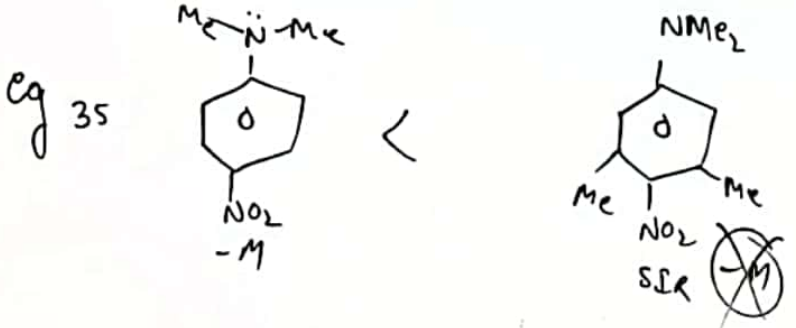
Basic strength of aromatic amine.

Eg 34

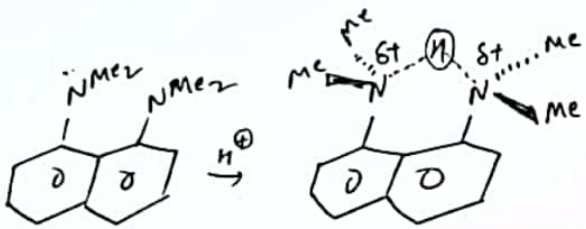


p > o > m

Eg 35

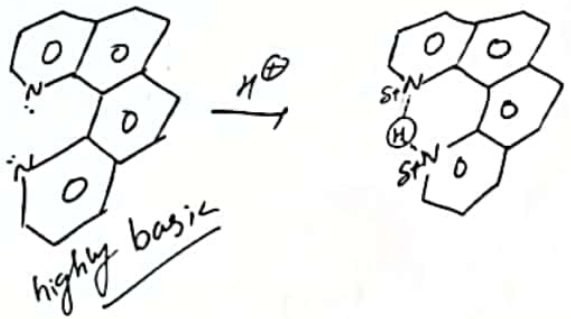


Eg 36



highly Basic

Eg 37



highly basic

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(IV) Acid-Base Reactions:-

Condition for acid-base reaction:- Acid of reactant side should be stronger than acid of product side/ base of reactant side should be stronger than base of product side.

$$\underset{\text{Acid}}{A-H} + \underset{\text{Base}}{B^{\ominus}} \longrightarrow \underset{\text{Acid}}{B-H} + \underset{\text{Base}}{A^{\ominus}}$$

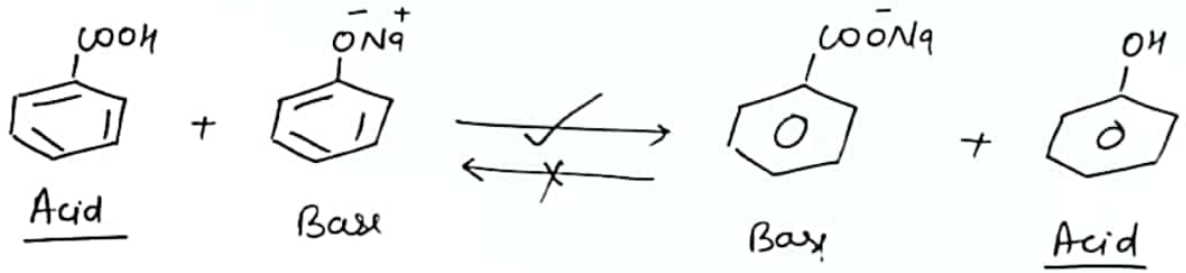
$A-H > B-H$ acidic Strength

$B^{\ominus} > A^{\ominus}$ Basic Strength

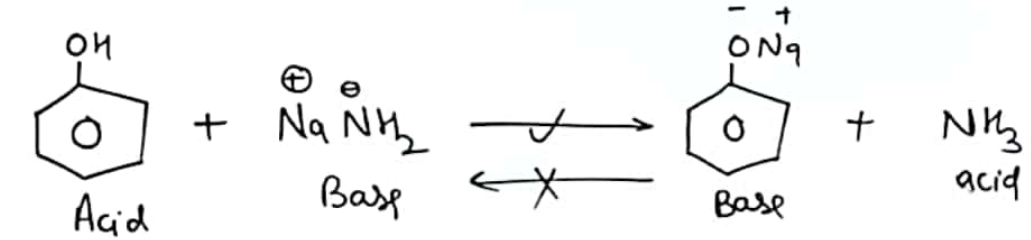
GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(IV) Acid-Base Reactions:-

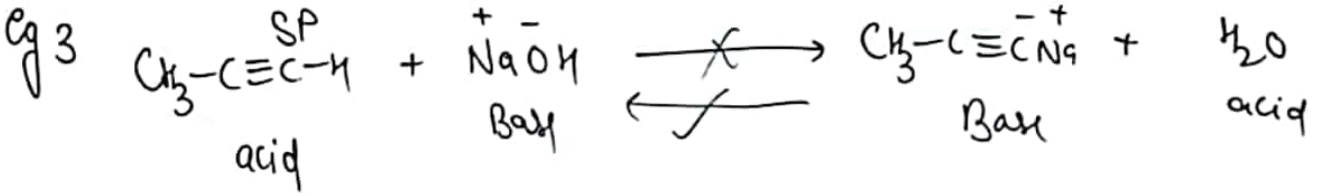
Eg 1



Eg 2



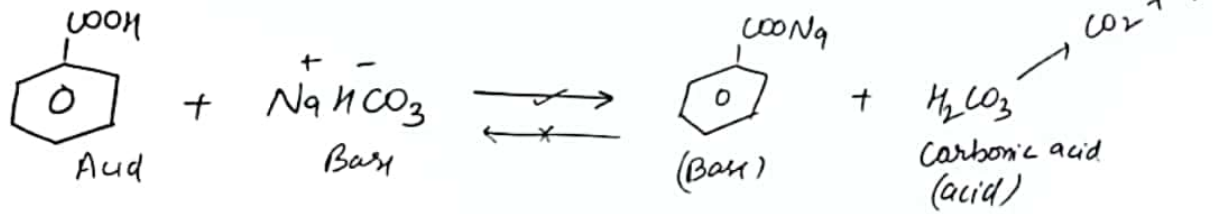
Eg 3



GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(IV) Acid-Base Reactions:-

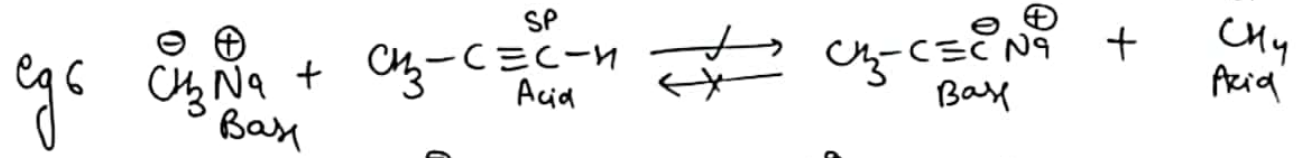
Eg 4



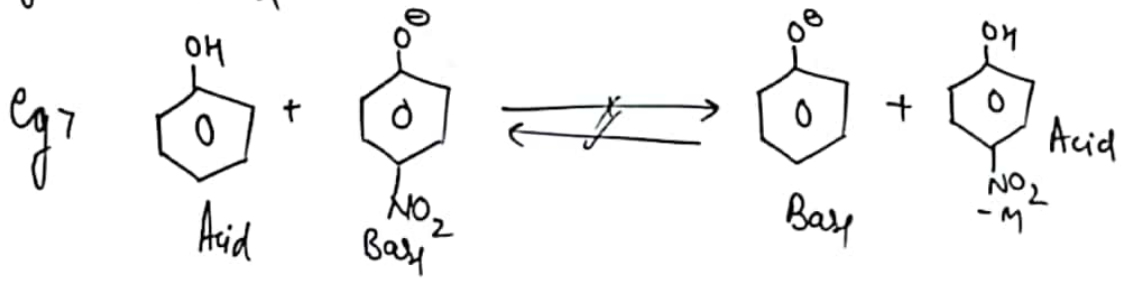
Eg 5



Eg 6



Eg 7

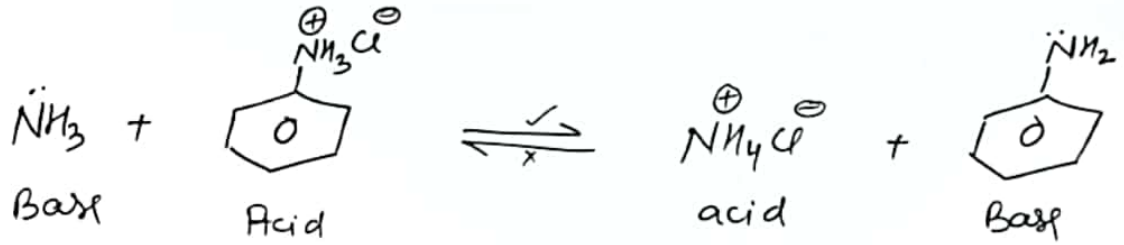


$R\text{-}COOH > H_2CO_3 > Ph\text{-}OH$

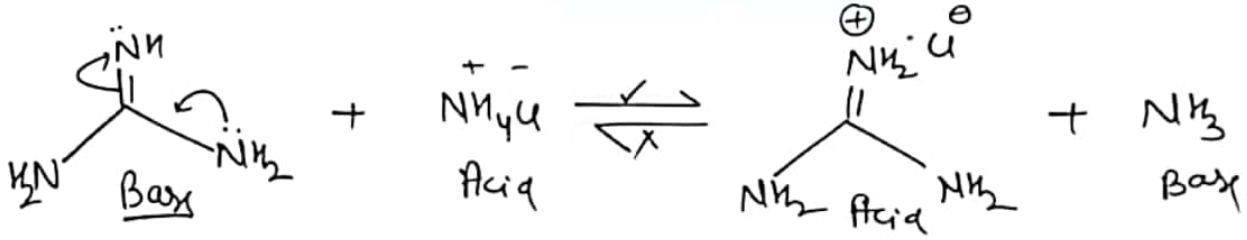
GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(IV) Acid-Base Reactions:-

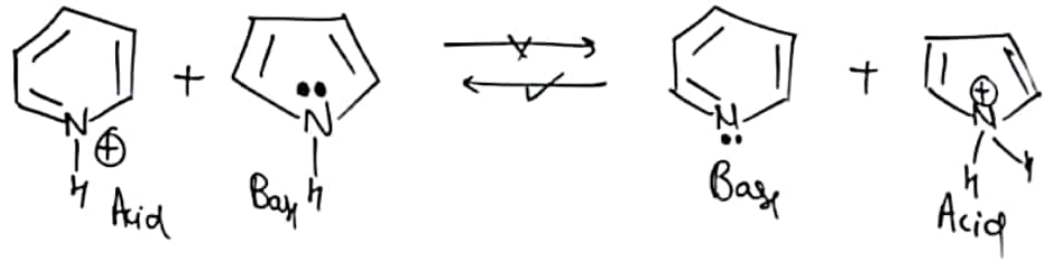
eg 8



eg 9



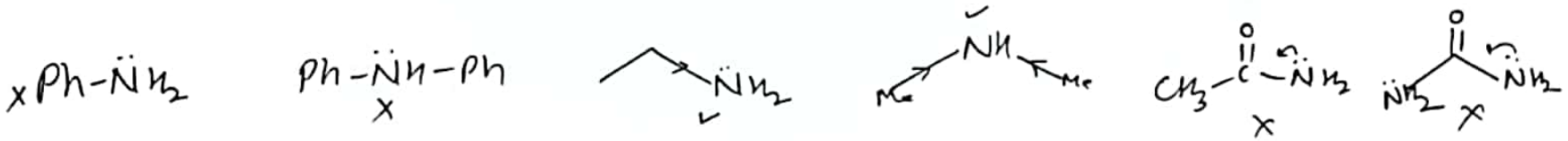
eg 10

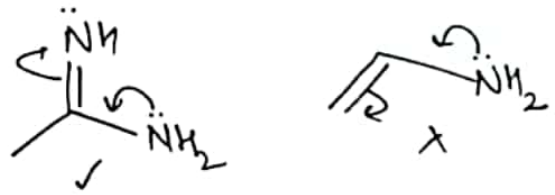


GENERAL ORGANIC CHEMISTRY - II (GOC-II)

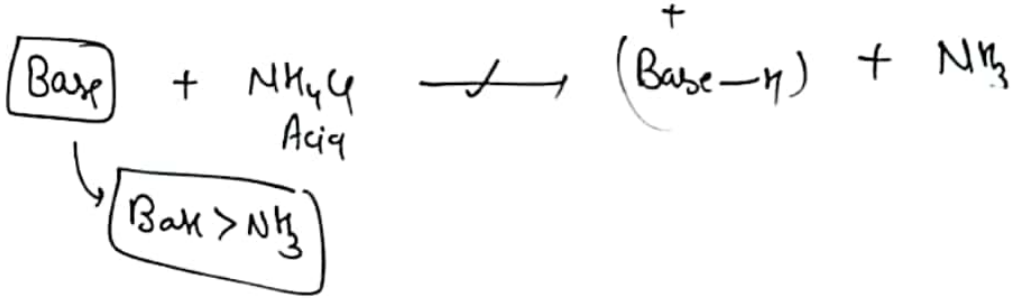
(IV) Acid-Base Reactions:-

Q Identify base with can form NH_3 with NH_4Cl





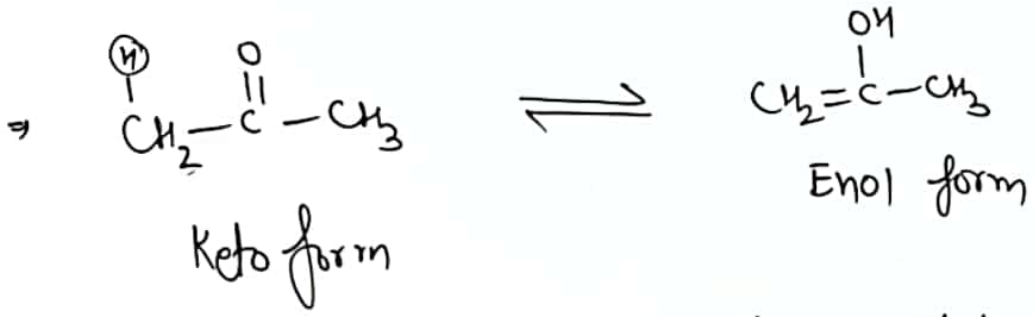
Sol.



GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(V) Tautomerism / Desmotropism / Alleloprotism / Kryptomerism etc.

Such type of structural isomers which are interconvertible into each other by changing position of an atom (generally hydrogen atom)



Keto form

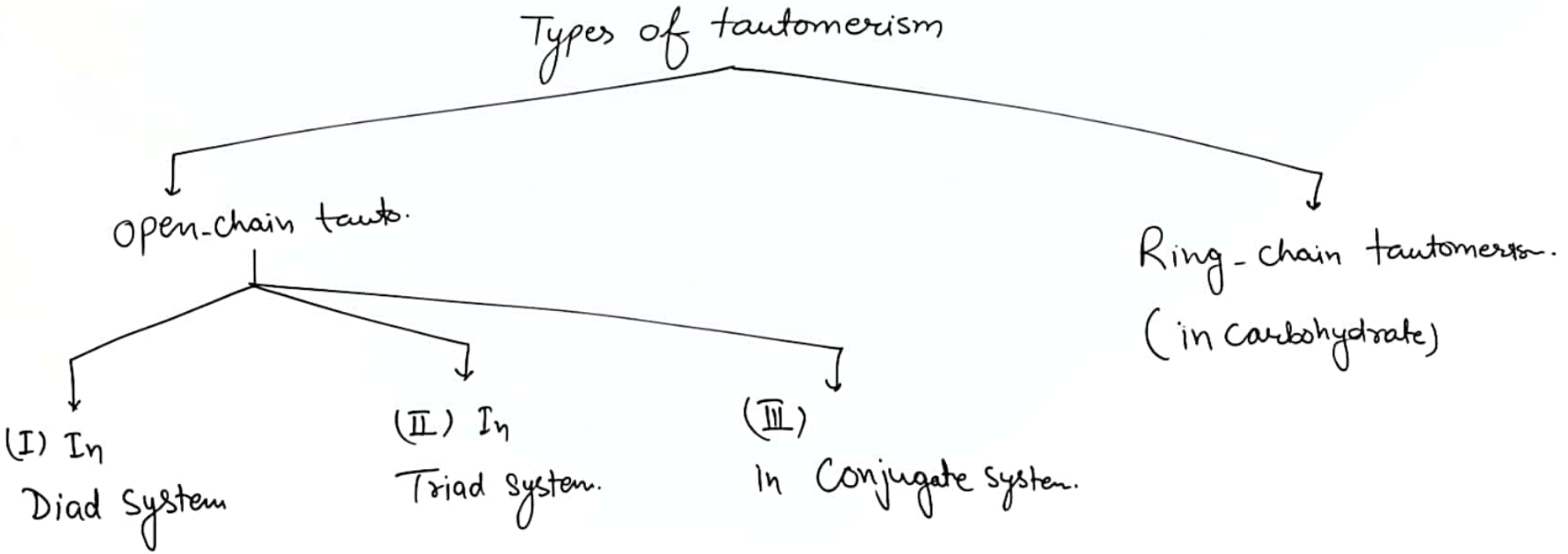
Enol form

⇒ Tautomers are exist in equilibrium state in gas, liq. and solution (in acid / Base)

⇒ Tautomers are isolable isomers.

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(V) Tautomerism / Desmotropism / Alleloprotism / Kryptomerism etc.



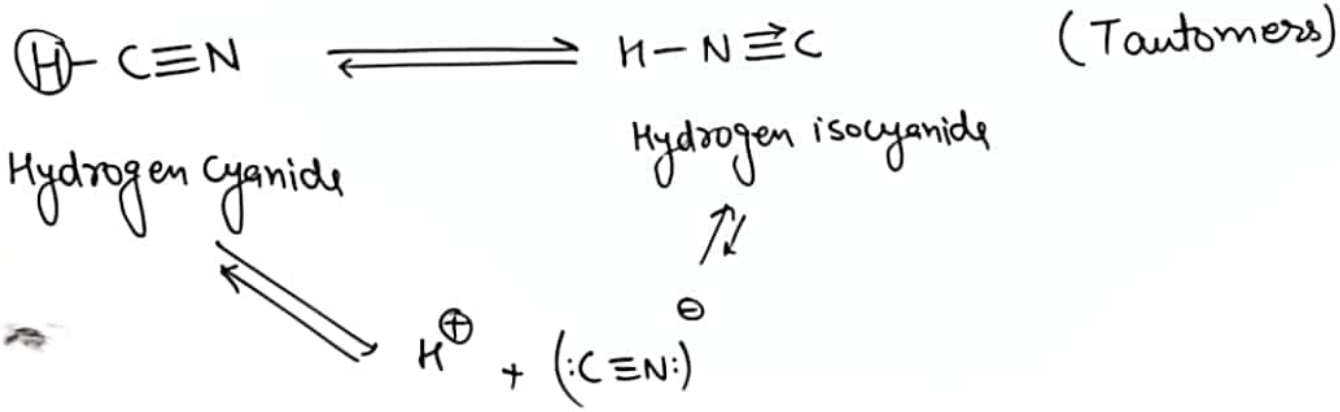
GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(V) Tautomerism / Desmotropism / Alleloprotism / Kryptomerism etc.

(I) Tautomerism in diad System.

When hydrogen shifts from 1st to 2nd position.

eg



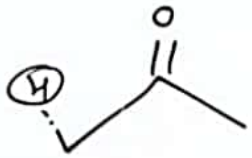
Note $R-C\equiv N \not\rightleftharpoons R-NC$ ⇒ Not Tautomers.

⇒ functional Isomers.

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(V) Tautomerism / Desmotropism / Alleloprotism / Kryptomerism etc.

(II) Tautomerism in Triad system.



When H/D shifts from 1st to 3rd ---





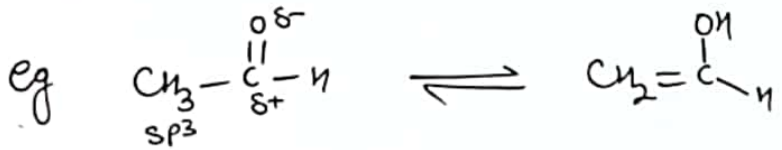


GENERAL ORGANIC CHEMISTRY - II (GOC-II)

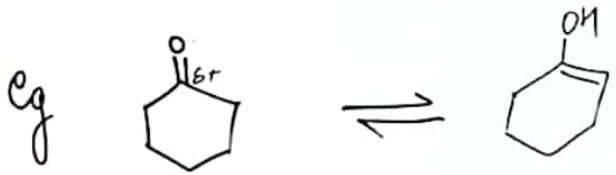
(V) Tautomerism / Desmotropism / Alleloprotism / Kryptomerism etc.

Examples of Triad .. tauto.

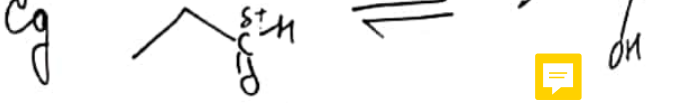
(A) Keto-enol tautomerism.

eg 

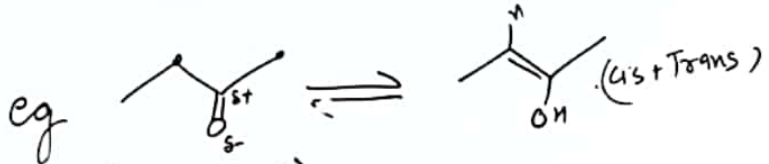
3. enolisable H

eg 

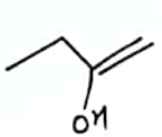
4 enolisable H

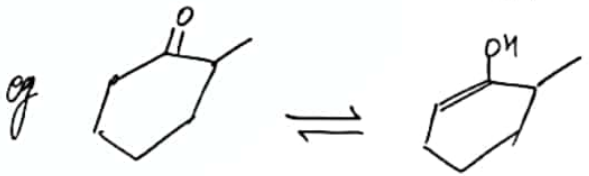
eg 

2 enolisable H

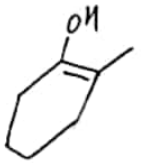
eg 

6 enolisable H



eg 

3 endism



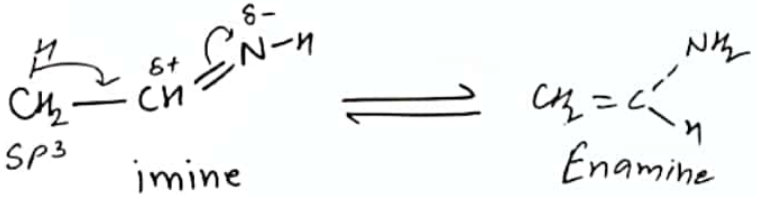
GENERAL ORGANIC CHEMISTRY - II (GOC-II)

(V) Tautomerism / Desmotropism / Alleloprotism / Kryptomerism etc.

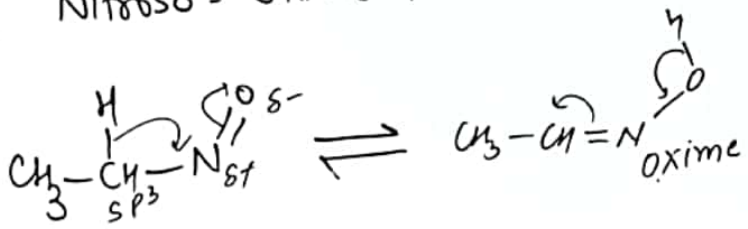
cyclic ester = Lactone
cyclic amide = Lactam

Examples of Triad .. tauto.

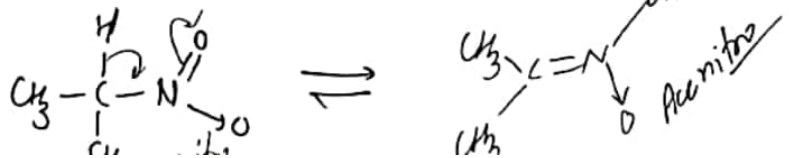
(B) Imine - Enamine tauto.



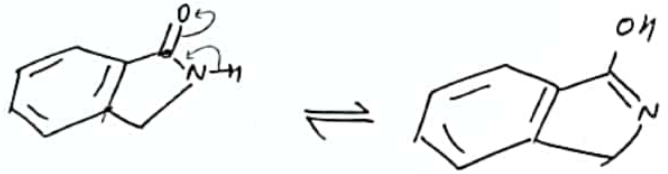
(C) Nitroso - Oxime tauto.

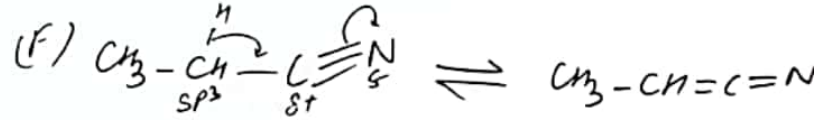


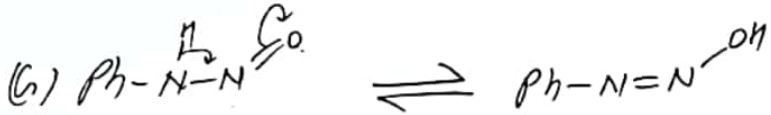
(D) Nitro - acenitro tauto.

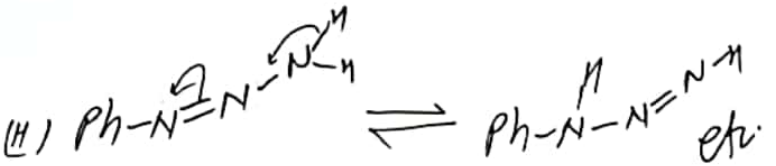


(E) Lactam - Lactim tauto.



(F) 

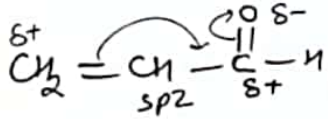
(G) 

(H) 

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

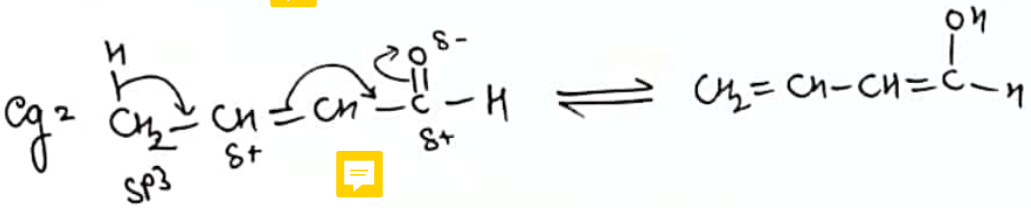
(V) Tautomerism / Desmotropism / Alleloprotism / Kryptomerism etc.

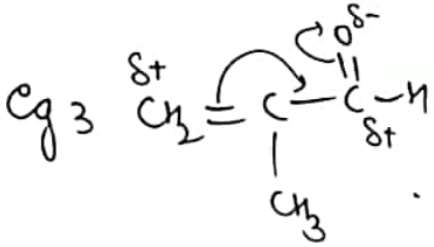
Examples of conjugate tauto.

Eg 1 

Reso. = ✓

Tauto = ✗

Eg 2 

Eg 3 

Reso = ✓

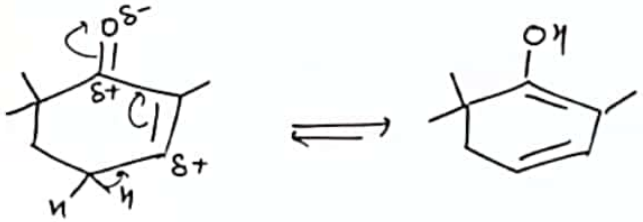
Tauto = ✗

GENERAL ORGANIC CHEMISTRY - II (GOC-II)

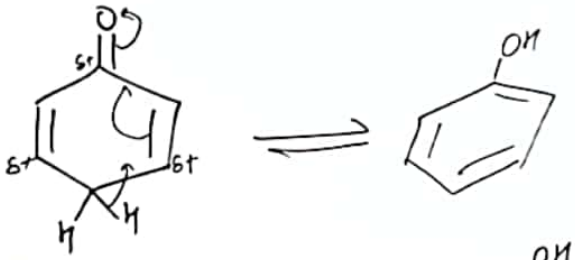
(V) Tautomerism / Desmotropism / Alleloprotism / Kryptomerism etc.

Examples of conjugate tauto.

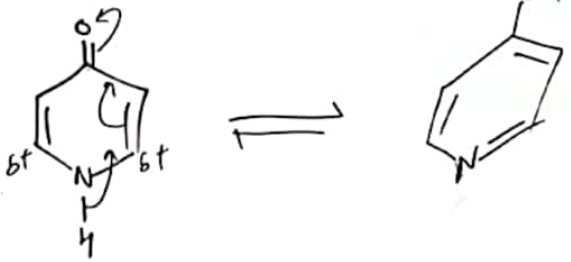
eg 4



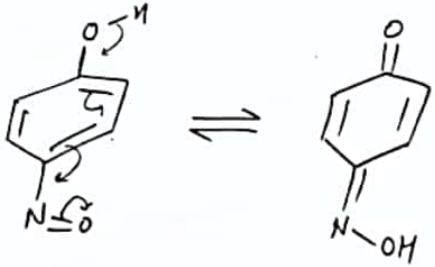
eg 5



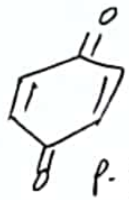
eg 6



eg 7



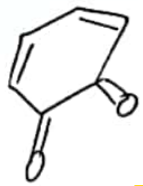
eg 8



No Tauto.

p. quinone

eg 9

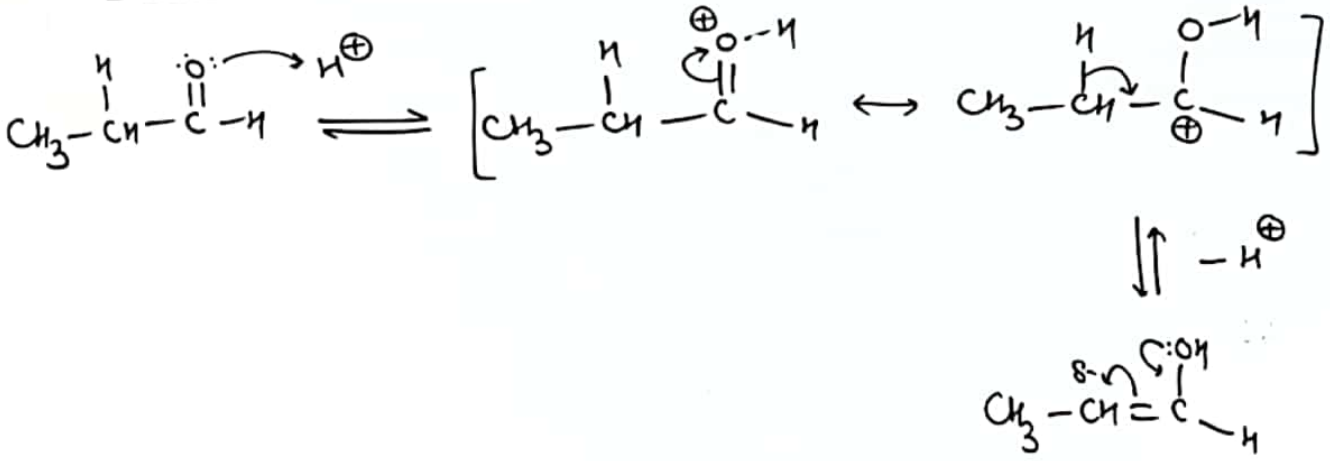


No Tauto

(V) Tautomerism :-

⇒ Mechanism of tauto (keto-enol tauto)

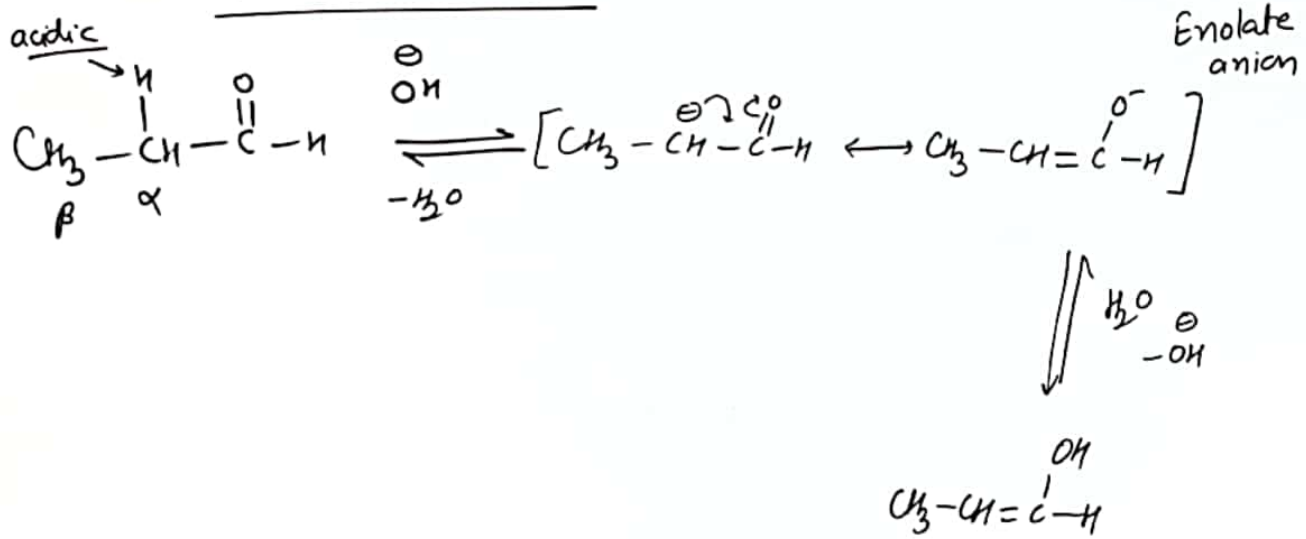
(i) In acidic medium :-



(V) Tautomerism :-

⇒ Mechanism of tauto (keto-enol tauto)

(ii) In basic medium :-

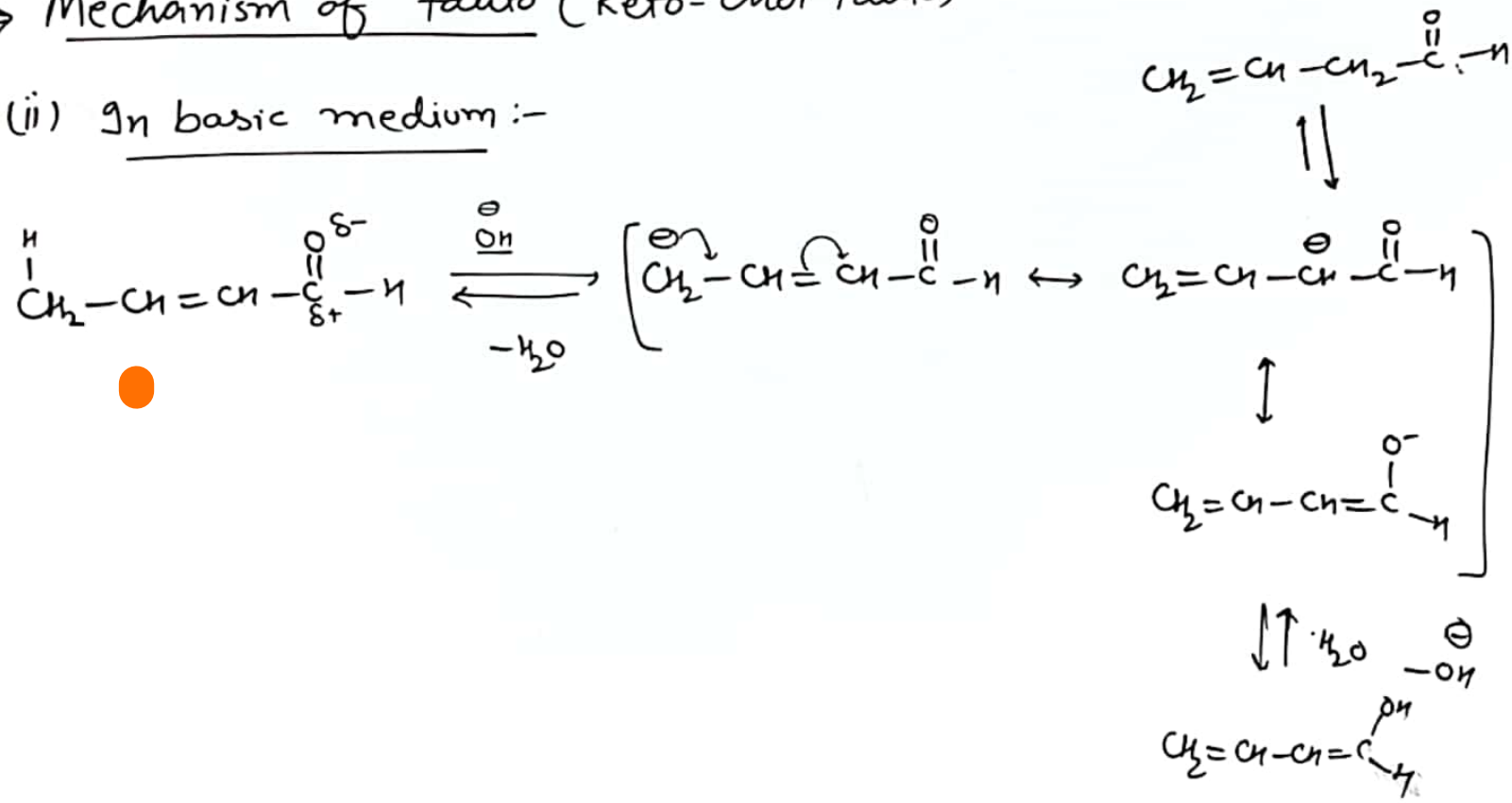


GENERAL ORGANIC CHEMISTRY - II GOC-II

(V) Tautomerism :-

⇒ Mechanism of tauto (keto-enol tauto)

(ii) In basic medium :-



GENERAL ORGANIC CHEMISTRY - II GOC-II

(V) Tautomerism :-

Note :-

① Generally Keto forms are more stable than enol form (Thermodynamically)

due to

⇒ Bond energy of keto form > Bond energy of enol form

⇒ Resonance energy of Keto form > Reso. energy of enol form.



(V) Tautomerism :-

Note :-

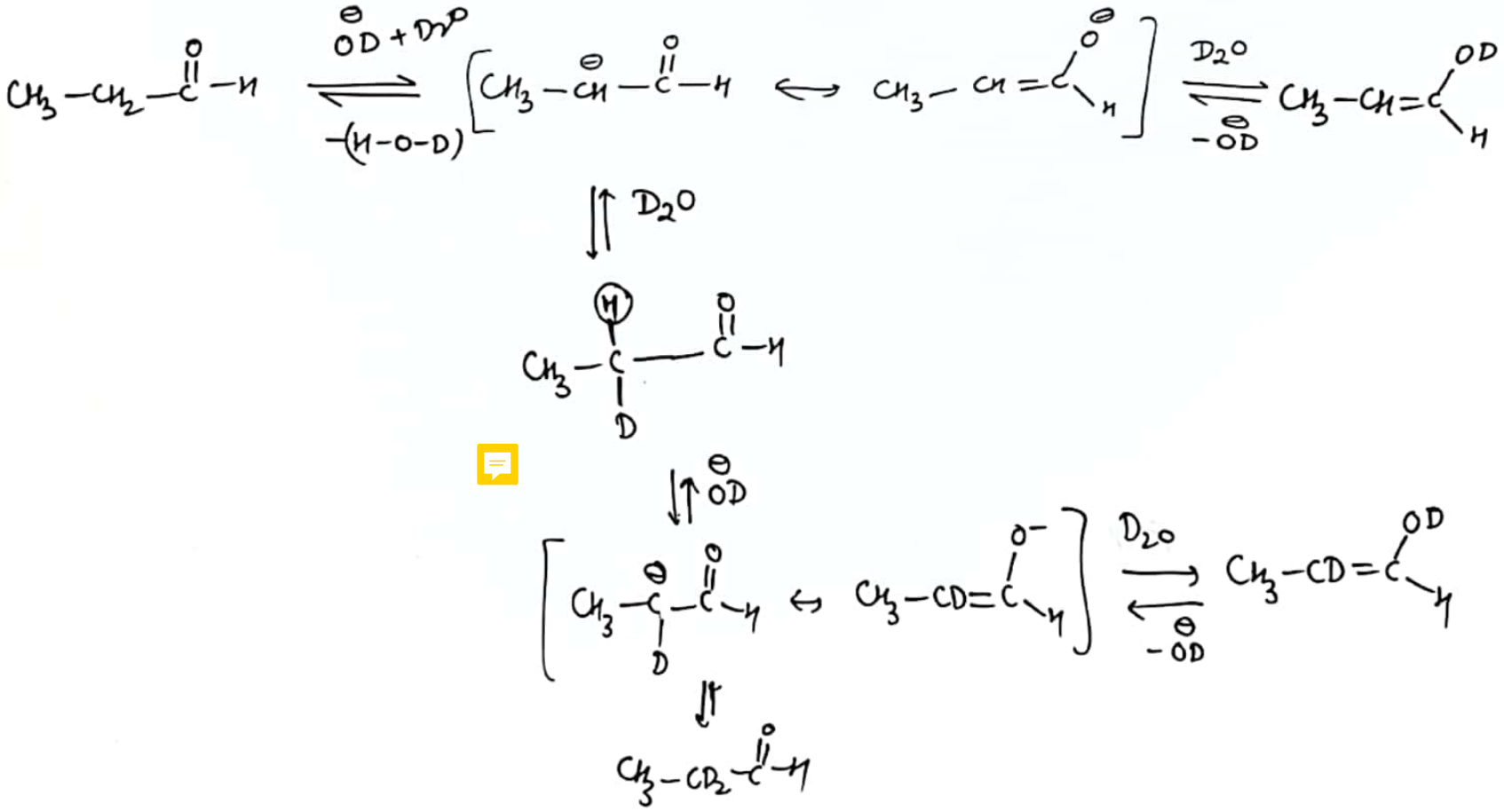
② All enolisable hydrogens are repleced by D in presence of $\overset{\ominus}{O}D + D_2O$ / $D^{\oplus} + D_2O$, It is called D-exchange.

Eg 1 $CH_3-\overset{O}{\overset{\|}{C}}-H \xrightarrow{D_2O + \overset{\ominus}{O}D} CD_3-\overset{O}{\overset{\|}{C}}-H$

Eg 2 $CH_3-\underset{-}{CH_2}-CHO \xrightarrow{D_2O + \overset{\ominus}{O}D} CH_3-CD_2-\overset{O}{\overset{\|}{C}}-H$

Eg 3 

(V) Tautomerism :-



GENERAL ORGANIC CHEMISTRY - II GOC-II

(V) Tautomerism :-

eg $CH_3-CH=CH-\overset{O}{\overset{\|}{C}}-H \xrightarrow{\overset{\ominus}{O}D + D_2O} CD_3-CH=CD-\overset{O}{\overset{\|}{C}}-H$

$\downarrow$

$\overset{\ominus}{C}H_2-CH=CH-\overset{O}{\overset{\|}{C}}-H$

$\updownarrow$

$CH_2=CH-\overset{\ominus}{C}H-\overset{O}{\overset{\|}{C}}-H$

$\updownarrow$

$CH_2=CH-CH=\overset{O^-}{\overset{|}{C}}-H$

---

eg $CH_2=CH-CH_2-\overset{O}{\overset{\|}{C}}-H \underset{\longleftarrow}{\overset{\ominus}{D}O + D_2O} CD_3-CH=CD-\overset{O}{\overset{\|}{C}}-H$

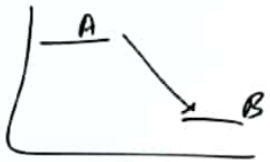
GENERAL ORGANIC CHEMISTRY-II (GOC-II)

(V) Tautomerism

Keto ⇌ enol

A ⇌ B

% Enol content in Keto-enol equilibrium:-



① % Enol content ∝ Stability of enol form

(due to resonance
" " aromatic
" " H-bonding etc.)

② % enol content ∝ acidic Strength of enolisable hydrogen

* ③ % enol " " ∝ Temp.

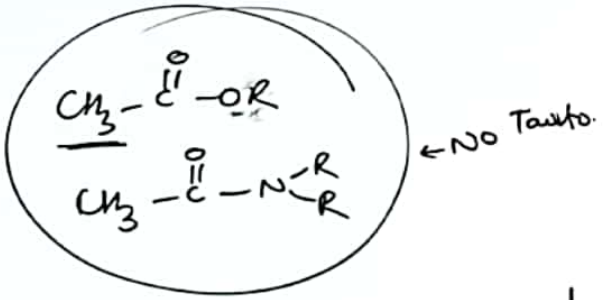
④ % enol " ∝ $\frac{1}{\text{Polarity of solvent}}$ (Polarity keto > enol)

$\chi=0$ $\chi=\infty$

GENERAL ORGANIC CHEMISTRY-II (GOC-II)

(V) Tautomerism

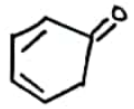
% Enol content in Keto-enol equilibrium:-

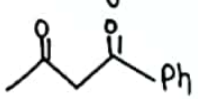


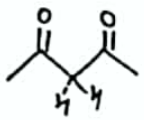
⇒ General order of % enol content:-

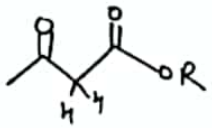
Aromatic (Phenol) > β-Diketones with aromatic ring > β-Diketones without aromatic ring > β-Keto-esters > β-Diester > Normal aldehyde and Ketone > ester

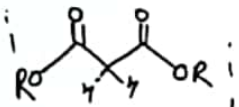
eg.

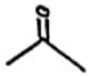


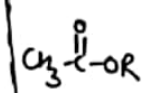










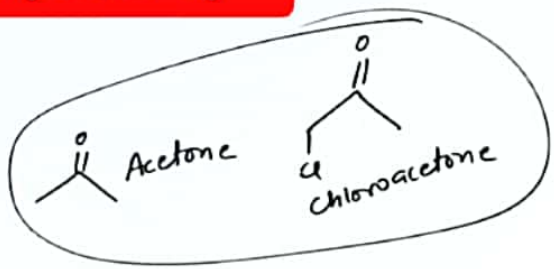


~100% enol | 90-99% | ~76% | ~7% | ·1% | 10^{-3} to 10^{-6} % | 0%

GENERAL ORGANIC CHEMISTRY-II (GOC-II)

(V) Tautomerism

% Enol content in Keto-enol equilibrium:-



eg 1



Keto form

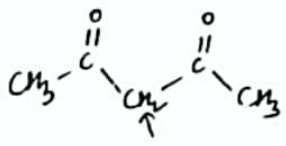
Non aro.

~ 100%

→ highly stable

→ (aromatic)

eg 2

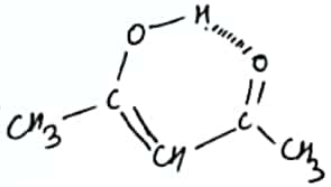


more acidic

Acetyl acetone

→ conjugation = X

Reso = X



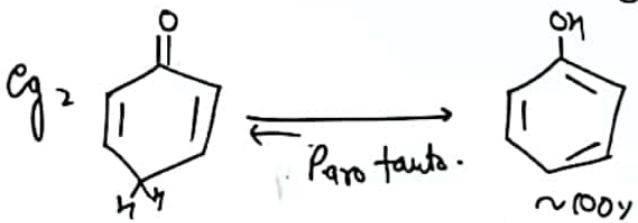
~ 76 %

conjugation = ✓

Reso = ✓

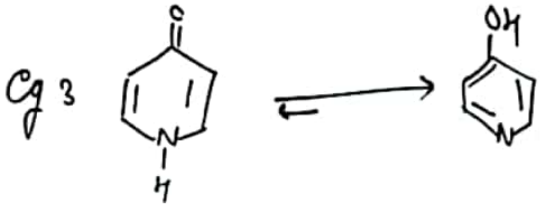
intramolecular H.Bonding = ✓

eg 2



~100%

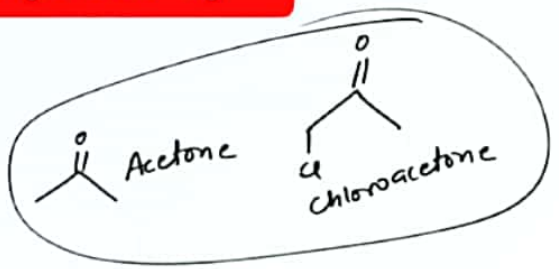
eg 3



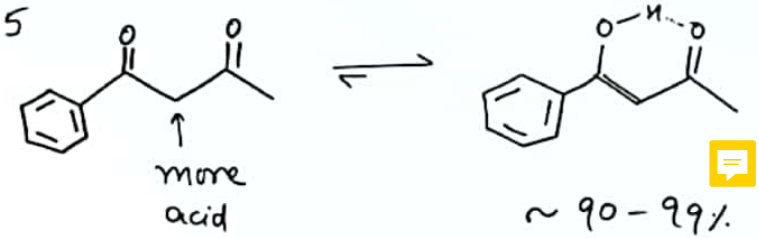
GENERAL ORGANIC CHEMISTRY-II (GOC-II)

(V) Tautomerism

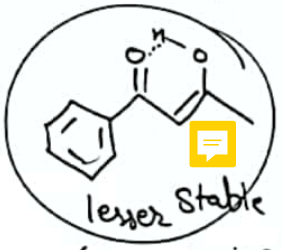
% Enol content in Keto-enol equilibrium:-



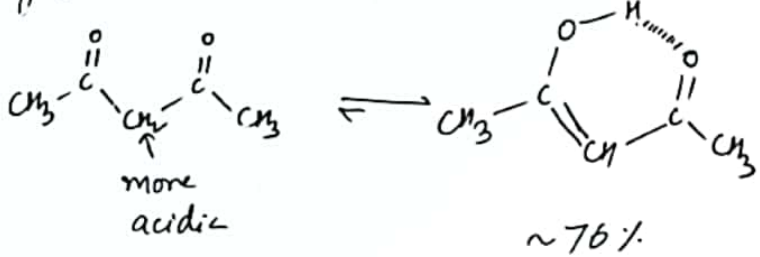
eg 5



~ 90 - 99%.



eg 4



~ 76%.

Acetyl acetone

→ conjugation = X

Reso = X

conjugation = ✓

Reso = ✓

intramolecular H.Bonding = ✓

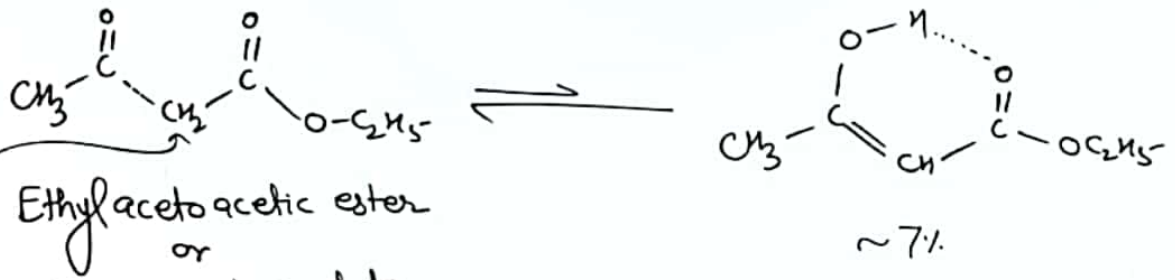
GENERAL ORGANIC CHEMISTRY-II (GOC-II)

(V) Tautomerism

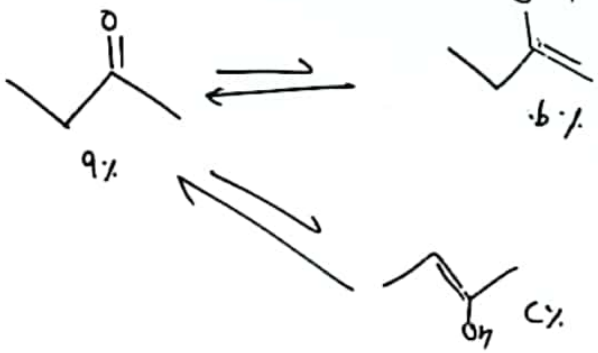
% Enol content in Keto-enol equilibrium:-



eg 6

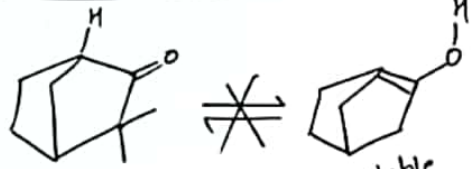


eg 7

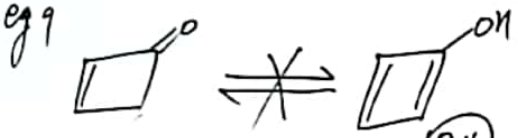


9% >>>> C > b

eg 8



eg 9

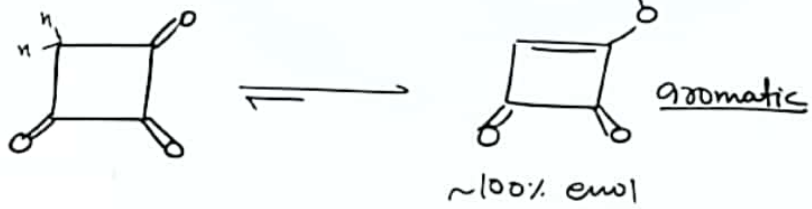


GENERAL ORGANIC CHEMISTRY-II (GOC-II)

(V) Tautomerism

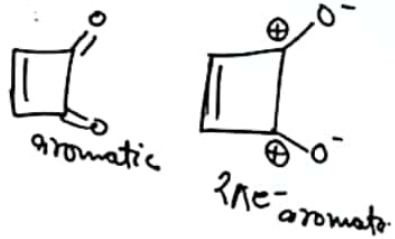
% Enol content in Keto-enol equilibrium:-

eg 10

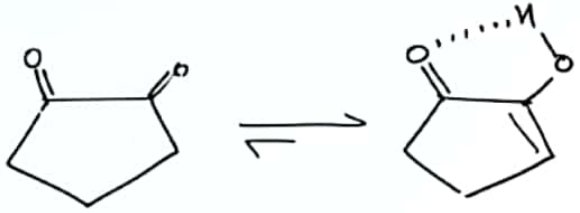


Hint

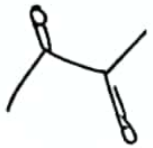


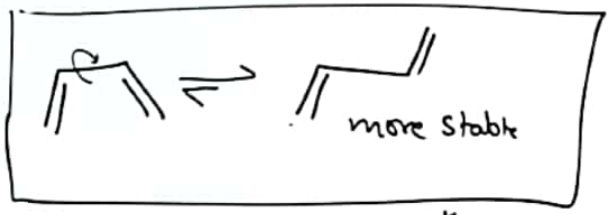


eg 11

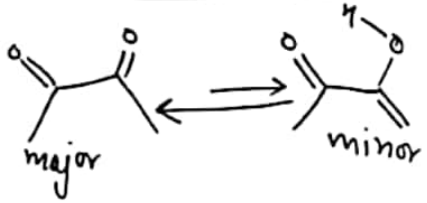


lesser π electronic repulsion





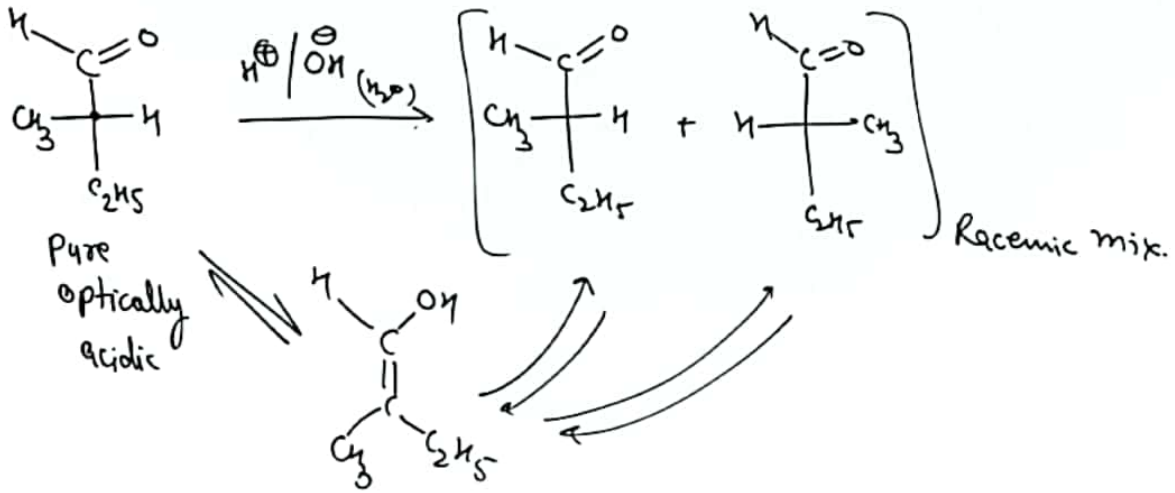
eg 12



GENERAL ORGANIC CHEMISTRY-II (GOC-II)

(V) Tautomerism

Note · When enolisable hydrogen is present on chiral carbon, then in acidic/Basic medium more than one Stereoisomeric products are form due to tautomerism.



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